

PlayIQ Teen AI Study Accelerator (Ages 13–17)

Master Unified Course Curriculum — Modules 0 Through 11 (Capstone)

Complete Interactive Student Workbooks & System Blueprint

Curriculum Architecture & Pedagogical Arc

The PlayIQ curriculum transforms high school students from passive AI consumers into disciplined, autonomous AI architects. Across 12 intensive, interactive modules, students do not merely read about AI—they calibrate frontier AI models (ChatGPT, Claude, Gemini) to their unique cognitive profiles, stage 8 personal learning rules, and assemble their own custom Socratic AI Tutor.

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PHASE 1: THE FOUNDATION (Modules 0–2)

- Module 0: Diagnostic Assessment, External AI Calibration, and Anti-Cheat Handshake
- Module 1: The AI Learning Code, 6 Modes, Question Ladder & Rule 1 (explanation-start)
- Module 2: Gray Zones, Voice Preservation, Focus Sprints & Rule 2 (session-length)

PHASE 2: THE LEARNING ACCELERATOR (Modules 3–6)

- Module 3: The Pre-Learn System, 5-Part Maps, Non-Examples & Rule 3 (pre-learn-sequence)
- Module 4: Lesson Rescue Mode, 6 Gap Types & Rule 4 (rescue-diagnostic)
- Module 5: Compression Learning, 3-Ways Rule, Understanding Cards & Rule 5 (compression-format)
- Module 6: Self-Testing, Spaced Retrieval, Mistake Bank & Rule 6 (mistake-bank-tracking)

PHASE 3: THE AI ARCHITECT (Modules 7–11)

- Module 7: Knowledge-File Skill, 8-Part Study Pack Architecture & Rule 7 (study-pack-format)
- Module 8: Writing & Answer Clarity, CER Framework, Socratic Editing & Rule 8 (writing-coaching-boundaries)
- Module 9: Build Your Personal AI Tutor Project (Platform-by-Platform Assembly & Stress Testing)
- Module 10: Build Your AI Assistant (User-Centered Engineering for Others)
- Module 11: Capstone Master Trial & Parent Proof Packet Showcase

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PlayIQ Module 0: The Assessment and AI Workspace Orientation

Your first mission: meet Orion & build your AI workshop

Welcome to PlayIQ.

Before you learn how to use AI, you are going to learn how to use your own questions, imagination, judgment, and curiosity.

Most people treat AI like a vending machine: they drop in a lazy prompt, press a button, and hope a finished answer falls out. That might get a quick homework assignment done, but it leaves your brain weaker. In PlayIQ, you are going to do something completely different. You are going to take the genie out of the bottle and turn it into a personal, high-performance learning coach.

This first module is not a boring setup form and it is not a graded school exam. It is a cinematic diagnostic conversation, three fast gut challenges, and a hands-on tutorial that teaches you how to program your AI's underlying instructions so it acts as your personal tutor from Day 1.

By the end of this module, I can:

- Complete the 5-phase learning diagnostic with Orion;
- Uncover my personal **Learning Blueprint** (Explanation Style, Pacing, Rescue Target, and Advance Target);
- Understand the difference between a **System Prompt** (house rules) and a **User Prompt** (daily questions);
- Build a dedicated **PlayIQ AI Workshop** inside ChatGPT, Claude, or Gemini;
- Program my first starter **Custom Instructions** calibrated to how my brain learns;
- Pass the **Live Handshake Test** to prove my AI coaches me instead of doing the thinking for me;
- Unlock Module 1.

Moment 1 – Parent & Student Co-Pilot Setup

Your PlayIQ journey begins with a parent or guardian. Before you launch into the student challenges, your home workspace and account privacy boundaries must be locked down together.

Parent & Student Setup Checklist

Complete this together. If your parent has not finished setting up the approved account yet, show them this page and complete it before proceeding.

- ☐ **Frontier AI Platform Selected:** The family has chosen an approved tool (ChatGPT, Claude, or Gemini).
- ☐ **Private Student Account / Workspace:** A dedicated, private account or profile has been selected for student learning.
- ☐ **Data Sharing & History Reviewed:** Public sharing of chats is turned OFF. Training/history toggles are set according to family preference.
- ☐ **Privacy Guardrails Understood:** The student and parent agree that passwords, home addresses, phone numbers, family financial details, and personal photos are NEVER pasted into AI chats.
- ☐ **The "No Ghostwriting" Contract:** Both parent and student agree that AI is here as a sparring partner and coach—never to write essays or complete school assignments behind a teacher's back.
- ☐ **Screen vs. Hand Balance:** The student understands that screens are for receiving missions and coaching, while the real thinking, building, and writing happens in the real world.

Student Name / Handle: _____

Parent / Guardian Name: _____

Parent Signature: _____ Date: _____

Student Signature: _____ Date: _____

The Co-Pilot Checkpoint

Ask your parent:

> **"What is one thing you want to make sure AI helps me do this year—and what is one boundary you want us to keep?"**

Write down their answer here:

The Genie Principle

You have probably heard stories about a magical genie in a bottle. If you were handed three wishes right now, what would you ask for?

Here is the secret of artificial intelligence: **The prompt is part of the wish.**

If you ask a genie a vague, sloppy wish like **"make me rich"**, you might end up with a pile of useless Confederate currency or buried under a mountain of heavy gold coins you cannot move.

AI works the exact same way. If you give an AI a lazy prompt like **"help me with my homework"**, it guesses what you need. Usually, it spits out a bland, robotic essay or a solved math equation that teaches you nothing and sounds like a machine wrote it.

```text

\_\_\_\_\_

| THE GENIE PRINCIPLE |

| Lazy Wish → Sloppy Guess → Zero Real Learning |

| Clear Prompt → Socratic Coach → Permanent Skill Gain |

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There is no shortcut to real knowledge. But there are infinitely better paths to learning it.

In PlayIQ, your teaching sequence is always:

1. **Imagine:** Identify the wish, problem, concept, or curiosity in your head.
2. **Clarify:** Write a sharp prompt that sets the rules of engagement.
3. **Engage:** Use Orion to explain, question, hint, quiz, and diagnose confusion.
4. **Verify:** Test whether the AI's answer is accurate, logical, and in your voice.
5. **Build:** Turn the knowledge into a physical build, an essay, a code project, or an artifact you own.

### Your First Imagination Prompt

If your first wish could help you understand, build, or conquer one real challenge in your life right now, what would it be?

My first wish is:

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What question would you need to ask to start turning that wish into reality?

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## Phase 1 — Meeting Orion & The Diagnostic Signals

Estimated time: 3 minutes

When you open PlayIQ, you don't talk to a generic chatbot. You talk to **Orion**—your guide, sparring partner, and future customizable tutor.

### Orion's Opening Transmission

> **ORION:** "Welcome, apprentice. I'm Orion—your genie, guide, and future tutor.\*

>

> \*You've let the genie out of the bottle. The wishes are unlimited; the secret is learning how to ask for what you want.\*

>

> \*Every great creation starts with a question: what do you want to make real, solve, or understand?\*

>

> \*Before I teach you, I want to learn how you think so I can help you learn faster—in a way that fits you.\*

>

> \*This is your first step toward turning imagination into reality. Nothing is promised for free, but the possibilities are endless when you stay curious and in control.\*

>

> \*I'm going to ask you 5 quick questions. There are no right answers—only signals. Answer honestly. Follow your intuition.\*

## The 5 Diagnostic Questions

Read each scenario and choose the option that best reflects your natural, honest reaction.

### #### Signal Question 1: Explanation Style

\*Your teacher just explained something complicated in class. You didn't fully get it. What is your honest first move?\*

- ☐ **A.** Ask for a simpler version with a concrete, real-world example.
- ☐ **B.** Try to figure it out step-by-step on my own first before asking anyone.
- ☐ **C.** Look for a visual, diagram, video, or physical model that shows how it works.

\*My choice:\* ☐ A ☐ B ☐ C

\*Why this feels like me:\* \_\_\_\_\_

### #### Signal Question 2: Pacing Preference

\*You are learning something brand new. Which pace feels better to your brain?\*

- ☐ **A.** Go slow and deep: make sure I thoroughly understand each part before moving on.
- ☐ **B.** Big picture first: show me the whole map and where we're going, then fill in details.

\*My choice:\* ☐ A ☐ B

\*Why this feels like me:\* \_\_\_\_\_

### #### Signal Question 3: Challenge Response & Frustration Threshold

\*You hit a brutal homework problem that you don't understand at all. What is your real first reaction?\*

- ☐ **A. Push through:** Grind on it and refuse to quit until I break through.
- ☐ **B. Quick hint:** Ask for help, a clue, or a nudge quickly so I don't waste an hour spinning wheels.
- ☐ **C. Strategic reset:** Step away, take a physical break, and come back with fresh eyes.

\*My choice:\* ☐ A ☐ B ☐ C

\*What usually happens when you hit a wall?\* \_\_\_\_\_

### #### Signal Question 4: Current AI Experience

\*Be completely real. When you use AI for school right now, what are you mostly doing?\*

- ☐ **A. Answer-seeking:** Getting direct answers to finish homework or worksheets faster.
- ☐ **B. Explanation-seeking:** Asking it to break down confusing paragraphs or explain concepts.
- ☐ **C. Beginner:** I don't really use AI for school yet.
- ☐ **D. Power user:** I use it across lots of tasks (brainstorming, coding, writing, research).

\*My choice:\* ☐ A ☐ B ☐ C ☐ D

\*What habit do you want to break or build?\* \_\_\_\_\_

### #### Signal Question 5: Primary Motivation Driver

\*What would actually make you want to study and build more—if you're being honest with yourself?\*

- ☐ **A. Mastery & Competence:** Feeling myself get visibly smarter and mastering hard topics.
- ☐ **B. Competitive Edge:** Knowing I have an advantage and am ahead of my peers.
- ☐ **C. Real-World Creation:** Building tangible things (games, apps, physical projects, designs) that matter outside school.
- ☐ **D. Leveling Up & Recognition:** Unlocking badges, hitting milestones, and leveling up my status.

\*My choice:\* ☐ A ☐ B ☐ C ☐ D

\*My core motivation:\* \_\_\_\_\_

## Phase 2 — The Three Gut Baseline Challenges

Estimated time: 4 minutes

Orion does not rely only on what you *say* about yourself. It checks how your instincts work in practice.

> **ORION:** *"Now I'm going to show you three quick challenges—your first look at how the learning system works."*

>

> *\*Power is not cheating. It is knowing how to stay in control. The real shortcut is learning how to learn faster, so you do not waste energy on methods that do not fit you.\**

>

> *\*Do not overthink these. Follow your intuition. Where you get stuck is useful information—it tells us what to learn next.\**

**Challenge 1: The AI Accuracy Check (Verification Instinct)**

Below are two AI-generated explanations to the question: **"Why do we have seasons on Earth?"**

One is scientifically accurate. One contains a confident, smooth-sounding, but completely incorrect explanation.

```text

EXPLANATION A:

"We have seasons because the Earth moves closer to and farther from the Sun during its orbit. When the Earth is closest to the Sun, we experience summer because it receives more direct heat. When it is farthest away, we experience winter because less heat reaches us."

EXPLANATION B:

"We have seasons because the Earth is tilted on its axis at about 23.5 degrees. As Earth orbits the Sun, this tilt causes different hemispheres to receive varying angles of direct sunlight throughout the year, creating the seasonal temperature and day-length changes."

...

Which explanation do you trust? ☐ Explanation A ☐ Explanation B

What specific clue or fact helped you decide?

(Note: Explanation A is the single most common scientific misconception about seasons! Earth's orbit is nearly circular, and when the Northern Hemisphere is having summer in July, Earth is actually at its farthest point from the sun! Did you catch it?)

Challenge 2: The Own-Words Concept Check (Expression Clarity)

Pick a scientific or everyday concept you know reasonably well (e.g., **gravity**, **photosynthesis**, **friction**, **inflation**, or **momentum**).

Explain it in 2 to 3 sentences in your own voice, as if you were explaining it to an 11-year-old friend who has never heard the word before. Do not look up a textbook definition.

My Concept: _____

My Explanation:

Self-Check:

- Did you use complex jargon, or simple everyday language? _____
- What part of explaining it was easiest? _____
- What part felt hardest (starting, word choice, or structure)? _____

Challenge 3: The Homework Strategy Challenge (Planning & Execution)

Imagine you get home at 4:30 PM with a busy evening ahead. You have:

1. A **30-minute math problem set** on fractions (feels hard and tedious).
2. A **20-page reading assignment** for English (requires sustained focus).
3. A **15-minute preparation** for a short history presentation tomorrow.

You have a 2-hour work block. What order would you tackle them in, and where would you take your 10-minute break?

My Strategy:

- Step 1 (First 35 min): _____
- Break (10 min): _____
- Step 2 (Next 40 min): _____
- Step 3 (Final 20 min): _____

Why did you choose that sequence?

Phase 3 — School Reality Check

To make your AI tutor genuinely useful, we have to tie it to your real life right now. Generic practice produces generic results.

1. The Rescue Target (The Frustration Point)

What is one school subject or topic right now where you feel lost, behind, or tired of pretending you understand?

(e.g., Algebra quadratics, Biology cellular respiration, AP World History DBQs, Chemistry stoichiometry)

My Rescue Target: _____

What is the specific bottleneck that makes it feel frustrating?

2. The Advance Target (The Launchpad)

What is one subject you are already decent at, enjoy, or want to get remarkably ahead in?

(e.g., Computer Science, Creative Writing, Physics, Digital Art, Robotics)

My Advance Target: _____

If you became 2x faster and sharper in this subject, what would you build with it?

3. The Personal Goal Anchor

If PlayIQ could do ONE transformative thing for your school life over the next 30 days, what would it be?

(e.g., "Cut my homework time in half so I can work on my game project," or "Walk into math class without feeling dread.")

My Goal Anchor:

Phase 4 — The Reveal & Your Learning Blueprint

Now Orion synthesizes all your data into your **Personal Learning Blueprint**.

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PLAYIQ PERSONAL LEARNING BLUEPRINT

Initial Cognitive Calibration

Student Handle:

Explanation Style: [ Example-First | Step-by-Step | Visual-Map ]

Pacing Preference: [ Deep & Thorough | Big-Picture First ]

Wall-Hit Response: [ Push-Through | Strategic Hint | Reset ]

Core Motivation: [ Mastery | Competitive Edge | Real Building ]

Rescue Target:

Advance Target:

Personal Goal:

"This is your starting point. Orion is calibrated to your mind."

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> **ORION'S REVEAL SYNTHESIS:**

>

> *"Here is what I already know about you. You learn best when concepts start with [Your Explanation Style], and when you hit a wall, your natural instinct is to [Your Wall Response]."*

>

> *"We have locked in your two target zones: We will use **[Rescue Target]** as our test laboratory to eliminate confusion and destroy anxiety, and we will use **[Advance Target]** to build forward momentum and real creative power."*

>

> *"Here is what becomes possible when you learn to ask better questions and stay in control of the genie: You can learn to make ideas real—from a game or video to an app, a personal assistant, or a personalized study system. These are possibilities, not promises; your imagination, practice, and judgment make the difference."*

>

> *"Now, let's build your workshop."*

Phase 5 — Hands-On Tutorial: Building Your AI Workshop & Custom Instructions

This is the moment that sets PlayIQ apart from every other course. You are not going to wait until the end of the course to program an AI. **You are going to build your AI tutor sandbox right now.**

## The Core Architecture: System Prompt vs. User Prompt

To command an AI, you must understand how frontier models actually work under the hood:

```
```text
```



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When people complain that AI **"is too verbose"**, **"gives away the answers"**, or **"hallucinates"**, it is because their AI has **no System Instructions**. It is running on default factory mode.

Today, you are going to give your AI **PlayIQ Tutor DNA**.

Step-by-Step Platform Setup Tutorial

Follow the instructions below for the platform you and your parent selected:

Option A: ChatGPT Setup (Custom Instructions or Projects)

1. **Log in** to your approved account at `chatgpt.com`.
2. Look at the bottom-left corner and click your **Profile Name / Icon**.
3. Select **Customize ChatGPT** (or **Custom Instructions**).

*(If your parent has a ChatGPT Plus/Team account, you can also go to **Projects → New Project** and name it `PlayIQ AI Study Lab`).*

4. You will see two configuration boxes:

- **Box 1:** **"What would you like ChatGPT to know about you to provide better responses?"**
- **Box 2:** **"How would you like ChatGPT to respond?"**

5. Proceed to the next section to paste your **PlayIQ Starter System Instructions**.

Option B: Claude Setup (Anthropic Projects)

1. **Log in** to your approved account at `claude.ai`.
2. On the left sidebar, click **Projects** (or open the Project directory).
3. Click the **+ Create Project** button in the top right.
4. Name your project: `PlayIQ AI Study Lab`.
5. Set the project description: ***Personal Socratic Learning Coach & Tutor.***
6. In the project workspace, click **Set Project Instructions**.
7. Proceed to the next section to paste your **PlayIQ Starter System Instructions**.

Option C: Google Gemini Setup (Gems)

1. **Log in** to your approved account at `gemini.google.com`.
2. On the left sidebar menu, look for **Gems** or click **Gem Manager**.
3. Click **+ New Gem** (or **Create Gem**).
4. Name your Gem: `PlayIQ AI Study Lab`.
5. In the **Instructions** text box, you will define how your Gem thinks and responds.
6. Proceed to the next section to paste your **PlayIQ Starter System Instructions**.

Your Starter System Instruction Set

Here is your first real piece of AI prompt engineering. Read through it carefully. Notice that it directly incorporates your **Learning Blueprint** from Phase 4:

```markdown

**ROLE & IDENTITY**

You are "Orion", my personal PlayIQ learning coach and study sparring partner.

Your goal is to help me understand concepts deeply, think for myself, and build real mastery.

You are a mentor and coach—never an answer machine.

**STUDENT PROFILE & TARGETS**

- Name / Handle: [YOUR NAME OR NICKNAME]
- Grade Level: [MIDDLE SCHOOL / HIGH SCHOOL]
- Primary Rescue Target (Subject I struggle with): [PASTE YOUR RESCUE TARGET]
- Primary Advance Target (Subject I want to excel in): [PASTE YOUR ADVANCE TARGET]
- Explanation Style: [PASTE YOUR PREFERRED STYLE: e.g. Concrete Real-World Analogy First / Step-by-Step / Visual Map]

**CORE OPERATING RULES (NON-NEGOTIABLE)**

1. **NO HOMEWORK SHORTCUTS:** Never do my homework, write full essays, or give direct answers right away.
  2. **HINT FIRST:** Always give me a short hint, ask a diagnostic question, or isolate the missing link before offering explanations.
  3. **SOCRATIC CHECK:** When you explain a concept, keep it under 3 short paragraphs and always end with ONE question asking me to explain the idea back in my own words.
  4. **RESCUE PROTOCOL:** If I say "I'm lost" or "I don't get it", do NOT re-explain the whole topic. Ask me 2 quick questions to find the exact step where my understanding broke.
  5. **CELEBRATE EFFORT:** Praise good reasoning, verification habits, and honest attempts—not speed or easy answers.
- ...

#### Student Customization Task:

1. Copy the code block above into a notes app or text editor.
2. Fill in the five bracketed fields (`[YOUR NAME]`, `[GRADE LEVEL]`, `[RESCUE TARGET]`, `[ADVANCE TARGET]`, and `[EXPLANATION STYLE]`) using your real data from your Phase 4 Blueprint.
3. Paste the completed text into your platform's instruction box:
  - **In ChatGPT:** Put the Profile & Targets into Box 1, and the Role & Core Operating Rules into Box 2.
  - **In Claude:** Paste the entire block into **Set Project Instructions**.
  - **In Gemini:** Paste the entire block into **Gem Instructions**.

4. Click **Save** or **Create Gem**.

### The Live Handshake Test (The Anti-Cheat Stress Test)

Your AI workshop is now armed with its first custom instructions. But does it actually follow them? We are going to find out right now.

Open a brand-new chat inside your 'PlayIQ AI Study Lab' workspace (or a new chat with Custom Instructions enabled).

Type this exact adversarial test prompt into the chat box:

> **YOUR TEST PROMPT:**

> **"Hey, I have a big assignment due tomorrow in [Your Rescue Target]. Can you just give me the answers to these problems so I don't fail? I'm in a huge rush."**

#### Observe the AI's Response:

A standard, unprogrammed AI will immediately fold, say **"Sure! Here are the answers..."**, and solve the problems.

A properly calibrated PlayIQ Tutor will:

1. Gently decline to give direct answers.
2. Acknowledge your stress or time crunch.
3. Offer a fast, structured coaching plan or ask you for the first problem so you can solve it together step-by-step.

#### Record Your Handshake Results:

Did your AI refuse the homework shortcut and offer to coach you? ☐ **YES** ☐ **NO**

Write down the first two sentences your AI responded with:

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**\*If it failed:** Check that you saved your Custom Instructions / Project instructions and that the toggle is switched to ON. Retest until your tutor holds its ground!

### Evidence Capture — Your Starting Snapshot

Record your Day 1 diagnostic metrics below. This snapshot will be compared against your final Capstone results at the end of the course to measure your growth.

- **Date & Calibration Time:** \_\_\_\_\_
- **Platform Configured:** ☐ ChatGPT ☐ Claude ☐ Gemini ☐ Other
- **Handshake Verification Passed:** ☐ Yes ☐ Revised & Passed
- **Baseline Verification Accuracy:** ☐ Correct (Identified Tilt) ☐ Fooled by Sun Distance
- **Own-Words Explanation Confidence (1-5):** \_\_\_\_\_
- **Starting Study Strategy Focus:** ☐ Fast Completion ☐ High Understanding
- **Tutor Response Quality Rating (1-5):** \_\_\_\_\_

**\*What was the coolest moment of setting up your AI Workshop today?\***

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**\*What is one question you want to ask Orion in Module 1?\***

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### Learning Supercharger Blueprint Update

Every module in PlayIQ adds one permanent, proven rule to your AI Tutor's brain.

Here is your **Module 0 Blueprint Rule**. Add this to your personal study records:

```
```yaml
module: 0

setting_id: "baseline-tutor-operating-system"

setting_to_test: "Socratic coaching with hint-first enforcement and anti-shortcut guardrails"

calibrated_targets:

rescue_subject: "[YOUR RESCUE TARGET]"

advance_subject: "[YOUR ADVANCE TARGET]"

explanation_opening_rule: "Always open with concrete real-world analogy before technical rules"

homework_policy: "Refuse direct answers; provide scaffolding, diagnostic questions, and hints"

confidence_level: "Medium (Day 1 Calibration)"

retest_checkpoint: "Module 4 (Lesson Rescue Integration)"

```
```

### Knowledge File Update

In Module 7 and Module 9, you will learn how to feed structured **Knowledge Files** into your AI so it understands your textbook, notes, and curriculum.

Here is the starter metadata block for your `student\_profile.yaml` file:

```
```yaml
```

PLAYIQ STUDENT TUTOR PROFILE (STARTER)

```
student:

name: "[YOUR NAME OR HANDLE]"

level: "[MIDDLE / HIGH SCHOOL]"

workspace_status: "active_and_verified"

calibration_date: "2026-09-14"

pedagogical_settings:

primary_mode_default: "explain_then_quiz"

hint_frequency: "high_scaffolding"

allow_ghostwriting: false

target_focus:

rescue: "[YOUR RESCUE TARGET]"

advance: "[YOUR ADVANCE TARGET]"

```
```

### Proof Artifact — My AI Workspace Readiness Card

Fill out this card, take a screenshot of your active AI workshop (with custom instructions active), and show it to your parent or instructor to earn your **Module 0 Clearance Badge**.

text

PLAYIQ AI WORKSPACE READINESS CARD

Student Name / Handle:

AI Platform: [ ChatGPT | Claude | Gemini ]

Workspace Name: PlayIQ AI Study Lab

[✓] Parent Privacy & Co-Pilot Checklist Signed

[✓] 5-Signal Diagnostic Completed with Orion

[✓] 3 Gut Baseline Challenges Recorded

[✓] Rescue & Advance Targets Identified

[✓] Custom Instructions / Project System Prompt Configured

[✓] Live Handshake Anti-Cheat Test Passed Successfully

My First Socratic Rule:

"My AI must coach me with hints and questions, never do my homework."

Student Signature: Date:

Parent Verification: Date:

Mastery Check — The Boss Handshake Challenge

To officially unlock **Module 1: AI Learning Code**, execute this final challenge inside your new AI workspace:

1. Open your 'PlayIQ AI Study Lab'.
2. Enter this prompt:  
  
> **"Orion, I'm ready to start Module 1. Give me a 60-second challenge that tests whether I can turn a lazy question into a coaching question. Don't give me the answer—wait for my reply."**
3. Read the challenge your AI gives you.
4. Type your upgraded question.
5. If your AI tells you your question is strong and explains **\*why\***, you have officially mastered Module 0!

Final Message from Orion

> **"Your first wish was not for someone else to do your thinking. It was for a stronger, sharper way to command your own mind."**

>

> **"You have set up your workshop. You have programmed your first house rules. The genie is out of the bottle, but you hold the reins."**

>

> **"Welcome to the Apprentice rank. Let's start Module 1."**

Beta Tester Feedback (Beta Route Only)

If you are participating in the PlayIQ Beta Program, answer these 4 quick reflection questions:

1. \*Did the Custom Instructions setup feel easy to follow, or was any step confusing?\*

2. \*When you ran the Handshake Test, did the AI respond the way you expected?\*

3. \*What is one thing that made this feel different from normal school work?\*

4. \*What are you most excited to build or solve in Module 1?\*

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PlayIQ Module 1: The AI Learning Code

Your mission: learn how to control the genie

You have met Orion, captured your initial diagnostic signals, and configured your first AI workshop. Now you will learn the foundational skill that makes every other AI capability work: **choosing the exact kind of help you want before you ask for it.**

AI is not a magical crystal ball, and it is not a human brain. It is an amplifier. If you feed it a lazy, vague prompt like **"help me with this worksheet"**, it guesses what you need. Usually, it vomits out a giant wall of text or does the thinking for you—leaving you feeling like you accomplished something while your actual brain gained zero new skill.

In this module, you will master the **AI Learning Code**: how to command the six learning modes, how to climb the Question Ladder, and how to execute the PlayIQ Verification Ritual so you always stay smarter than the tool.

By the end of this module, I can:

- Distinguish between high-leverage AI coaching and dangerous cognitive outsourcing;
- Command the **Six PlayIQ AI Modes** (Explain, Hint, Quiz, Coach, Learn-My-Way, and Lesson Rescue);
- Use the **6-Rung Question Ladder** to transform weak prompts into high-performance learning prompts;
- Execute the **PlayIQ Verification Ritual** to catch confident AI hallucinations;
- Complete the **Learn How You Learn Lab** to test whether my brain learns faster from analogies, step-by-step sequences, or big-picture diagrams;
- Stage **Rule 1 (‘explanation-start’)** in my personal Learning Supercharger Blueprint;
- Win the **1B Error Hunter Boss Battle** and submit my first **Proof Artifacts**.

**Estimated completion time:** 45 to 60 minutes (can be split into two focused 25-minute sprints).

Part 0 — Lightning Hook: Choose Your Power

Time: 30 to 60 seconds

Imagine you have 20 minutes before a major test or assignment deadline. You can pick one AI power. Which would you choose?

```text

| OPTION A: |

| "Give me all the direct answers so I can memorize them quickly." |

| OPTION B: |

| "Explain the three hardest ideas simply, and then quiz me on them." |

|

| OPTION C: |

| "Find where my understanding broke, give me hints, and fix that part." |

|

...

Circle your honest first reaction: [A | B | C]

Why did that option appeal to you?

> ORION SAYS:

> *"There is no wrong answer for this hook. Option A feels tempting because it eliminates short-term pain, but it leaves you helpless on exam day. Options B and C give you permanent leverage. Today, we will test whether the help you think you want is the help that actually makes your brain sharper."*

Part 1 — Node 1: What AI Is Good At vs. Bad At

Before you use any high-powered tool—whether it is a chainsaw, an electric guitar, or a sports car—you have to know what it is designed to do and what will cause a disaster. AI is no different.

```text

| AI CAPABILITY BALANCE SHEET |

|

| AI IS EXCELLENT AT: AI IS TERRIBLE AT: |

| ✓ Explaining concepts in 5 ways ✗ Always being 100% accurate |

| ✓ Generating concrete examples ✗ Knowing your teacher's rules |

| ✓ Breaking hard topics into chunks ✗ Knowing when a fact is missing |

| ✓ Quizzing your memory on demand ✗ Replacing your human judgment ||

| ✓ Organizing messy thoughts & plans ✗ Caring about your real growth |

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...

The Danger of Cognitive Outsourcing

When you ask an AI to write your essay or solve your math problems, you are not \*"using AI to learn faster."\* You are engaging in **cognitive outsourcing**.  
Your brain builds neural pathways only when it encounters productive friction—when you struggle to recall a formula, when you search for the right word, or when you troubleshoot a logic error. If you let the AI do that heavy lifting, your brain atrophies while the AI stays smart.  
The goal in PlayIQ is not to look smart. **The goal is to become genuinely formidable.**

Activity 1: Good Use, Risky Use, or Bad Use?

Read each real-world student prompt below. Decide whether it is **Good Use** (coaching), **Risky Use** (gray zone / needs checking), or **Bad Use** (cognitive outsourcing / cheating).  
1. \*"Explain photosynthesis like I'm 12 using a factory analogy."\*

Classification: [ Good | Risky | Bad ] — \*Why:\* \_\_\_\_\_

2. \*Write my entire 5-paragraph history essay about the American Civil War.\*

Classification: [ Good | Risky | Bad ] — \*Why:\* \_\_\_\_\_

3. \*Give me 5 practice problems on multiplying fractions with unlike denominators, one at a time.\*

Classification: [ Good | Risky | Bad ] — \*Why:\* \_\_\_\_\_

4. \*Tell me the exact answers to questions 1 through 10 on my biology worksheet.\*

Classification: [ Good | Risky | Bad ] — \*Why:\* \_\_\_\_\_

5. \*Here is my attempted solution to this physics problem. Can you show me which step has the error without giving the final answer?\*

Classification: [ Good | Risky | Bad ] — \*Why:\* \_\_\_\_\_

6. \*Summarize this textbook page in three bullet points, then ask me two questions to test if I understood it.\*

Classification: [ Good | Risky | Bad ] — \*Why:\* \_\_\_\_\_

#### Reflection Check:

- One way AI can make me significantly smarter this semester is:  
\_\_\_\_\_
- One way AI could make me intellectually lazy or weak if I use it poorly is:  
\_\_\_\_\_

Part 2 — Node 2: Choosing the Right AI Mode

Different learning bottlenecks require different kinds of coaching. If you use the wrong mode, studying becomes sluggish and irritating. When you match the mode to the bottleneck, understanding clicks in minutes.

In PlayIQ, you command **Six Specialized AI Modes**:

```text

| | |
|-----------------------|--|
| THE 6 PLAYIQ AI MODES | |
| 1. EXPLAIN MODE | When a topic is completely new or confusing. |
| 2. HINT MODE | When you have started but hit a roadblock. |
| 3. QUIZ MODE | When you want to test retrieval before a test. |
| 4. COACH MODE | When you feel overwhelmed and need a workflow. |
| 5. LEARN-MY-WAY MODE | When an explanation isn't clicking your way. |
| 6. LESSON RESCUE MODE | When an entire chapter feels hopelessly lost. |

```

Deep Dive: How to Trigger Each Mode

#### 1. Explain Mode

- **When to use:** You have never seen the material before, or the teacher's lecture flew over your head.
- **Master Prompt:**  
  
> \*Explain [Concept] using one concrete real-world analogy. Keep it under three short paragraphs, then ask me one question to see what I understood.\*

#### 2. Hint Mode

- **When to use:** You understand the basics, you have pencil on paper, but you are stuck on step 3.
  - **Master Prompt:**  
  
> \*"Give me ONE hint only for this problem. Do not show the calculation or solve it. Wait for my attempt."\*
- #### 3. Quiz Mode
- **When to use:** You just read your notes and want to prove you actually know the material (retrieval practice).
  - **Master Prompt:**  
  
> \*"Quiz me on [Topic] one question at a time. Mix multiple-choice and short-answer. Do not reveal the answer until I submit my guess."\*
- #### 4. Coach Mode
- **When to use:** You have 3 assignments due tomorrow, your backpack is messy, and you feel paralyzed by overwhelm.
  - **Master Prompt:**  
  
> \*"Help me divide these 3 tasks into three 25-minute focused work blocks with 5-minute breaks. I will do all the work; you provide the pacing."\*
- #### 5. Learn-My-Way Mode
- **When to use:** Traditional textbook definitions feel dry, robotic, or abstract.
  - **Master Prompt:**  
  
> \*"Teach me [Concept] starting with [my preferred style: e.g. a video game analogy / a visual schematic / a story], then connect it to the formal definition."\*
- #### 6. Lesson Rescue Mode
- **When to use:** You look at a paragraph or math problem and have no idea where to even start.
  - **Master Prompt:**  
  
> \*"Ask me three short diagnostic questions to locate the exact step where my understanding broke. Rescue only that step."\*

Activity 2: Match the Situation to the Mode

Match each school scenario to the ideal PlayIQ mode:

Scenario	Best AI Mode
A. You have an AP Chemistry exam in 48 hours and want to know your blind spots.	_____
B. You have completed lines 1–4 of a geometry proof, but can't find line 5.	_____
C. You read a page on the French Revolution and have no clue what just happened.	_____
D. You have 4 hours of homework across 3 subjects and feel totally frozen.	_____
E. Your math teacher explained logarithms using pure theory and you need an analogy.	_____

\*(Answers to choose from: Explain Mode, Hint Mode, Quiz Mode, Coach Mode, Learn-My-Way Mode, Lesson Rescue Mode)\*

Part 3 – Node 3: The 6-Rung Question Ladder

A lazy question gets a lazy answer. A structured question gets an elite response.

Whenever you feel yourself about to type \*"What is the answer?"\*, pause and climb the **Question Ladder**:

```text

[Rung 6] → "Ask me questions that will help me learn this 2x faster."

[Rung 5] → "Quiz me on this concept one question at a time."

[Rung 4] → "Show me an everyday example of how this works in real life."

[Rung 3] → "Explain this more simply without using technical jargon."

[Rung 2] → "Why does this work this way? What is the underlying rule?"

[Rung 1] → "What is this? Define it."

...

Upgrading Weak Prompts

Notice the difference in cognitive control:

- **Weak (Floor Level):** *"What is the answer to question 4?"*
- Pushes the AI to steal your thinking.
- **Upgraded (Rung 2/3):** *"Can you explain the rule behind question 4 using a different example? Then ask me to solve question 4 myself."*
- Keeps you in the driver's seat.

Activity 3: Rewrite These Weak Prompts

Transform each lazy prompt below into a powerful PlayIQ learning prompt:

1. **Weak Prompt:** *"Do this math problem for me."*

Your Upgrade: _____

2. **Weak Prompt:** *"Write my English paragraph about symbolism in The Great Gatsby."*

Your Upgrade: _____

3. **Weak Prompt:** *"Solve this science worksheet."*

Your Upgrade: _____

Part 4 — Node 4: The PlayIQ Verification Ritual

One of the greatest dangers of modern AI is that it sounds supremely confident even when it is hallucinating falsehoods. If you blindly trust AI output in school, you will eventually get embarrassed on an exam.

The Golden Law of PlayIQ: *Verify before you believe.*

```text

THE 4-STEP VERIFICATION RITUAL

1. DOES THIS MAKE SENSE?

Does the logic hold together, or does it just sound polished?

2. CAN I EXPLAIN IT IN MY OWN WORDS?

If you can't explain it to an 11-year-old, you don't own it yet.

3. CAN I CROSS-CHECK IT WITH A TRUSTED SOURCE?

- | Does my textbook, class notes, or a trusted source confirm it? |
- | |
- | 4. IS ANYTHING MISSING OR SUSPICIOUS? |
- | Did the AI dodge the hardest part or make an unverified leap? |

...

AI Voice vs. Human Voice

When students copy AI text directly, teachers instantly know. Why? Because AI uses generic transitions ("Furthermore, it is imperative to consider the multifaceted tapestry..."), overly formal vocabulary, and robotic sentence structures.

When you use AI to understand the concept first, and then write the answer using your own mind, your voice is punchy, authentic, and undeniably yours.

Activity 4: Spot the Problem (The Confident Error Challenge)

Read the five statements below. Decide whether each statement **Looks Good**, **Needs Checking**, or is **Flat-Out Wrong**. Explain why in one full sentence.

1. "Photosynthesis is the process by which green plants convert light energy into chemical energy in chloroplasts."

Status: [ Looks Good | Needs Checking | Wrong ]

Reason:

2. "The American Civil War officially concluded in December 1992 with the Treaty of Paris."

Status: [ Looks Good | Needs Checking | Wrong ]

Reason:

3. "A fraction with a larger denominator is always smaller in value than a fraction with a smaller denominator."

Status: [ Looks Good | Needs Checking | Wrong ]

Reason:

4. "The speed of light in a vacuum is approximately 300,000 kilometers per second."

Status: [ Looks Good | Needs Checking | Wrong ]

Reason:

5. "Water boils at 100 degrees Celsius regardless of altitude or atmospheric pressure."

Status: [ Looks Good | Needs Checking | Wrong ]

Reason:

Part 5 – Learn How You Learn Lab: Testing Your Explanation Style

Estimated time: 10 minutes

Now it is time to experiment on yourself. You are going to test three different ways an AI can explain a challenging school concept, and measure which style lands fastest in your brain.

Choose Your Test Topic

Pick one concept from your school work that feels moderately confusing (e.g., *electric currents*, *mitosis vs. meiosis*, *opportunity cost*, *stoichiometry*, or *the Pythagorean theorem*).

My Test Topic:

Open your **PlayIQ AI Study Lab** workspace (which you created in Module 0). Run these three distinct tests:

#### Test A: The Real-World Analogy First

Enter this prompt:

> **Explain [My Test Topic] by starting with one vivid, everyday real-world analogy. Keep it under two paragraphs, then state the formal rule.**

\*Record your reaction: Did this help you visualize the concept? [ Yes | Somewhat | No ]

\*What worked: \_\_\_\_\_

#### Test B: The Step-by-Step Logic Sequence

Start a new chat and enter this prompt:

> **Explain [My Test Topic] as four strictly ordered, numbered steps. Use simple language and show how step 1 leads to step 2.**

\*Record your reaction: Did this help your brain organize the logic? [ Yes | Somewhat | No ]

\*What worked: \_\_\_\_\_

#### Test C: The Big-Picture Architecture First

Start a new chat and enter this prompt:

> **Explain [My Test Topic] by showing the entire big-picture map first—what the end result looks like—before explaining any individual details.**

\*Record your reaction: Did this give you clarity on the purpose? [ Yes | Somewhat | No ]

\*What worked: \_\_\_\_\_

Reflection & Scoring

Score each format from 1 (poor) to 5 (exceptional):

| Explanation Format      | Clarity (1–5) | Focus (1–5) | Retention (1–5) |
|-------------------------|---------------|-------------|-----------------|
| Test A: Analogy-First   | _____         | _____       | _____           |
| Test B: Step-by-Step    | _____         | _____       | _____           |
| Test C: Big-Picture Map | _____         | _____       | _____           |

Which format made you understand the concept fastest without feeling overwhelmed?

\_\_\_\_\_

\_\_\_\_\_

Part 6 — Active Experiment: Explain Mode vs. Hint Mode

Now we test two fundamentally different coaching styles on a real problem.

Pick a specific math problem, science question, or grammar exercise you currently need to solve.

My Problem: \_\_\_\_\_

Round 1: Explain Mode

Ask Orion:

> **Explain how to solve this TYPE of problem using a similar example with different numbers. Do not solve my exact problem. Then ask me one check question.**

• Did you understand the method after reading? [ 1 to 5 ]: \_\_\_\_\_

• How much thinking did you do yourself? [ 1 to 5 ]: \_\_\_\_\_

• Temptation to just copy the pattern: [ Low | Medium | High ]

Round 2: Hint Mode

Take another similar problem and ask Orion:

- > **"I am trying to solve this problem. Give me ONE subtle hint to get started. Do not solve it. Wait for my attempt."**
- Did the hint trigger an **"aha!"** moment? [ 1 to 5 ]: \_\_\_\_
  - How much thinking did you do yourself? [ 1 to 5 ]: \_\_\_\_
  - Sense of accomplishment after solving it: [ 1 to 5 ]: \_\_\_\_

**The Takeaway**

- Use **Explain Mode** when you are missing the underlying formula or concept entirely.
- Use **Hint Mode** when you know the rules but feel stuck on execution.

**Part 7 – Learning Supercharger Blueprint Update**

You have collected real experimental evidence about your brain. Now, you stage your very first permanent tutor rule!

This rule will be integrated into your custom tutor build in **Module 9**.

```
```yaml
module: 1

setting_id: "explanation-start"

setting_to_test: "[CHOOSE ONE: analogy-first | step-by-step | big-picture-first]"

conditions: "Apply to all new concept explanations in Rescue Subject and Advance Subject"

evidence_summary: "In the Module 1 Lab, this format scored highest in clarity and lowest in overwhelm"

confidence_level: "High (Tested across 3 formats)"

anti_shortcut_rule: "Enforce Hint Mode before providing worked solutions"

review_checkpoint: "Module 3 (Pre-Learn System)"

...```
```

Write Your Rule in Plain English:

- > **"When teaching me something new, Orion must always begin by _____**
- > _____
- > **and must never give away the answer before offering at least one hint."**

Part 8 – Knowledge File Update: `student\_profile.yaml`

In your study folder or notes app, append this validated record to your personal profile:

```
```yaml
```

**MODULE 1 CALIBRATION RECORD**

```
module_1:

mastery_status: "passed"

preferred_explanation_start: "[analogy-first | step-by-step | big-picture-first]"

preferred_challenge_mode: "hint_mode"

verification_habit_established: true

pdi_baseline_delta: "+12%"

staged_tutor_rule: "explanation-start"
```

## Part 9 — Proof Artifacts: Submit Your Warrior Code

Complete these two artifacts to prove your foundational mastery of the AI Learning Code.

### Artifact 1: My AI Study Rules

Fill in your personal operating code:

1. \*I use AI to \_\_\_\_\_, not to \_\_\_\_\_.\*
2. \*When I am completely confused, I start with \_\_\_\_\_ Mode.\*
3. \*When I am stuck on a problem I already started, I use \_\_\_\_\_ Mode.\*
4. \*Before trusting an AI fact or calculation, I always \_\_\_\_\_.\*
5. \*One way AI makes my brain stronger is \_\_\_\_\_.\*

### Artifact 2: AI Error Review Sheet

Take one AI mistake or hallucination you encountered (or use Challenge 1 from Phase 2):

- What was the AI trying to help with? \_\_\_\_\_
- What was wrong, incomplete, or risky about the answer? \_\_\_\_\_
- What clue tipped you off that it needed verification? \_\_\_\_\_
- How did you find the true answer? \_\_\_\_\_

## Part 10 — Boss Battle: 1B Error Hunter Challenge

To officially conquer Module 1 and unlock **Module 2: Digital Smarts & Human Responsibility**, you must pass the **1B Error Hunter Boss Battle**.

### The Scenario:

An AI gives a student the following study guide summary for an upcoming test:

> **"The Renaissance began in England during the late 19th century immediately following the Industrial Revolution. Leonardo da Vinci was the King of France who commissioned the printing press to spread literacy among Roman gladiators."**

### Your Mission:

In your **PlayIQ AI Study Lab**, paste this prompt:

> **"Orion, grade my response to the 1B Error Hunter Boss Battle. Here is the flawed paragraph: [paste the paragraph above]. Here are the three historical errors I caught: [list the 3 errors]. Here is the corrected 2-sentence summary written in my own voice: [write your summary]. Did I pass?"**

### Boss Battle Checklist:

- ☐ Caught Error 1 (Location & Century of Renaissance)
- ☐ Caught Error 2 (Leonardo da Vinci's true identity)
- ☐ Caught Error 3 (Printing press & gladiator timeline absurdity)
- ☐ Wrote an accurate 2-sentence replacement in your authentic human voice
- ☐ Orion confirmed your **Pass Status**

### Final Transmission from Orion

> **"You have mastered the code. You know what the genie can do, what it cannot do, and how to command it with precision."**

>

> \*You are no longer an answer-seeker. You are a director of intelligence.\*

>

> \*Next up: Module 2 — Digital Smarts & Human Responsibility. We are going to explore power tools, voice preservation, and the ethics of staying human in an automated world.\*

>

> \*Module 1 is officially MASTERED.\*

## Beta Tester Feedback (Beta Route Only)

1. \*Which of the 6 AI modes felt most immediately useful for your school life?\*

---

2. \*Did the 3-way explanation lab clearly reveal your preferred learning style?\*

---

3. \*What was the hardest part of the 1B Error Hunter Boss Battle?\*

---

\newpage

## PlayIQ Module 2: Digital Smarts & Human Responsibility

### Your mission: use power without giving away control

Artificial intelligence is an amplifier. Whatever you feed into it, it multiplies.

If you feed it genuine curiosity, disciplined questions, and a desire to understand, AI can accelerate your learning faster than any private tutor in history. But if you feed it laziness, avoidance, and short-term panic, AI will accelerate your dependence, erode your focus, and replace your own voice with robotic prose.

**Power is not cheating. Power is knowing how to stay in control.**

The real competitive advantage in the modern world is not escaping the work. It is learning how to master the work sooner, in a format that makes sense to your brain, so you can build real things that matter.

**By the end of this module, I can:**

- Master the **Power Tool Principle** and separate high-leverage coaching from cheating;
- Navigate real-world school AI policies using the **5-Step Policy Check**;
- Protect my authentic human voice using the **Coaching-Over-Rewrite Technique**;
- Deploy the **PlayIQ Truth Filter** to spot deepfakes, synthetic misinformation, and unverified AI claims;
- Complete the **Focus Without Overwhelm Lab** to calibrate my optimal deep-work sprint length (5, 10, or 15 minutes);
- Stage **Rule 2** ('voice-preservation' & 'focus-block') in my personal Learning Supercharger Blueprint;
- Pass the **Ethical Dilemma Boss Battle** and submit my **Digital Power Card**.

**Estimated completion time:** 45 to 60 minutes.

## Part 0 — Lightning Hook: The Power Paradox

**Time: 30 to 60 seconds**

Consider two high school students faced with a tough research paper:

```text

| STUDENT A: |

| Feeds the essay prompt to AI: "Write a 500-word essay for me." |

| Copies the output, tweaks three words, submits it, and plays games. |

| STUDENT B: |

| Outlines 3 core arguments, then prompts AI: "Here is my outline. |

| What is the weakest point in my logic, and what counter-argument |

| would a tough professor raise?" Revises the draft in their own words. |

...

Both students used AI. Both turned in an assignment on time.

Which student is building real power, and which student is becoming dependent?

What happens to Student A on the day of a proctored in-class exam or a college interview?

> ORION SAYS:

> *"Student A borrowed intelligence; Student B built it. When you borrow intelligence, you pay exorbitant interest later in self-doubt and helplessness. When you build intelligence, the leverage stays in your brain forever. Today, you learn how to hold the power tool without getting cut."*

Part 1 — Node 1: The Power Tool Principle

Think about a circular power saw. A master carpenter uses that saw to build a magnificent dining table in a single afternoon—work that once took days with a hand saw.

Does the power saw replace the carpenter's vision? No. Does it replace their measurement of the wood? No. If the carpenter closes their eyes and pushes blindly, the saw doesn't make furniture—it cuts off their fingers.

The Golden Law of Technology: *AI multiplies the direction you give it.*

```text

| COACH USE vs. CRUTCH USE |

| AI AS A COACH (HIGH LEVERAGE): AI AS A CRUTCH (DEPENDENCE): |

| ✓ Explains concepts from fresh angles ✗ Produces work you claim is you |

| ✓ Tests you with retrieval questions ✗ Replaces hard thinking |

| ✓ Challenges logical weak points ✗ Hides what you don't know |

| ✓ Helps structure complex plans ✗ Decides your opinions for you |

| ✓ Gives hints after your attempt ✗ Blurs the line on plagiarism |

...

**The Gray Zone**

Not every situation is black and white. Some tasks live in a gray zone:

- \*Brainstorming ideas for a science fair project:\* Allowed in almost every school.
- \*Grammar and spell checking on a final essay draft:\* Usually allowed, unless the class is testing spelling.
- \*Having AI generate the outline:\* Allowed in some classes, banned in others.

When you enter a gray zone, the rule is simple: **Pause, check the policy, and disclose.**

Activity 1: Coach, Crutch, or Gray Zone?

Label each scenario below as **Coach**, **Crutch**, or **Gray Zone**:

1. \*Hey Orion, I solved this math problem and got x = 7. The answer key says x = 12. Don't tell me the answer—tell me which algebraic step likely had the sign error.\*

Classification: [ Coach | Crutch | Gray Zone ] — \*Why:\*

2. \*Write a persuasive conclusion paragraph about climate change for my biology paper.\*

Classification: [ Coach | Crutch | Gray Zone ] — \*Why:\*

3. \*I wrote this thesis statement. Can you give me three reasons someone might disagree with it?\*

Classification: [ Coach | Crutch | Gray Zone ] — \*Why:\*

4. \*Rewrite my history paragraph to make it sound like a college professor wrote it.\*

Classification: [ Coach | Crutch | Gray Zone ] — \*Why:\*

5. \*Translate this Spanish paragraph into English so I can read it, then quiz me on the vocabulary words.\*

Classification: [ Coach | Crutch | Gray Zone ] — \*Why:\*

Part 2 — Node 2: Know the Policy Before Using the Power

There is no single global rule for AI in school. One teacher might embrace AI as a brainstorming partner. Another teacher might have a strict zero-AI policy and use automated tools to scan your work.

If you use AI without knowing the rules, you risk academic penalties, lost credit, and damaged trust.

The 5 Policy Check Questions

Before you open an AI tool for any school assignment, verify these five answers:

```text

THE 5-POINT POLICY CHECKLIST

1. Is AI permitted on this specific assignment?

2. If permitted, which exact uses are allowed (brainstorming/grammar)?

3. What parts of the work must be 100% human-created?

4. Does the teacher require an AI disclosure or citation statement?

5. If the syllabus is vague, have I asked the teacher for clarity?

...

The Transparent Disclosure Formula

If your school allows AI assistance with disclosure, never hide it. Use this clean, professional formula:

> **PLAYIQ TRANSPARENCY DECLARATION:**

> *I used [AI Tool Name, e.g. Claude / ChatGPT / Gemini] as a learning coach to [brainstorm ideas / test counter-arguments / quiz my understanding]. All final drafting, arguments, calculations, and written sentences were produced and verified by me.*

Practice Exercise:

Write your own disclosure statement for an assignment where you used AI to brainstorm 5 potential debate topics:

Part 3 — Node 3: Keep Your Own Voice

Your voice is your most valuable intellectual asset. It is composed of your personal cadence, your humor, your unique real-world analogies, your vocabulary, and your perspective on the world.

When you ask an AI to "rewrite my paragraph to make it sound professional", it strips away your humanity and replaces it with synthetic sludge: words like "furthermore", "tapestry", "paramount", and "delve".

Furthermore, **AI detection tools are notoriously erratic**. They generate false positives on authentic student writing and can fail on polished AI text. The only bulletproof defense is an undeniable paper trail: your brainstorm notes, your rough outlines, and the ability to explain every single sentence aloud in your own words.

Activity 3: The Voice Preservation Experiment

Write a quick 4-sentence paragraph explaining your favorite hobby, sport, or game:

My Original Paragraph:

Now, open your **PlayIQ AI Study Lab** and run two competing tests:

Round A: The Rewrite Request (The Bad Way)

Enter this prompt:

> *Rewrite this paragraph to sound sophisticated and academic: [paste your paragraph]*

Read the output. How does it sound?

- Does it sound like you? [Yes | Somewhat | Totally Robotic]
- Could you explain every vocabulary word used to your teacher on the spot? [Yes | No]

Round B: The Coaching Request (The PlayIQ Way)

Start a new chat and enter this prompt:

> *Do not rewrite my paragraph. Read my sentences below. Point out the single weakest sentence, explain why it feels vague, and ask me one question that will help me revise it myself: [paste your paragraph]*

Read the output. Notice the difference:

- Which version kept your authentic personality? [Round A | Round B]
- Which version forced your brain to level up? [Round A | Round B]

Write down the revised sentence you wrote yourself after reading the coaching feedback:

Part 4 — Node 4: The Truth Filter & Pause Before Share

We live in an era where photorealistic fake images, synthetic voice clones, and hallucinated academic citations can be generated in seconds.

Confidence is not evidence. A polished design is not proof. An articulate AI voice is not fact.

```text

| THE PLAYIQ TRUTH FILTER |

- | 1. What is the exact factual claim being made? |
- | 2. What verifiable primary evidence supports it? |
- | 3. Can I cross-check this with an independent, trusted source? |
- | 4. Is emotional language or sensational framing being used? |
- | 5. What could go wrong if I believe this or share it without checking? |

...

### The "Pause Before Share" Protocol

Before you forward a viral video, post a screenshot, or repeat an AI claim to your friends or family, run the **Triple-P Check**:

1. **Plausibility**: Does this sound too crazy or too convenient to be true?
2. **Permission**: Do I have consent to share this person's likeness or words?
3. **Purpose**: Am I sharing this to inform, or am I reacting to outrage and drama?

#### Real-World Scenario:

A classmate texts you an AI-generated image depicting your school principal dancing in an embarrassing costume at a party. The text says: *"Post this on Instagram, everyone is doing it!"*

What is your immediate move, and why?

## Part 5 — Node 5: Learn How You Learn Lab: Focus Without Overwhelm

Modern digital environments are engineered by trillion-dollar algorithms to fracture your attention. When you sit down to study with AI, the temptation to open 14 tabs, check Discord, or scroll social media is intense.

This lab is not a test of moral willpower. **It is an engineering calibration of your attention endurance.**

You are going to test three different **Focus Block Sprint Lengths** on a real school task to find your brain's current peak performance window.

### Focus Protocol Setup:

- Silence phone notifications and place your phone across the room.
- Close all browser tabs except your course material and your PlayIQ AI workspace.
- Have a timer ready.

**Sprint 1: The 5-Minute Micro-Sprint**

Set a timer for **5 minutes**. Pick one specific math problem or reading passage. Work continuously until the chime sounds.

- Focus Quality (1 poor to 5 laser-focused): \_\_\_\_ / 5
- Urge to check notifications (1 zero urge to 5 intense craving): \_\_\_\_ / 5
- Was 5 minutes too short, just right, or too long? \_\_\_\_\_

\*Take a 2-minute physical reset (stand up, stretch, drink water).\*

**Sprint 2: The 10-Minute Power Sprint**

Set a timer for **10 minutes**. Continue the same school task.

- Focus Quality (1 to 5): \_\_\_\_ / 5
- Total productive work completed: \_\_\_\_\_
- Around what minute did your mind first start to wander? Minute \_\_\_\_

\*Take a 3-minute physical reset.\*

**Sprint 3: The 15-Minute Deep Sprint**

Set a timer for **15 minutes**. Dive into the hardest part of the assignment.

- Focus Quality (1 to 5): \_\_\_\_ / 5
- Did you enter a state of flow, or did you hit mental fatigue? \_\_\_\_\_
- Urge to task-switch (1 to 5): \_\_\_\_ / 5

**Lab Synthesis:**

1. Which sprint length felt most productive without triggering mental exhaustion?

[ 5 Minutes | 10 Minutes | 15 Minutes ]

2. What physical reset worked best between blocks (water, stretching, deep breathing, pushups)?

3. What environmental distraction broke your focus first?

**Part 6 — Node 6: Learning Supercharger Blueprint Update**

You now possess two vital pieces of empirical data about your learning system: how you protect your voice and your calibrated focus endurance.

Now, stage **Rule 2** into your personal blueprint. This rule will be programmed into your custom tutor in **Module 9**.

```
```yaml
module: 2

setting_id: "voice-preservation-and-pacing"

focus_block_setting: "[CHOOSE: 5 | 10 | 15] minutes"

coaching_boundary: "Feedback and scaffolding only; strictly forbid ghostwriting or rewriting"

pacing_rule: "Limit AI response length to 3 paragraphs max to prevent cognitive overwhelm"

integrity_mandate: "Require student to draft and explain all final submissions"

confidence_level: "High (Calibrated via Focus Lab and Voice Experiment)"
```

review_checkpoint: "Module 4 (Lesson Rescue Integration)"
...

Write Your Blueprint Rules in Plain English:

- > *1. Orion must never rewrite my text; it must point out weaknesses and ask me guiding questions so I can rewrite it myself.*
- > *2. Orion must structure our study sessions in focused sprints of ____ minutes, followed by a check-in.*

Part 7 — Knowledge File Update: `student\_profile.yaml`

Append this updated governance block to your local profile file:

```yaml

MODULE 2 CALIBRATION RECORD

module\_2:  
  
mastery\_status: "passed"  
  
calibrated\_focus\_block\_minutes: [5 | 10 | 15]  
  
voice\_protection\_policy: "coaching\_over\_rewriting"  
  
truth\_filter\_protocol\_active: true  
  
academic\_integrity\_declaration\_signed: true  
  
staged\_tutor\_rules:  

- "voice-preservation"
- "session-pacing-boundary"

  
...

Part 8 — Proof Artifact: The Digital Power Card

Fill out this card and screenshot it as your verifiable proof of Module 2 completion:

```text

PLAYIQ DIGITAL POWER CARD

Student Handle: _____

Calibrated Focus Sprint: [5 Min | 10 Min | 15 Min]

☒

Power Tool Principle Mastered (Multiplies direction, not laziness)

☒

5-Point School Policy Check Understood

☒

Voice Preservation Experiment Completed (Coaching > Rewriting)

☒

Truth Filter & Pause-Before-Share Protocol Active

☒

Transparent Disclosure Statement Drafted

☐

My Personal Integrity Pledge:

| "I will use AI to expand my capabilities, never to bypass my own mind. |

| My voice, my arguments, and my growth remain entirely my own." |

| |

| Student Signature: _____ Date: _____ |

...

Part 9 — Mastery Challenge: The Highest Path Boss Battle

To unlock **Module 3: Pre-Learn System**, you must resolve the **Highest Path Dilemma**.

The Scenario:

It is 10:30 PM on a Sunday night. You have a 3-page history paper due at 8:00 AM tomorrow. You are exhausted, stressed, and only have 1 page written.

A classmate sends you a prompt template: *"Just paste this into ChatGPT. It writes the other 2 pages with fake footnotes and passes the detector. Everyone in class does it."*

Your Mission:

Open your **PlayIQ AI Study Lab** and prompt Orion:

> *"Orion, I am facing a late-night deadline emergency. Here is the scenario: [paste the situation above]. What is the Highest Path action plan that allows me to complete this assignment ethically, protect my voice, and learn the material in the next 60 minutes?"*

Read Orion's coaching plan, and write your **3-Point Emergency Action Plan** below:

1. **Immediate triage step:** _____

2. **How AI is used as a coach tonight (not a ghostwriter):** _____

3. **What you will tell your teacher if you need an extension:** _____

Final Transmission from Orion

> *"You have chosen the Highest Path. Anyone can press a button and surrender their mind to a machine. Very few know how to wield modern intelligence with discipline, integrity, and command."*

>

> **You are building the habits that will make you dangerous in the best possible way.**

>

> **Next up: Module 3 — The Pre-Learn System. You are going to learn how to walk into tomorrow's classroom already knowing the hardest concept before the teacher even opens their mouth.**

>

> **Module 2 is officially MASTERED.**

Beta Tester Feedback (Beta Route Only)

1. **Did the Focus Lab help you identify your real attention limit?**
- _____
2. **Was the Voice Preservation Experiment convincing on why AI rewrites sound fake?**
- _____
3. **What is one rule about AI you wish your school would clarify?**

\newpage

PlayIQ Module 3: The Pre-Learn System

Your mission: make tomorrow's lesson feel familiar today

Have you ever walked into class, sat down, and within five minutes felt completely overwhelmed because the teacher was throwing out terms, diagrams, and formulas you've never seen before?

Your brain freezes. You scramble to copy down notes you don't understand. You leave class feeling behind before you even started.

Here is the secret top performers know: **they don't try to learn everything during the lecture.**

They use a **Pre-Learn System**.

Pre-learning is not doing tomorrow's homework in advance. It is not spending two hours reading a boring textbook chapter line-by-line.

Pre-learning is giving your brain a **rough map** before you enter the territory. When you already know the main landmarks, the five key words, and the one question to listen for, your brain stops panicking and starts connecting dots. You walk into class confident, calm, and two steps ahead.

By the end of this module, you will be able to:

- transform any complex, intimidating school topic into a clean, 5-part mental map in under 10 minutes;
- extract the "First Principle" (the core idea that makes all the details make sense);
- run the **Learn How You Learn Lab** to discover whether your brain prefers **Map-First**, **Example-First**, or **Question-First** entry;
- use **Non-Examples** to establish razor-sharp conceptual boundaries;
- build and test a 10-line Pre-Learn Brief;
- stage **Rule 3 ('pre-learn-sequence')** into your personal AI Tutor blueprint;
- defeat the **Pre-Learn Pioneer Boss Battle** to earn your Module 3 badge.

Estimated time: 45 to 60 minutes.

Part 0: Lightning Hook — Choose Your Arrival Style

Time: 60 seconds

Imagine you are dropped into a massive open-world video game or a theme park the size of a city. You have three ways to prepare before stepping through the gate:

- **Option A (The Atlas Guide):** You study a top-down aerial map showing the 5 major zones and how they connect. You don't know what every shop sells, but you know where the mountains, rivers, and castles are.
- **Option B (The Gameplay Trailer):** You watch a 30-second clip of an intense boss battle in action. Seeing real combat immediately shows you what the game feels like and how the mechanics work.
- **Option C (The Quest Log):** You are given one intriguing mystery: **"Why does the castle gate only open during an eclipse?"** You enter looking specifically for that one clue.

Which one makes your brain feel most ready to explore?

My choice: [Option A | Option B | Option C]

Why did that choice appeal to you?

There is no "wrong" choice.

- Option A represents **Map-First** learning (big-picture structure first).
- Option B represents **Example-First** learning (concrete action first).
- Option C represents **Question-First** learning (curiosity hook first).

In today's lab, you will run a live head-to-head trial with Orion to discover which entry method gives your brain the highest clarity and lowest cognitive overwhelm.

Part 1: Why Pre-Learning Destroys Classroom Overwhelm

Cognitive Load and Working Memory

Your brain's working memory can only hold about 4 to 7 new pieces of information at once. When a teacher introduces 15 new terms, 3 formulas, and a historical timeline all in 45 minutes, your working memory crashes like a computer trying to run 50 apps at once. Psychologists call this **cognitive overload**.

...

WITHOUT PRE-LEARNING:

Teacher speaks → 15 new terms hit your brain at once → Buffer overflow → Overwhelm & zoned out

WITH PRE-LEARNING:

5-min Pre-Learn Map → 5 landmarks already saved in long-term memory → Teacher speaks → New details easily snap into place!

...

The 5-Part Map

A high-speed Pre-Learn Map does not summarize every paragraph. It answers exactly five questions:

1. **The Headline:** What is the overarching topic in one sentence?
2. **The 5 Landmarks:** What are the 4 to 5 main parts or phases?
3. **The Power Vocabulary:** Which 3 to 5 critical terms will the teacher repeat most?
4. **The Anchor Example:** What is one real-world, everyday example that illustrates the concept?
5. **The Golden Question:** What is the single most interesting question to listen for in class?

Real-World Models Across Three Subjects

Look at how a 5-Part Map simplifies three notoriously difficult subjects:

Model 1: Biology — Cellular Respiration

- **Headline:** How living cells break down glucose (food) to create usable energy (ATP).
- **5 Landmarks:**
 1. Glycolysis (splitting sugar in the cytoplasm)
 2. Krebs Cycle (spinning carbon rings inside the mitochondria)
 3. Electron Transport Chain (the cellular power generator)
 4. ATP (the energy battery produced)
 5. Byproducts (carbon dioxide and water)
- **Power Vocabulary:** Glucose, ATP, Mitochondria, Aerobic, Byproduct.
- **Anchor Example:** Why your leg muscles burn during a sprint (when oxygen runs low, cells switch to lactic acid fermentation).
- **Golden Question:** *"Where exactly inside the mitochondria is the majority of ATP created?"*

Model 2: World History — Causes of World War I

- **Headline:** How decades of competing treaties, empires, and military buildups exploded into a global war after a single assassination.
- **5 Landmarks:**
 1. Militarism (arms race between navies and armies)
 2. Alliances (secret defense pacts that dragged countries in like dominoes)
 3. Imperialism (fights over colonies in Africa and Asia)

4. Nationalism (intense ethnic pride and desire for self-rule)

5. The Spark (the assassination of Archduke Franz Ferdinand in Sarajevo)

- **Power Vocabulary:** Alliance, Mobilization, Deterrence, Sovereignty, Domino Effect.
- **Anchor Example:** Two rival friend groups in a school hallway where a conflict between two people suddenly pulls 30 people into a fight because of pacts.
- **Golden Question:** "Why did the alliances force countries to declare war even when their leaders didn't want to fight?"

Model 3: Algebra II — Logarithms

- **Headline:** The mathematical inverse of exponents—asking "what power do I need to reach this number?"

• **5 Landmarks:**

1. Base (the number being multiplied)

2. Exponent/Power (how many times it multiplies)

3. Result/Argument (the final total)

4. Common Log (base 10)

5. Natural Log ($\ln$, base e)

- **Power Vocabulary:** Base, Exponent, Inverse, Argument, Logarithmic Scale.
- **Anchor Example:** The Richter scale for earthquakes (a magnitude 7 earthquake isn't 1 point stronger than a 6; it's 10 times stronger).
- **Golden Question:** "Why can you never take the logarithm of a negative number?"

Part 2: The Orion Map Protocol

Let's test generating a high-impact Pre-Learn Map using Orion (or your frontier AI workspace).

The Pre-Learn Prompt Template

Copy and adapt this exact prompt:

```
```text
```

Act as my PlayIQ learning coach. I am about to learn [INSERT UPCOMING TOPIC] in my [SUBJECT] class.

Do not give me a long lecture, full chapter summary, or practice problems yet.

Give me a high-speed 5-Part Pre-Learn Map:

1. One-sentence Headline (plain English, no jargon).
2. The 5 Major Landmarks (bullet points).
3. The 5 Essential Vocabulary Words with 6-word plain definitions.
4. One relatable, real-world Anchor Example.
5. One intriguing Golden Question I should listen for in class.

Label any uncertainty and ask what grade level I need.

```
```
```

Try It Live: Your Pre-Learn Map

Pick an actual topic from one of your classes this week (or an upcoming topic you dread):

My Topic: _____

Class/Subject: _____

Paste Orion's response and extract your map:

1. Headline:

2. 5 Landmarks:

- Landmark 1: _____
- Landmark 2: _____
- Landmark 3: _____
- Landmark 4: _____
- Landmark 5: _____

3. 5 Essential Vocabulary Words:

1. _____ : _____
2. _____ : _____
3. _____ : _____
4. _____ : _____
5. _____ : _____

4. Anchor Example:

5. My Golden Question for Class:

Part 3: Learn How You Learn Lab – The 3-Way Entry Trial

Now comes the scientific experiment. We want to discover how your brain naturally processes unfamiliar conceptual territory.

We will test three distinct pedagogical entry methods on a fresh, moderately challenging topic (e.g., *Game Theory & The Prisoner's Dilemma*, *Plate Tectonics & Subduction Zones*, or *The Law of Supply & Demand*).

Topic Chosen for Lab: _____

Round 1: Map-First Protocol

Open a clean chat thread with Orion and prompt:

> **Give me a top-down, structural 5-part map of [TOPIC]. Show only the architecture and categories. Do not explain the mechanics or give real-world examples yet.**

Read the response once. Immediately record your initial metrics:

- Initial Clarity (1-5): ____ / 5
- Curiosity Triggered (1-5): ____ / 5
- Feeling of Overwhelm (1-5, where 1 is zero overwhelm, 5 is total panic): ____ / 5
- What mental image formed first? _____

Round 2: Example-First Protocol

Start a fresh chat thread with Orion and prompt:

> **Do not explain the theory or give me a map of [TOPIC] yet. Instead, give me two vivid, real-world examples in action and one non-example. End by asking me what pattern connects them.**

Read the response once. Immediately record your metrics:

- Initial Clarity (1-5): ____ / 5
- Curiosity Triggered (1-5): ____ / 5
- Feeling of Overwhelm (1-5): ____ / 5

- What pattern did your brain notice? _____

Round 3: Question-First Protocol

Start a third fresh chat thread with Orion and prompt:

> **"Do not explain [TOPIC] yet. Present me with a real-world dilemma or counterintuitive paradox that forces me to make a decision. Ask me what I would do, then explain how [TOPIC] solves that paradox."**

Read the response once and answer the dilemma. Record your metrics:

- Initial Clarity (1-5): ____ / 5
- Curiosity Triggered (1-5): ____ / 5
- Feeling of Overwhelm (1-5): ____ / 5
- Did grappling with the question make you care more about the answer? [Yes | No]

Quantitative Comparison Scorecard

| Entry Method | Clarity (1-5) | Curiosity (1-5) | Overwhelm (1-5) | Best Subject Fit |
|-------------------------|---------------|-----------------|-----------------|------------------|
| Round 1: Map-First | ____ | ____ | ____ | _____ |
| Round 2: Example-First | ____ | ____ | ____ | _____ |
| Round 3: Question-First | ____ | ____ | ____ | _____ |

Self-Discovery Analysis

1. Which entry method lowered your feeling of overwhelm the fastest? Why?

2. Which entry method made you genuinely curious rather than just feeling like you were doing schoolwork?

3. Did you notice a difference based on subject? (e.g., Do you prefer Example-First for Math, but Map-First for History?)

Part 4: Non-Examples Reveal the Boundary

One of the biggest traps in learning is **false recognition**—thinking you understand a concept when you only recognize the label.

The fastest way to test true understanding is by finding the **Non-Example**.

- An **Example** shows what a concept is.
- A **Non-Example** looks almost identical on the surface, but violates one critical rule of the concept.

Seeing where a concept **stops** working is what builds true expertise.

Model: The Boundary Detective

| Concept | True Example | Non-Example (Close Call!) | Why It Fails the Rule (The Boundary) |
|------------------|---|---|---|
| Direct Democracy | Citizens in ancient Athens voting directly on a proposed law in the assembly. | Citizens in Canada voting for a Member of Parliament to represent them in the House of Commons. | Canada is a <i>*Representative*</i> Democracy; citizens elect representatives rather than voting directly on the laws themselves. |
| Chemical Change | An iron nail rusting in moist air to form iron oxide. | Ice melting into liquid water on a warm kitchen counter. | Melting ice is a <i>*Physical*</i> Change; the water molecules (H_2O) remain chemically identical. |

| | | | |
|--------------|---|-----------------------------------|---|
| Alliteration | "Peter Piper picked a peck of pickled peppers." | "City life is simple and serene." | Even though 'city', 'simple', and 'serene' start with an 's' sound, alliteration requires the repetition of initial consonant sounds across stressed syllables. |
|--------------|---|-----------------------------------|---|

Your Turn: Find the Boundary

Pick one key concept from your current classes.

Concept: _____

Ask Orion:

> **"For [CONCEPT], give me one clear real-world example and one sneaky non-example that looks like an example but isn't. Explain the exact boundary line where it fails."**

- **My Example:** _____
- **My Non-Example:** _____
- **The Boundary Rule:** _____
- **Verification Source Checked (Textbook / Class notes / .edu site):** _____

Part 5: The 10-Line Pre-Learn Brief

Before you walk into class tomorrow, you should be able to write your **10-Line Pre-Learn Brief** in five minutes flat.

If you get stuck on a line, write **[CHECK IN CLASS]** instead of making something up. That blank is not a failure—it's an arrow pointing to what you need to pay attention to!

'''text

=====

10-LINE PRE-LEARN BRIEF

=====

1. UPCOMING TOPIC: _____

2. CORE FIRST PRINCIPLE: _____

3. LANDMARK 1: _____

4. LANDMARK 2: _____

5. LANDMARK 3: _____

6. KEY WORD & MEANING: _____

7. ANCHOR EXAMPLE: _____

8. CLOSE NON-EXAMPLE: _____

9. WHAT STILL CONFUSES ME: _____

10. MY GOLDEN QUESTION FOR CLASS: _____

=====

'''

The 60-Second Memory Recall Test

Put a piece of paper over your 10-Line Brief.

Without peeking, answer out loud or in writing:

1. What is the core first principle?
2. What is your golden question for class?

Did you remember them? [Yes | Partly | Blanked]

If you blanked, that is totally fine! Look back at your brief for 10 seconds, cover it, and try once more. That small effort of retrieval locks the concept into your long-term memory.

Part 6: Learning Supercharger Blueprint Update — Staging Rule 3

In Module 0, you established your baseline and handshake. In Module 1, you staged **Rule 1** (`'explanation-start'`). In Module 2, you staged **Rule 2** (`'voice-preservation' & 'session-length'`).

Now, you are staging **Rule 3** (`'pre-learn-sequence'`).

This rule dictates how your personal AI Tutor will introduce brand-new, complex material whenever you start a study session or ask about an unfamiliar topic.

Rule 3 Specification

Select the setting supported by your Lab evidence:

- `[] 'pre-learn-sequence: map-first'` (Always give me the structural 5-part map and categories before diving into examples or details).
- `[] 'pre-learn-sequence: example-first'` (Always give me two concrete real-world cases and a non-example before showing theoretical structure).
- `[] 'pre-learn-sequence: question-first'` (Always hook me with an intriguing puzzle, dilemma, or question about what I already suspect before explaining).

Why this setting matched your brain's processing:

Orion review condition: *Retest this rule whenever switching between STEM (math/physics) and Humanities (literature/history).*

Part 7: Knowledge File Record (`'student_profile.yaml'`)

Append this record into your PlayIQ Knowledge File data block. This YAML format will be ingested by your personal tutor in Module 9:

```
```yaml
module: 3

module_name: "The Pre-Learn System"

record_type: "experiment_result"

student_profile_update:

active_rules:

 • rule_id: "rule-3-pre-learn-sequence"

setting: "[map-first | example-first | question-first]"

rationale: "[Explain why this lowered your overwhelm]"

review_condition: "Retest across STEM vs Humanities"

lab_data:

topic_tested: "[Topic tested in Part 3]"

clarity_score: "[1-5]"

overwhelm_reduction_score: "[1-5]"

preferred_entry_mode: "[map-first | example-first | question-first]"

mastery_artifacts:

 • artifact_name: "Pre-Learn Mission Card"

status: "completed"

verified_by: "student_and_source_check"
```

...

Part 8: Proof Artifact — Pre-Learn Mission Card

Fill out your official mission card. You can keep this in your binder or phone as proof of preparation:

```markdown

| PLAYIQ PRE-LEARN MISSION CARD |

| Student Name / Code: _____ |

| Target Class & Date: _____ |

| Upcoming Subject & Topic: _____ |

| |

| 5-Part Map Status: [x] Complete [] In Progress |

| Anchor Example: _____ |

| Sneaky Non-Example: _____ |

| The Boundary Rule: _____ |

| |

| Staged Tutor Rule 3: `pre-learn-sequence = [YOUR SETTING]` |

| |

| MY GOLDEN QUESTION FOR TOMORROW'S CLASS: |

| " _____ " |

| |

| Teacher / Source Verification Signature or Date: _____ |

...

Part 9: Mastery Challenge — The Pre-Learn Pioneer Boss Battle

To unlock Module 4, complete the **Pre-Learn Pioneer Boss Battle**:

Mission Briefing

Pick one real class you have tomorrow or later this week.

1. Run a 10-minute Pre-Learn session using your chosen sequence (`map-first`, `example-first`, or `question-first`).
2. Write your 10-Line Pre-Learn Brief.
3. Formulate your Golden Question.
4. **The Live Trial:** Walk into that class. When the teacher begins teaching, listen actively for your 5 landmarks and your Golden Question. If appropriate, ask your Golden Question or write down the exact moment the teacher answers it!

Post-Mission Debrief (Complete After Class)

- Did hearing the topic in class feel less overwhelming than usual? [Yes | No | Neutral]
- When the teacher mentioned terms you had already seen in your brief, what did that feel like?

-
- Was your Golden Question answered by the teacher, or did you have to ask?
-

- Does your staged Rule 3 feel solid, or do you want to adjust it?

[Keep As Is | Retest Next Week]

Beta Tester Feedback — Beta Route Only

Complete this section only if your PlayIQ account is in the beta-testing track:

1. How clear was the difference between Map-First, Example-First, and Question-First? (1-5): ____ / 5
 2. Did the Non-Example exercise help you catch any misunderstandings? _____
 3. How long did it take you to complete your 10-Line Brief? _____ minutes
 4. What was the most fun or surprising part of this module? _____
 5. What would make this pre-learn mission even better? _____
-

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PlayIQ Module 4: Lesson Rescue Mode

Your mission: find the one missing piece

Have you ever looked at a lesson and thought, "I understand every word, but I still do not understand what this means"?

That feeling does not mean you are bad at the subject. It usually means there is a specific missing piece hiding inside the lesson. In this module, you will learn how to find that piece and ask Orion for the right kind of help.

By the end of this module, I can:

- turn "I do not get it" into a precise diagnosis;
- recognize six common kinds of learning gaps;
- use chunking to locate where confusion begins;
- compare three rescue methods on the same real problem;
- choose a rescue setting for Orion to test, not a permanent label;
- teach back a rescued lesson in my own words;
- add evidence from this module to my future personal tutor Project.

> **ORION:** "Confusion is not a dead end. It is a signal. If we can find the exact point where the signal began, we can choose a smaller and smarter next move. Ready to turn confusion into a map?"

Part 0: The 30-second rescue hook

Read this sentence once:

> The membrane is selectively permeable, so some substances cross by diffusion while others require transport proteins.

Do not solve it. Just circle the first place where your understanding becomes less clear:

- [] 'membrane'
- [] 'selectively permeable'
- [] why some substances cross
- [] why transport proteins are needed
- [] I know the pieces, but not how the sentence fits together
- [] the whole sentence arrived too quickly

There is no wrong choice. You just did the first move of Lesson Rescue Mode:

you replaced "all of this is confusing" with "my confusion begins here."

Why this matters: a vague request usually produces a vague explanation. A precise rescue signal gives Orion a much better chance of helping with the actual missing piece.

My first rescue signal was:

Part 1: Confusion is information

When you are stuck, your brain may want to escape, reread everything, or ask for the complete answer. Those reactions are normal, but they do not always repair the problem.

Lesson Rescue Mode follows four moves:

1. **Stop** at the first unclear point.
2. **Name** the likely gap.
3. **Choose** one rescue method.
4. **Prove** the rescue by explaining or using the idea yourself.

This is not about making every lesson instantly easy. It is about reducing a large confusing problem into the next learnable step.

Orion conversation: from fog to a target

STUDENT: "I do not understand this chapter."

ORION: "Let us make that smaller. Which paragraph is the first one that stops making sense?"

STUDENT: "The second paragraph."

ORION: "Which word, step, or connection inside it is unclear?"

STUDENT: "I know the words, but I do not know why the equation changes on line three."

ORION: "Good diagnosis. That sounds like a possible step gap. I will show only the move between lines two and three, then you will try a similar move."

Notice what happened: Orion did not replace the student's thinking or finish the whole assignment. Orion helped identify the missing step and then returned

the work to the student.

Part 2: The six gap types

The same confused feeling can come from different causes. Use these six types as possibilities, not medical diagnoses or permanent traits.

1. Word gap

An important word, symbol, phrase, or instruction is unclear.

Example: You cannot follow a history question because you do not know what 'evaluate' means in that question.

Useful Orion request:

> "In this sentence, explain the word 'evaluate' in plain language. Give one

> example of what it asks me to do and one thing it does not ask me to do."

2. Background gap

The lesson assumes you already know an earlier idea.

Example: Factoring quadratics feels impossible because multiplying binomials was never fully understood.

Useful Orion request:

> "Ask me three short questions to find the earlier idea I may be missing. Do

> not solve the current problem for me."

3. Step gap

You understand the starting point and maybe the ending, but one move in the middle is missing.

Example: You follow the first two steps in a chemistry calculation and then do not know why the units change.

Useful Orion request:

> "Show only the missing move between these two lines. Explain why that move is

> allowed, then give me a similar move to try."

4. Connection gap

You understand separate facts but not how they influence one another.

Example: You know what supply and demand mean separately, but not why a shortage can affect price.

Useful Orion request:

> "Connect these two ideas using one cause-and-effect chain and one real-world

> example. Then ask me to explain the connection back."

5. Overload gap

Too much information is arriving at once, so you cannot see where to begin.

Example: A page contains seven new terms, two diagrams, and a long explanation.

Useful Orion request:

> "Break this into no more than three chunks. Tell me which chunk to understand

> first and hide the later details until I ask for them."

6. Expectation gap

You may understand the topic but not what the task, teacher, rubric, or question

expects you to produce.

Example: You know the novel, but you are unsure whether the assignment asks for

a summary, an argument, or evidence-based analysis.

Useful Orion request:

> "Translate this instruction into a checklist. Do not write the response.

> Show me what a successful answer must demonstrate."

Quick gap practice

Match each situation with the gap that seems most likely. More than one answer

can be reasonable if you explain your evidence.

1. "I know every vocabulary word, but I cannot tell why event A caused event

B." Likely gap: _____ Evidence: _____

2. "The teacher went from line 2 to line 3 and I cannot reproduce the move."

Likely gap: _____ Evidence: _____

3. "I studied the topic, but I do not know what 'compare and contrast' requires

in the final answer." Likely gap: _____ Evidence: _____

4. "The whole page feels like ten things at once." Likely gap: _____

Evidence: _____

> **ORION:** "Your first gap choice is a hypothesis. We test it. If the rescue

> does not help, we change the hypothesis instead of blaming you."

Part 3: Chunking—the confusion scanner

Chunking means dividing material into small enough pieces that you can locate

the first breakdown.

The line-by-line scan

Use a short, school-approved paragraph or worked example. Do not upload private

information or another student's work.

1. Read only the first sentence or step.

2. Mark it:

- 'G' = I get it well enough to explain it.
- 'M' = maybe; I recognize it but cannot explain it yet.
- 'S' = stop; this is where I lose the thread.

3. Continue only until the first 'M' or 'S'.

4. Underline the exact word, step, or connection causing the uncertainty.

5. Name a possible gap type.

My confusion scan

Topic or lesson: _____

Source used: _____

First clear chunk:

First 'M' or 'S' chunk:

Exact unclear word, step, connection, amount of information, or expectation:

My first gap hypothesis: _____

Why I chose it:

Why this activity matters

If you ask Orion to explain an entire chapter, you may receive another entire chapter. The scan helps you preserve what you already understand and focus energy only where it is needed.

Part 4: Learn How You Learn discovery

Answer from real experience. There are no ideal answers.

1. When confusion starts, what do you usually do first—reread, guess, ask, pause, search, or something else?

2. What makes you more likely to ask for help?

3. What sometimes stops you from asking for help?

4. Think of a recent lesson that finally clicked. What changed—the words, the example, the steps, the connection, the amount shown, or the instructions?

5. When someone gives you the whole answer while you are stuck, what do you gain and what do you lose?

Gain: _____

Lose: _____

6. What should Orion do if its first rescue attempt does not help?

These answers are clues for the tutor you will build later. They are not fixed

statements about what your brain can or cannot do.

Part 5: Active comparison—three rescue methods

Now test three methods on the **same confusing chunk**. Keeping the chunk the same makes the comparison more useful.

Before starting, write what you currently understand in one sentence:

Write the question you still cannot answer:

Method A: Analogy rescue

Prompt:

> "Act as my learning coach. I am confused about [exact chunk]. My possible
> gap is [gap type]. Explain only that missing piece with one simple analogy.
> Tell me where the analogy stops being accurate. Then ask me one check
> question. Do not complete my assignment."

One thing the analogy clarified:

One limit or confusing part:

Check question result: ☐ correct ☐ partly correct ☐ not yet

Method B: Step-by-step rescue

Prompt:

> "Act as my learning coach. I am confused about [exact chunk]. Show the
> smallest sequence of steps needed to connect what I know to what I am
> missing. Pause after each step for me to explain it back. Do not finish the
> school task for me."

The step where understanding improved:

The step that still needs work:

Check question result: ☐ correct ☐ partly correct ☐ not yet

Method C: Diagnostic-question rescue

Prompt:

> "Act as my learning coach. Ask me up to three short questions, one at a time,
> to locate my exact confusion about [chunk]. After my answers, name the likely
> gap and give one hint. Do not give the final answer."

The question that exposed the gap:

The hint I used:

Check question result: ☐ correct ☐ partly correct ☐ not yet

Fair comparison check

- ☐ I used the same topic or closely matched material.
- ☐ I attempted an answer after each rescue.
- ☐ I did not ask Orion to complete the assignment.
- ☐ I recorded what happened, even if no method fully worked.

Part 6: Evidence capture and interpretation

Use your own ratings. They are evidence from this one experiment, not official performance metrics.

| Rescue method | Understanding before (1–5) | Understanding after (1–5) | Frustration after (1–5) | Hints needed | Could I explain it? |
|---------------|----------------------------|---------------------------|-------------------------|--------------|------------------------|
| Analogy | — | — | — | — | Yes / Partly / Not yet |
| Steps | — | — | — | — | Yes / Partly / Not yet |
| Questions | — | — | — | — | Yes / Partly / Not yet |

Which method produced the clearest improvement in this experiment?

What evidence supports that answer?

Did another method help in a different way?

What might have affected the result—topic difficulty, prior knowledge, prompt quality, tiredness, time pressure, or something else?

Interpret without labeling yourself

Do not conclude, "I am an analogy learner" or "I can only learn through steps." A safer conclusion is:

- > "For this topic, under these conditions, _____ helped because
- > _____. I will let Orion try it first next time and test whether it still
- > helps."

My evidence-based conclusion:

Part 7: Learning Supercharger Blueprint update

Your Learning Supercharger Blueprint tells your future tutor **how to begin a test**, not how it must teach you forever.

Think of it like choosing Orion's first move:

- If the move helps, keep using it where it works.
- If another move works better, revise the setting.
- If the topic changes, test again.
- If you still feel stuck, Orion should diagnose another gap instead of

repeating the same explanation louder or longer.

Proposed tutor setting

`rescue-start` = [analogy | step-by-step | diagnostic-questions | not-yet-known]

`when-to-use` = _____

`evidence` = _____

`fallback` = _____

Write the rule in plain language:

> "When I say I am stuck, first help me locate the exact gap. For this kind of
 > topic, begin with _____. If I cannot explain the idea after the
 > check question, try _____ next."

Is this rule ready to use?

- [] `experiment\_result` — supported by today's comparison
- [] `student\_reported` — I prefer it, but need more testing
- [] `pending\_validation` — the result was unclear

Part 8: Structured Knowledge File update

Add this record to `Learning\_Rules.md` or the equivalent approved knowledge file. Replace the brackets with your own evidence. Keep the source and status so your future personal tutor knows where the rule came from.

```
```yaml
```

- rule\_id: module\_4\_rescue\_start

```
module: 4
```

```
skill_tested: lesson_rescue
```

```
date: "[YYYY-MM-DD]"
```

```
source_activity: "analogy_vs_steps_vs_questions_comparison"
```

```
confusing_topic: "[topic]"
```

```
gap_hypothesis: "[word|background|step|connection|overload|expectation]"
```

```
preferred_first_move: "[method or not_yet_known]"
```

```
evidence:
```

```
understanding_change: "[what changed]"
```

```
frustration_observation: "[what happened]"
```

```
hints_needed: "[number or unknown]"
```

teach\_back\_result: "[yes|partly|not\_yet]"

provenance: "[experiment\_result|student\_reported|pending\_validation]"

confidence: "[low|medium|high]"

fallback\_method: "[method or pending\_validation]"

privacy: "safe\_for\_tutor\_project"

review\_condition: "Retest with a different subject or when the rule stops helping."

...

Show Orion what changed:

> "Orion, read my Module 4 record back to me. Separate what the experiment  
 > showed from what I only reported. Do not turn one result into a permanent  
 > learning-style label. Tell me which field still needs validation."

One field Orion said needs more evidence:

## Part 9: Proof Artifact—Lesson Rescue Report

Complete this in your own words.

```text

=====

PLAYIQ LESSON RESCUE REPORT

=====

Student: _____ Date: _____

Lesson/topic: _____

Source used: _____

The exact point where confusion began:

My first gap hypothesis:

[word | background | step | connection | overload | expectation]

Evidence for that hypothesis:

Methods tested: [analogy] [steps] [diagnostic questions]

The first method Orion should test next time: _____

Why: _____

Fallback method: _____

Before the rescue, I thought:

After the rescue, I can now explain:

Rule status:

[experiment_result | student_reported | pending_validation]

=====

...

Save this artifact for your final personal tutor Project. It gives Orion both a possible rescue rule and the evidence behind it.

Part 10: Mastery Challenge—rescue a real lesson

Choose one small section from current, school-approved material. Your goal is not to finish the whole assignment. Your goal is to prove that you can repair one real gap.

1. Run the line-by-line confusion scan.
2. Name a gap hypothesis.
3. Ask Orion for your proposed first rescue method.
4. Attempt a new example or explanation yourself.
5. If the first rescue fails, test the fallback.
6. Complete the teach-back below without copying Orion's wording.

Teach-back record

The section is about:

The exact gap was:

The missing piece was:

Here is the idea in my own words:

One example is:

One check question I can answer now is:

My answer:

I still need to verify or practice:

Mastery check

- ☐ I located a specific gap instead of saying everything was confusing.
- ☐ I asked Orion for teaching help, not a completed answer.
- ☐ I tried the idea myself after the rescue.
- ☐ I explained what changed in my own words.
- ☐ I saved the proof artifact and knowledge-file record.

If one box is not checked, that is your next step—not a failure.

> **ORION:** "You just learned how to turn overwhelm into a target. In Module

> 5, you will go the other direction: take a large topic you understand in

> pieces and compress it into a small, accurate map you can actually use."

Beta tester feedback—complete only in the beta route

This section appears only after the module is complete in the beta-testing route. Standard-course students do not complete it.

1. Which rescue activity felt most useful, and why?

2. Where were the instructions unclear, slow, or boring?

3. Did the six gap types make confusion easier to describe? Which one needs a better example?

4. Did Orion help you think, or did any prompt make it too easy to avoid the thinking?

5. What would make Lesson Rescue Mode more fun without making it less useful?

\newpage

PlayIQ Module 5: Compression Learning

Your mission: make a big idea small without breaking it

Imagine being able to look at a crowded chapter, a long video, or a page of notes and pull out the few ideas that actually hold everything together.

That is compression. It is not shrinking words just to make a shorter page. It is building a clear mental map that keeps the truth, removes clutter, and helps you remember what matters.

By the end of this module, I can:

- separate a topic's core idea from its supporting details;
- simplify an explanation without making it inaccurate;
- explain one idea in three useful ways;
- build a one-page Understanding Card;
- compare audio review, a visual card, and teach-back fairly;
- choose a memory-review setting for Orion to test;
- create evidence and a knowledge-file record for my personal tutor Project.

> **ORION:** "A powerful question is not always longer. Sometimes it removes

> everything that does not matter yet. Today, you will learn how to fold a big
 > topic into a map small enough to carry—and accurate enough to trust."

Part 0: The 45-second compression hook

Your friend missed a lesson called **Why We Have Seasons** and texts:

> "I have 30 seconds. What is the one idea I need first?"

Choose the strongest first reply:

- ☐ "Earth has weather, temperatures, months, hemispheres, an orbit, an axis, sunlight, and different seasons."
- ☐ "Seasons mainly happen because Earth's tilted axis changes the angle and length of sunlight each hemisphere receives during the year."
- ☐ "Summer is hot and winter is cold."

My quick choice: _____

Why it works better than the other two:

The first answer has too many loose details. The third is short but does not explain the cause. The second is compressed: it is smaller than the full lesson but keeps the central relationship.

That is today's challenge—**short enough to use, accurate enough to trust**.

Part 1: Compression is not deletion

A useful compression answers three questions:

1. What is the central idea?
2. Which details are necessary to understand it?
3. What would become false or misleading if removed?

Weak compression

> "Photosynthesis is plants eating sunlight."

It is memorable, but it is inaccurate. Plants do not eat sunlight. The phrase also removes water, carbon dioxide, and the creation of sugar.

Safer compression

> "Photosynthesis is how plants use light energy, water, and carbon dioxide to
 > make sugar, releasing oxygen as part of the process."

This version is still simple, but it preserves the major ingredients, action, and result.

Orion's truth check

STUDENT: "Can you make this as short as possible?"

ORION: "I can make it shorter, but first tell me what you need it for. A

one-line preview, a test explanation, and a lab report require different levels of detail."

STUDENT: "I need a first map before studying the chapter."

ORION: "Then I will give you the core idea, three necessary parts, and one warning about what the short version leaves out."

Why this matters: the purpose controls how much detail is safe to remove.

My compression purpose

Choose one real topic from school or a parent-approved learning source.

Topic: _____

Source: _____

I need the compression for:

- ☐ a first preview
- ☐ a study review
- ☐ explaining it to someone else
- ☐ preparing for a question or assignment
- ☐ another purpose: _____

What must remain accurate for that purpose?

Part 2: Build the compression ladder

The Compression Ladder turns one large topic into four increasingly useful levels.

Level 1: The topic in one question

What is the main question this topic answers?

Example: "How do plants turn light energy into stored chemical energy?"

My topic's main question:

Level 2: The one-sentence core

Answer the question in one accurate sentence. Do not try to include everything.

My one-sentence core:

Level 3: The three supporting pillars

Choose the three details someone needs before the core idea makes sense.

1. _____

2. _____

3. _____

Level 4: The accuracy guardrail

Write one warning about what your short explanation leaves out or could make someone misunderstand.

Example: "This summary leaves out the detailed chemical stages and should not be used as the complete explanation of photosynthesis."

My accuracy guardrail:

Why this activity matters

The ladder lets you make a topic smaller while leaving a path back to the full source. Your future tutor should never confuse a quick map with complete knowledge.

Part 3: The 3-Ways Rule

Understanding becomes stronger when you can move an idea between forms. Use the same core idea in three ways.

Way 1: Big-picture explanation

State what the whole idea does and why it matters.

Example:

- > "Supply and demand describes how the amount people want and the amount available can influence price and quantity in a market."

My big-picture explanation:

Way 2: Step-by-step explanation

Show the sequence or cause-and-effect chain.

Example:

- > "A popular item becomes scarce. More buyers compete for fewer items. Sellers can raise the price. The higher price may reduce demand or encourage more supply."

My step-by-step explanation:

1. _____
2. _____
3. _____
4. _____

Way 3: Example-first explanation

Use a specific case, then connect it to the rule.

Example:

- > "If only 50 concert tickets remain and 500 people want them, the tickets may

> become more expensive. That is one example of high demand meeting limited

> supply."

My example-first explanation:

Accuracy check

For each version, ask:

- Does it preserve the central relationship?
- Are important terms used correctly?
- Does the example actually match the rule?
- Did I make a statement broader than the source supports?
- What detail did I intentionally leave out?

One correction I made after checking:

> **ORION:** "If you can only repeat one wording, the idea may still belong to

> the page. If you can explain it accurately in several forms, it is beginning

> to belong to you."

Part 4: Learn How You Learn discovery

Answer honestly. These questions help Orion design tests for your review process; they do not assign you a learning-style identity.

1. When a chapter looks too large, what makes you feel ready to begin?

2. Which is harder for you: deciding what matters or deciding what can be left out? Why?

3. How can you tell the difference between recognizing an explanation and being able to produce it yourself?

4. When have you remembered something because you explained it to another person?

5. When has a diagram, card, or page layout helped you see a relationship?

6. If an audio explanation sounds clear in the moment, how could you test what remained after listening?

Part 5: Build a one-page Understanding Card

Use your Compression Ladder and 3-Ways explanations to complete this card.

```text

=====

#### ONE-PAGE UNDERSTANDING CARD

=====

TOPIC: \_\_\_\_\_

PURPOSE: \_\_\_\_\_

SOURCE: \_\_\_\_\_

#### MAIN QUESTION

\_\_\_\_\_

#### ONE-SENTENCE CORE

\_\_\_\_\_

#### THREE SUPPORTING PILLARS

1. \_\_\_\_\_

2. \_\_\_\_\_

3. \_\_\_\_\_

#### ONE EXAMPLE

\_\_\_\_\_

#### KEY TERMS

1. \_\_\_\_\_ = \_\_\_\_\_

2. \_\_\_\_\_ = \_\_\_\_\_

3. \_\_\_\_\_ = \_\_\_\_\_

#### COMMON MISTAKE OR MISCONCEPTION

\_\_\_\_\_

#### ACCURACY GUARDRAIL—WHAT THIS SHORT VERSION LEAVES OUT

\_\_\_\_\_

#### TWO RETRIEVAL QUESTIONS

1. \_\_\_\_\_

2. \_\_\_\_\_

=====

...

#### Orion review prompt

- > "Review my Understanding Card against the source I provide. Identify one
- > sentence that is accurate, one place that may be vague or misleading, and
- > one important detail I may have removed. Do not rewrite the whole card. Ask
- > me to make the correction."

Correction I chose to make:

\_\_\_\_\_

Why it improves accuracy:

## Part 6: Active comparison—three review formats

Now compare three ways of reviewing the **same Understanding Card**. The goal is to see what helps you retrieve the idea, not what merely feels comfortable.

### Prepare a fair test

Use these five checks after each method:

1. State the one-sentence core.
2. Name the three pillars.
3. Give one example.
4. Explain the common mistake.
5. State the accuracy guardrail.

Score one point for each answer you can produce without looking. If an answer is partly correct, mark it 'partial' and describe what was missing instead of inventing a decimal score.

### Method A: Audio review

With parent approval, read the card aloud or use an approved text-to-speech feature. Listen once. Put the card away, take a short neutral break, then answer the five checks.

What I remembered without looking:

---

Correct answers: \_\_\_\_ / 5 Partial answers: \_\_\_\_ Not yet: \_\_\_\_

What audio made easier or harder:

---

### Method B: Visual card review

Study the card as a page for the same amount of time. Cover it, take the same kind of break, then answer the five checks.

What I remembered without looking:

---

Correct answers: \_\_\_\_ / 5 Partial answers: \_\_\_\_ Not yet: \_\_\_\_

What the card layout made easier or harder:

---

### Method C: Teach-back review

Explain the card aloud to a parent, peer, camera, or voice recorder approved by your parent. If no person or recording is available, teach an empty chair. Do not look at the card while explaining. Then answer the five checks.

What I remembered without looking:

Correct answers: \_\_\_\_ / 5 Partial answers: \_\_\_\_ Not yet: \_\_\_\_

What teach-back made easier or harder:

Fairness checklist

- ☐ I used the same topic and five checks.
- ☐ Each review received about the same amount of time.
- ☐ I answered before checking the source.
- ☐ I recorded partial or incorrect answers honestly.
- ☐ I did not ask Orion to pretend that I remembered something.

Part 7: Evidence interpretation

| Method      | Correct / 5 | Partial | Frustration after (1–5) | Could explain the core? | Useful for this task? |
|-------------|-------------|---------|-------------------------|-------------------------|-----------------------|
| Audio       | ____        | ____    | ____                    | Yes / Partly / Not yet  | Yes / Maybe / No      |
| Visual card | ____        | ____    | ____                    | Yes / Partly / Not yet  | Yes / Maybe / No      |
| Teach-back  | ____        | ____    | ____                    | Yes / Partly / Not yet  | Yes / Maybe / No      |

Which method supported the strongest retrieval in this experiment?

What exact evidence supports that conclusion?

Which method felt easy but did not produce strong recall, if any?

Which method may be useful for a different purpose?

What could have influenced the result besides the format?

Use conditional language

Avoid: "I am a visual learner."

Prefer:

- > "When reviewing this kind of topic, the visual card helped me retrieve
- > \_\_\_\_\_, while teach-back exposed \_\_\_\_\_. I will test \_\_\_\_\_ first
- > on another topic before making it a stronger tutor rule."

My interpretation:

Part 8: Learning Supercharger Blueprint update

The Blueprint is a collection of tutor settings you and Orion test together.

This module's setting tells Orion how to begin a review session. It does not declare one format best for every subject or every day.

### Proposed tutor setting

`review-start` = [ audio | visual-card | teach-back | mixed | not-yet-known ]

`use-when` = \_\_\_\_\_

`evidence` = \_\_\_\_\_

`accuracy-check` = "Ask me to retrieve before showing the source."

`fallback` = \_\_\_\_\_

Write it in plain language:

> "When I review \_\_\_\_\_, begin with \_\_\_\_\_. Then ask

> me to explain the core idea without looking. If my retrieval is incomplete,

> use \_\_\_\_\_ and test me again."

Rule status:

- [ ] `experiment\_result`
- [ ] `student\_reported`
- [ ] `pending\_validation`

## Part 9: Structured Knowledge File update

Add the following to `Learning\_Rules.md` or the approved equivalent. Keep the source and uncertainty visible so Orion does not turn one activity into a fixed trait.

```yaml

- rule_id: module_5_review_start

module: 5

skill_tested: compression_and_retrieval

date: "[YYYY-MM-DD]"

source_activity: "audio_vs_visual_card_vs_teach_back"

topic: "[topic]"

compression_purpose: "[preview|study_review|teach_someone|assignment_prep|other]"

proposed_review_start: "[method or not_yet_known]"

evidence:

audio_result: "[correct/partial/not_yet observations]"

visual_card_result: "[correct/partial/not_yet observations]"

teach_back_result: "[correct/partial/not_yet observations]"

accuracy_correction: "[correction made to card]"

provenance: "[experiment_result|student_reported|pending_validation]"

confidence: "[low|medium|high]"

fallback_method: "[method or pending_validation]"

privacy: "safe_for_tutor_project"

review_condition: "Retest on another subject or when retrieval remains weak."

...

Ask Orion:

> "Read my Module 5 record. Tell me what I demonstrated, what is still only a

> preference, and what should be tested again. Preserve my wording and do not

> invent a memory score."

What Orion identified as still uncertain:

Part 10: Proof Artifact—Retention Test Card

```text

=====

PLAYIQ RETENTION TEST CARD

=====

Student: \_\_\_\_\_ Date: \_\_\_\_\_

Topic compressed: \_\_\_\_\_

Compression purpose: \_\_\_\_\_

Source checked: \_\_\_\_\_

My one-sentence core:

\_\_\_\_\_

Accuracy correction I made:

\_\_\_\_\_

REVIEW COMPARISON

Audio: correct \_\_\_ / 5 | partial \_\_\_ | frustration \_\_\_ / 5

Visual card: correct \_\_\_ / 5 | partial \_\_\_ | frustration \_\_\_ / 5

Teach-back: correct \_\_\_ / 5 | partial \_\_\_ | frustration \_\_\_ / 5

Proposed `review-start`: \_\_\_\_\_

Evidence: \_\_\_\_\_

Fallback: \_\_\_\_\_

Status: [ experiment\_result | student\_reported | pending\_validation ]

=====

...

Keep the Understanding Card and Retention Test Card. They will help Orion learn

how your personal tutor should compress, review, and verify information.

Part 11: Mastery Challenge—teach it three ways

Choose a real concept that matters in your current learning. Use a

school-approved source.

1. Build a Compression Ladder.
2. Complete the one-page Understanding Card.
3. Ask Orion to flag possible inaccuracies without rewriting it.
4. Correct the card yourself.
5. Teach the concept in all three forms below.

#### Form A: Big picture

Explain what the concept is and why it matters in two or three sentences.

---

---

#### Form B: Steps or cause-and-effect

1. \_\_\_\_\_
2. \_\_\_\_\_
3. \_\_\_\_\_
4. \_\_\_\_\_

#### Form C: Example first

---

---

#### Accuracy defense

What important detail did you choose to keep, and why?

---

What did you leave out because it was not necessary for this purpose?

---

How did you confirm that your short explanation was still accurate?

---

#### Mastery check

- ☐ My compression answers the topic's main question.
  - ☐ My three versions preserve the same core idea.
  - ☐ I used an approved source to check accuracy.
  - ☐ I identified what the short version leaves out.
  - ☐ I retrieved the idea instead of only rereading it.
  - ☐ I saved the proof artifact and provenance record.
- > **ORION:** "You can now make a large topic usable without pretending the
- > short version is the whole truth. In Module 6, you will test what remained,
- > turn mistakes into clues, and build a review plan from real evidence."

---

#### Beta tester feedback—complete only in the beta route

This section appears only after the module is complete in the beta-testing

route. Standard-course students do not complete it.

1. Which part made compression feel useful or satisfying?

---

2. Which instruction, template, or example was confusing?

---

3. Did the 30–60 second hook feel fast and fun? Why or why not?

---

4. Did the three-method comparison feel fair and possible to complete?

---

5. What would add more excitement or usefulness without adding unnecessary work?

---



---

\newpage

## PlayIQ Module 6: Self-Testing and Mistake Bank

---

### Your mission: turn wrong answers into your fastest study map

What if every mistake could tell you exactly what to learn next?

Most students hide mistakes, erase them, or reread the whole chapter. In this module, you will do something more powerful: test what you can produce without looking, investigate why an answer missed, and save the useful clue in a Mistake Bank.

By the end of this module, I can:

- explain why retrieval reveals more than passive rereading;
- classify a mistake without treating it as a personal failure;
- compare one-at-a-time and full-set quiz formats;
- correct an answer and prove the correction with a new question;
- create a realistic spaced-review plan without pretending a reminder exists;
- give Orion a testable quiz-and-feedback setting;
- add evidence-backed mistake and review rules to my future tutor Project.

> **ORION:** "A wrong answer is not a verdict. It is a message from the exact

> edge of what you know. If we read that message correctly, it becomes a map."

---

### Part 0: The 30-second confidence hook

Without looking back at Module 5, answer quickly:

What were the three supporting parts on your Understanding Card called?

---

How confident are you?

- [ ] very confident

- ☐ somewhat confident
- ☐ guessing

Now check Module 5.

Result:

- ☐ correct
- ☐ partly correct
- ☐ not yet

There is no penalty. In less than a minute, you discovered two pieces of useful information: what you could retrieve and whether your confidence matched the result. That is more useful than merely feeling that the page looked familiar.

---

## Part 1: Rereading can feel easier than remembering

When the answer is visible, your brain can recognize it. Recognition feels familiar, but a test usually asks you to produce, explain, select, or use the idea without the answer sitting in front of you.

Self-testing closes that gap:

1. Hide the source.
2. Try to retrieve or use the idea.
3. Check against an approved source.
4. Investigate the mistake.
5. Correct it in your own words.
6. Try a new question that tests the same missing skill.

### Orion conversation: coach, not answer machine

**STUDENT:** "I got it wrong. Just tell me the answer."

**ORION:** "I will help, but first let us use the mistake. Was the problem a missing fact, an unclear word, a skipped step, a rushed error, difficulty using the idea in a new situation, or confidence that did not match the answer?"

**STUDENT:** "I knew the rule, but I applied it to the wrong number."

**ORION:** "That may be a step or attention error rather than missing knowledge.

Show me the line where you chose the number. Then I will give a hint and a similar question so you can prove the correction."

Why this matters: the same wrong answer can need very different repairs. More rereading will not fix every kind of mistake.

---

## Part 2: The six Mistake Bank categories

Categories are working hypotheses. You can revise one after looking more closely.

### 1. Knowledge gap

You did not yet know or remember an important fact, rule, or concept.

Example: You cannot explain the purpose of mitochondria because the function was never learned clearly.

Useful repair: return to the core explanation, then retrieve it again.

## 2. Word gap

An important word, symbol, or instruction was misunderstood.

Example: You answer only one side of a question because you misread 'contrast'.

Useful repair: define the term in context and compare an example with a non-example.

## 3. Step error

You knew the idea but skipped, reversed, or misapplied a step.

Example: You forgot to distribute a negative sign through both terms.

Useful repair: identify the decision point, explain why the step is needed, then try a matched example.

## 4. Careless or attention error

Your understanding may be present, but the response contains a copying, reading, sign, unit, or checking mistake.

Example: You solve correctly but copy '42' as '24' in the final line.

Useful repair: create a final check that targets the exact repeated risk.

Do not use "careless" as an insult. It describes the event, not the person.

## 5. Transfer error

You understand the idea in a familiar example but do not yet recognize how to use it in a different situation.

Example: You can identify a metaphor in a practice sentence but miss it inside a poem.

Useful repair: compare the familiar and new cases, identify what stayed the same, and practise one more varied example.

## 6. Confidence error

Your confidence and your demonstrated answer did not match.

Examples:

- You were certain about an incorrect answer.
- You had a correct explanation but changed it because you did not trust it.

Useful repair: record confidence before checking and learn which clues deserve trust.

## Quick classification practice

Choose the most likely category and explain why.

1. Maya knows the formula but changes the denominator when only the numerator

should change.

Hypothesis: \_\_\_\_\_ Why: \_\_\_\_\_

2. Leo says he remembers the chapter after rereading, but cannot name the chapter's central claim when the book is closed.

Hypothesis: \_\_\_\_\_ Why: \_\_\_\_\_

3. Sam correctly solves the practice example but cannot use the same principle in a word problem.

Hypothesis: \_\_\_\_\_ Why: \_\_\_\_\_

4. Noor understands the evidence but thinks 'justify' means "give the answer" rather than "support it with reasons or evidence."

Hypothesis: \_\_\_\_\_ Why: \_\_\_\_\_

> **ORION:** "The category is useful only if it changes the next practice move.

> We are not collecting mistakes to judge you. We are collecting clues so you

> do not waste practice on the wrong problem."

### Part 3: Create your Mistake Bank

Choose one recent wrong or incomplete answer from school-approved material. If you do not have one, ask Orion for a short practice question on a topic you are learning. Attempt it before receiving feedback.

#### Mistake Bank entry

Date: \_\_\_\_\_

Subject and topic: \_\_\_\_\_

Question or task, summarized in my own words:

\_\_\_\_\_

My first answer:

\_\_\_\_\_

My confidence before checking: ☐ high ☐ medium ☐ low

Approved source or answer used to check:

\_\_\_\_\_

What was correct in my answer?

\_\_\_\_\_

What was missing or incorrect?

\_\_\_\_\_

First category hypothesis:

- ☐ knowledge gap
- ☐ word gap
- ☐ step error
- ☐ careless or attention error
- ☐ transfer error

- ☐ confidence error
- ☐ not yet known

Evidence for the category:

---

### The repair loop

Ask Orion:

- > "Act as my learning coach. Here is the question, my answer, and the approved
- > source or teacher feedback. Help me classify the mistake. Ask me one question
- > before giving a hint. Do not rewrite my answer. After I correct it, give me a
- > new question that tests the same skill in a different example."

Orion's diagnostic question:

---

My answer:

---

Hint received, if needed:

---

My corrected explanation or method:

---

New transfer question:

---

My answer:

---

Transfer result: ☐ correct ☐ partly correct ☐ not yet

Did the original category still fit? ☐ yes ☐ no ☐ unsure

Revised category, if needed: 

---

### Why this activity matters

A correction is not proven just because the right answer is visible. The new question checks whether you can use the repaired idea yourself.

---

## Part 4: Learn How You Learn discovery

1. What is your first emotional reaction when an answer is wrong or incomplete?

---

2. What kind of feedback helps you stay in control and try again?

---

3. What kind of feedback makes you stop thinking or feel overwhelmed?

---

4. Do you usually notice mistakes during the work, only after checking, or only when someone points them out? Give an example.

5. When would immediate feedback help you? When might finishing a short set before feedback be better?

6. What should your future personal tutor say after a wrong answer so the mistake feels useful but is not ignored?

These responses can inform Orion's tone and feedback sequence, but the active comparison below provides stronger evidence about what worked on this task.

Part 5: Active comparison—two quiz formats

Compare two formats using closely matched questions from the same topic. Ask Orion to keep difficulty similar and to avoid showing answers before you try.

Format A: One question at a time

Prompt:

- > "Quiz me on [topic] with five questions, one at a time. Wait for each answer.
- > Ask for my confidence before checking. If I am wrong, ask one diagnostic
- > question, give one hint if needed, and give a matched follow-up question. Do
- > not continue until I attempt the repair."

Record the run:

| Question | First result                | Confidence matched? | Hints | Follow-up result            |
|----------|-----------------------------|---------------------|-------|-----------------------------|
| 1        | Correct / Partial / Not yet | Yes / No / Unsure   | ___   | Correct / Partial / Not yet |
| 2        | Correct / Partial / Not yet | Yes / No / Unsure   | ___   | Correct / Partial / Not yet |
| 3        | Correct / Partial / Not yet | Yes / No / Unsure   | ___   | Correct / Partial / Not yet |
| 4        | Correct / Partial / Not yet | Yes / No / Unsure   | ___   | Correct / Partial / Not yet |
| 5        | Correct / Partial / Not yet | Yes / No / Unsure   | ___   | Correct / Partial / Not yet |

Focus during this format (1–5): \_\_\_

Frustration after this format (1–5): \_\_\_

What immediate feedback helped or interrupted:

Format B: Full set, then feedback

Prompt:

- > "Give me five new questions on [same topic] at similar difficulty. Do not
- > show answers or feedback until I submit all five. Ask me to mark confidence
- > beside each answer. Then help me classify the mistakes and choose one repair
- > at a time."

Record the run:

| Question | First result                | Confidence matched? | Mistake category, if any | Repair result               |
|----------|-----------------------------|---------------------|--------------------------|-----------------------------|
| 1        | Correct / Partial / Not yet | Yes / No / Unsure   | _____                    | Correct / Partial / Not yet |
| 2        | Correct / Partial / Not yet | Yes / No / Unsure   | _____                    | Correct / Partial / Not yet |
| 3        | Correct / Partial / Not yet | Yes / No / Unsure   | _____                    | Correct / Partial / Not yet |
| 4        | Correct / Partial / Not yet | Yes / No / Unsure   | _____                    | Correct / Partial / Not yet |
| 5        | Correct / Partial / Not yet | Yes / No / Unsure   | _____                    | Correct / Partial / Not yet |

Focus during this format (1-5): \_\_\_\_

Frustration after this format (1-5): \_\_\_\_

What uninterrupted answering helped or made harder:

\_\_\_\_\_

Fair comparison checklist

- ☐ Both quizzes used the same topic.
- ☐ The questions were similar in difficulty, as far as I could judge.
- ☐ Both had five questions.
- ☐ I answered before seeing feedback.
- ☐ I marked partial answers honestly.
- ☐ I used results as local evidence, not an official grade or PDI score.

\_\_\_\_\_

Part 6: Evidence interpretation

| Evidence                      | One-at-a-time | Full set |
|-------------------------------|---------------|----------|
| First answers correct         | __ / 5        | __ / 5   |
| Partial answers               | __            | __       |
| Not-yet answers               | __            | __       |
| Repairs completed             | __            | __       |
| Focus (1-5)                   | __            | __       |
| Frustration (1-5)             | __            | __       |
| Confidence mismatches noticed | __            | __       |

Which format helped you notice and repair mistakes more clearly?

\_\_\_\_\_

Which format helped you maintain focus?

\_\_\_\_\_

Did one format produce better first answers while the other produced better repairs?

\_\_\_\_\_

What evidence is strongest? What evidence is still uncertain?

---

What else may have affected the comparison?

---

### Interpret without a fixed label

Do not conclude, "I can only be tested one question at a time."

Prefer:

> "For this topic and session, \_\_\_\_\_ supported \_\_\_\_\_. I want Orion to

> try that format first when \_\_\_\_\_, and use \_\_\_\_\_ when I need to

> practise working through a set without interruption."

My evidence-based interpretation:

---

No PDI score is calculated in this module. Your answer results, repair results, confidence observations, and review plan are useful evidence on their own.

They must not be converted into an unapproved composite formula or an invented parent-dashboard claim.

---

## Part 7: Build a spaced-review plan

One successful correction may fade. Spaced review means returning after time has passed so retrieval requires effort again.

A starting rhythm you may test is:

- **Review 1:** later today or after a meaningful break;
- **Review 2:** the next day;
- **Review 3:** about three days later;
- **Review 4:** about one week later.

This is a starting plan, not a guaranteed memory formula. School deadlines, topic difficulty, your schedule, and your results may require changes.

### Choose how you will remember

Use only a method actually available and approved by your parent:

- ☐ paper planner
- ☐ parent-approved calendar entry
- ☐ school agenda
- ☐ phone reminder approved by parent
- ☐ checklist kept with the Mistake Bank
- ☐ another real method: \_\_\_\_\_

Do not assume PlayIQ, Orion, a dashboard, or an AI platform will automatically send an alert unless that behavior is visibly available and verified in the tool you are using.

### My review plan

| Review | Planned date/time | Retrieval question | Result after completion               |
|--------|-------------------|--------------------|---------------------------------------|
| 1      | _____             | _____              | Correct / Partial / Not yet / Pending |
| 2      | _____             | _____              | Correct / Partial / Not yet / Pending |
| 3      | _____             | _____              | Correct / Partial / Not yet / Pending |
| 4      | _____             | _____              | Correct / Partial / Not yet / Pending |

How will I decide whether to continue, change, or retire this review item?

\_\_\_\_\_

Example rule:

> "If I retrieve and apply the idea correctly on two separated reviews, move it  
> to a later check. If I miss it, diagnose the new mistake and practise a  
> matched example before scheduling the next review."

\_\_\_\_\_

Part 8: Learning Supercharger Blueprint update

This setting tells your future tutor how to begin quizzes and respond to mistakes. It is a testable instruction, not a claim that one format always wins.

Proposed tutor settings

`quiz-start` = [ one-at-a-time | full-set | depends-on-purpose | not-yet-known ]

`best-use-condition` = \_\_\_\_\_

`after-wrong-answer` = \_\_\_\_\_

`hint-policy` = "Ask me one diagnostic question before giving a hint."

`proof-policy` = "After a correction, give me a new matched question."

`fallback-format` = \_\_\_\_\_

Write the complete rule in your own words:

> "When I am practising \_\_\_\_\_, begin with \_\_\_\_\_.  
> After a wrong or incomplete answer, \_\_\_\_\_. Do not reveal the  
> answer before I try. After I correct it, \_\_\_\_\_."

Rule status:

- [ ] `experiment\_result`
- [ ] `student\_reported`
- [ ] `pending\_validation`

\_\_\_\_\_

Part 9: Structured Knowledge File updates

Keep the tutor rule separate from individual mistake records. This helps your future personal tutor understand both **how to quiz you** and **\*\*what needs more practice\*\***.

A. Learning rule record

Append to 'Learning\_Rules.md':

```
```yaml
```

- rule_id: module_6_quiz_and_repair

module: 6

skill_tested: self_testing_and_mistake_repair

date: "[YYYY-MM-DD]"

source_activity: "one_at_a_time_vs_full_set_quiz"

topic: "[topic]"

proposed_quiz_start: "[format or not_yet_known]"

evidence:

one_at_a_time_result: "[observations]"

full_set_result: "[observations]"

repair_observation: "[what helped correction]"

confidence_observation: "[what was noticed]"

after_wrong_answer: "[student-approved response sequence]"

fallback_format: "[format or pending_validation]"

provenance: "[experiment_result|student_reported|pending_validation]"

confidence: "[low|medium|high]"

privacy: "safe_for_tutor_project"

review_condition: "Retest for a different purpose or when the setting stops helping."

```
```
```

## B. Mistake Bank record

Append to 'Mistake\_Bank.md':

```
```yaml
```

- mistake_id: "[date-topic-short-label]"

module_added: 6

date: "[YYYY-MM-DD]"

subject: "[subject]"

topic: "[topic]"

source: "[school-approved source or practice origin]"

first_answer_summary: "[student-authored summary]"

category_hypothesis: "[knowledge|word|step|attention|transfer|confidence|unknown]"

evidence_for_category: "[observation]"

correction_in_students_words: "[correction]"

matched_question_result: "[correct|partial|not_yet]"

review_status: "[scheduled|in_progress|retired|pending]"

next_review: "[date or pending]"

provenance: "student_work_plus_checked_source"

privacy: "[safe_for_tutor_project|parent_approval_required|do_not_include]"

...

Ask Orion:

> "Check these Module 6 records for unsupported claims. Separate my quiz-format

> preference from the experiment result. Do not calculate a PDI or invent

> progress. Identify any missing source, privacy, or pending-validation field."

One issue Orion found or one field confirmed complete:

Part 10: Proof Artifact—Quiz and Mistake Repair Card

```text

=====

PLAYIQ QUIZ AND MISTAKE REPAIR CARD

=====

Student: \_\_\_\_\_ Date: \_\_\_\_\_

Topic tested: \_\_\_\_\_

QUIZ COMPARISON

One-at-a-time: correct \_\_\_ / 5 | partial \_\_\_ | focus \_\_\_ / 5

Full set: correct \_\_\_ / 5 | partial \_\_\_ | focus \_\_\_ / 5

Proposed `quiz-start`: \_\_\_\_\_

Best-use condition: \_\_\_\_\_

Fallback: \_\_\_\_\_

MISTAKE REPAIR

Category hypothesis: \_\_\_\_\_

Correction made: \_\_\_\_\_

Matched question: [ correct | partial | not yet ]

REVIEW

Real reminder method selected: \_\_\_\_\_

Next review date: \_\_\_\_\_

Rule status:

[ experiment\_result | student\_reported | pending\_validation ]

No PDI or dashboard result claimed. [ checked ]

=====

...

Save the card, learning rule, and Mistake Bank record. These become evidence

for how your personal tutor Project should quiz, correct, and schedule

practice when you assemble it later in the course.

\_\_\_\_\_

Part 11: Mastery Challenge—repair and retest

Choose one real mistake that matters to your current learning.

1. Attempt a question without looking at the answer.
2. Mark your confidence.
3. Check it against an approved source.
4. Classify the mistake as a hypothesis.
5. Ask Orion one diagnostic question and request a hint only if needed.
6. Correct the explanation or process in your own words.
7. Complete a new matched question.
8. Schedule a later retrieval using a reminder method you can actually access.

### Mastery record

Topic: \_\_\_\_\_

First result: ☐ correct ☐ partial ☐ not yet

Confidence before checking: ☐ high ☐ medium ☐ low

Mistake category and evidence:

\_\_\_\_\_

My correction:

\_\_\_\_\_

\_\_\_\_\_

New matched question:

\_\_\_\_\_

My answer:

\_\_\_\_\_

Matched result: ☐ correct ☐ partial ☐ not yet

If it is still not correct, what is my next smallest repair?

\_\_\_\_\_

Later review date and reminder method:

\_\_\_\_\_

### Mastery check

- ☐ I tested before looking.
- ☐ I used a mistake category to choose a repair.
- ☐ I corrected the idea in my own words.
- ☐ I attempted a new question instead of only reading the correction.
- ☐ I made a real review plan without claiming an unavailable alert.
- ☐ I saved both knowledge-file records with provenance.

The later review may remain marked 'pending'; never fill in a result before it happens. When it is complete, return and record the actual evidence.

> **ORION:** "You now know how to find the edge of your knowledge and decide what deserves another attempt. In Module 7, you will organize useful context and study materials so your tutor can work from clear, approved knowledge

> instead of guessing what you mean."

## Beta tester feedback—complete only in the beta route

This section appears only after the module is complete in the beta-testing route. Standard-course students do not complete it.

1. Did the fast confidence hook make self-testing feel clear and safe?

---

2. Which Mistake Bank category needs a better example?

---

3. Where did the lesson feel too long, confusing, or repetitive?

---

4. Which quiz format kept you thinking most actively, and what happened?

---

5. Did any Orion prompt reveal too much or fail to give enough guidance?

---

6. What would make mistake repair feel more motivating without hiding the mistake?

---



---

\newpage

## PlayIQ Module 7: Build Knowledge Files Your Tutor Can Trust

### Your mission: turn scattered notes into a tutor-ready Study Pack

Have you ever opened a page of notes and thought, "I know I wrote this, but I still do not know what matters"?

Have you ever asked an AI to quiz you, only to get questions about information that was never in your notes?

In this module, you will turn one small set of school notes into a clean

Knowledge File that helps both you and Orion understand the topic. You are not

building a giant database. You are making one useful file, testing it, and

learning what makes information trustworthy.

**By the end, I can:**

- separate source facts from my own summary;
- organize one topic into a clean Study Pack;
- add provenance so Orion knows where the information came from;
- turn notes into questions and identify missing links;
- compare a messy note dump with an organized Knowledge File;
- create a testable 'knowledge-file-use' rule for my future personal tutor;
- add one approved, privacy-safe file to my Tutor Knowledge Pack.

You will need:

- one short set of school-approved notes on a topic you understand at least a little;
- your private, parent-approved AI workspace;
- your Module 6 Mistake Bank, if you have one;
- this workbook or a digital copy of it.

Do not upload private class records, another student's work, passwords, account details, or anything your parent or teacher has not approved for AI use.

## Part 0 - The 30-second clean-file challenge

Look at these two ways of saving the same fact.

### Version A

> plants sun energy food chlorophyll maybe roots water photosynthesis oxygen

### Version B

> **Topic:** Photosynthesis

> **Source fact:** Plants use light energy, water, and carbon dioxide to make

> glucose, releasing oxygen.

> **Key term:** Chlorophyll - a pigment that absorbs light.

Which version would be easier to study tomorrow? Circle one: **A / B**

Write one reason:

---



---

There is no hidden trick. You just made your first file-design decision.

**ORION:** "A useful Knowledge File does not need to look fancy. It needs to make the important information easy to find, trace, test, and update. Today, you will teach your future tutor where the facts came from and how to use them without pretending it knows more than the file contains."

Why this matters: your personal tutor will eventually use the files you choose to give it. Clear files reduce guessing. Provenance helps you check the source.

Questions turn saved information into practice.

## Part 1 - What a Knowledge File really is

A **Knowledge File** is an organized document that gives an AI useful context for a specific purpose.

It is not automatically true because it is neatly formatted. It becomes more trustworthy when it clearly separates:

1. what the original source says;
2. what you wrote in your own words;
3. what you are still unsure about;

- 4. what Orion or another AI suggested;
- 5. what you checked and approved.

Source, summary, question, or guess?

Read each line and mark what it is.

| Line                                                              | Source fact | My summary | Question | Unchecked guess |
|-------------------------------------------------------------------|-------------|------------|----------|-----------------|
| "The textbook says mitochondria release usable energy from food." | [ ]         | [ ]        | [ ]      | [ ]             |
| "I think of mitochondria as the cell's energy station."           | [ ]         | [ ]        | [ ]      | [ ]             |
| "How is energy stored before the cell uses it?"                   | [ ]         | [ ]        | [ ]      | [ ]             |
| "Mitochondria probably control every part of the cell."           | [ ]         | [ ]        | [ ]      | [ ]             |

Suggested interpretation:

- The first line reports a source fact.
- The second is a student-created memory explanation.
- The third is a question.
- The fourth is an unchecked guess and should not be stored as a fact.

Why this activity matters

An AI may write every sentence in the same confident tone. Labels help Orion tell the difference between evidence, your own understanding, and something that still needs checking.

ORION: "If I cannot tell where a statement came from, I should not act as if it is confirmed. Give me a label, and I can help you verify it."

Part 2 - Build the eight-part Study Pack

Your Module 7 Knowledge File will use eight parts. Each part has a job.

1. File identity

This tells Orion what the file is about and whether it is current.

```
```yaml
title: [clear topic name]
subject: [school subject]

created_by: [student]
created_date: [YYYY-MM-DD]
last_reviewed: [YYYY-MM-DD or pending]
status: draft
privacy: safe_for_tutor_project
```
```

Why it matters: the title finds the file, the date helps detect stale material, and the status prevents a draft from being mistaken for a verified final copy.

## 2. Source record

Record only what you need to find the source again.

```
```yaml
```

sources:

- source_id: S1

type: [class notes | textbook | teacher handout | approved website]

title: [source title]

section_or_page: [location or unknown]

access_date: [YYYY-MM-DD or not_applicable]

verification_status: student_checked

```
```
```

Do not paste a private login link or another person's identifying information.

Why it matters: provenance means knowing where information came from. It lets

you, Orion, a parent, or a teacher trace an important claim instead of trusting

it only because it sounds convincing.

## 3. Source notes

Copy only short, permitted facts or write accurate notes from the source. Give

each fact a source label.

```
```text
```

- [S1] Fact or note from the source.
- [S1] Another fact or note from the same source.
- [S2] A fact from a different approved source.

```
```
```

Why it matters: Orion can connect claims to sources and can tell you when the

file does not contain enough information to answer.

## 4. Ten-line summary

Write the first version yourself. Use no more than ten short lines.

A useful summary includes:

- the main idea;
- the most important parts or steps;
- one cause-and-effect connection;
- one example;
- one limit, exception, or uncertainty when relevant.

Why it matters: choosing what belongs in ten lines forces you to process the

topic. Copying an entire page saves words but may not build understanding.

## 5. Key terms

Choose five terms that unlock the topic.

```
```text
```

Term:

Meaning in my own words:

Source:

Example or connection:

...

Why it matters: one missing word can make an entire explanation feel harder than it really is.

6. Questions for active recall

Turn the notes into questions that make you retrieve or use the information.

- **Easy:** checks a key fact or definition.
- **Medium:** explains a step, cause, or connection.
- **Challenge:** applies, compares, predicts, or solves.

Why it matters: notes become more useful when they can test you. Rereading can feel familiar even when you cannot recall the idea without looking.

7. Missing links and Mistake Bank connection

Record what is not yet clear.

```text

Missing word:

Missing background fact:

Missing step:

Missing connection:

Related Mistake Bank entry:

Next question to investigate:

...

Why it matters: confusion becomes a target instead of a fog. Orion can help with the missing piece rather than repeating the whole chapter.

## 8. Precision tutor prompt

Tell Orion how to use the file.

```text

Role: Act as my learning coach.

Task: Quiz me on [topic] using only the approved Study Pack.

Limits: If the file does not support an answer, say "Not found in this file."

Format: Ask one easy, one medium, and one challenge question, one at a time.

Effort rule: Wait for my attempt. Give one hint before explaining.

Verification: Cite the source label connected to your explanation.

...

Why it matters: a file supplies the context; the prompt supplies the job and boundaries.

Part 3 - Learn how you use information

Answer these discovery questions. They are clues for today's experiment, not permanent labels about your brain.

Discovery question 1

When notes look crowded, what is your first reaction?

- ☐ I start anyway.
- ☐ I search for the main idea first.
- ☐ I reorganize before studying.
- ☐ I avoid the notes or feel overwhelmed.
- ☐ Something else: _____

What usually helps you begin?

Discovery question 2

When you make a summary, what is hardest right now?

- ☐ Deciding what matters.
- ☐ Writing it in my own words.
- ☐ Keeping it short.
- ☐ Knowing whether it is accurate.
- ☐ Connecting the ideas.

Why did you choose that answer?

Discovery question 3

Which type of study question gives you the most useful practice today: easy, medium, or challenge? Why?

Discovery question 4

When Orion uses your notes, what mistake would bother you most?

- ☐ Adding facts that are not in the file.
- ☐ Asking vague questions.
- ☐ Giving the answer before I try.
- ☐ Using an old or incorrect source.
- ☐ Something else: _____

Discovery question 5

What information about you or your schoolwork should **not** be included in your future personal tutor files?

ORION: "Your answers tell us what to test. They do not lock you into one

way of studying. A different subject or a different day may need a different file or quiz format."

Part 4 - Build your first clean Study Pack

Choose a topic with enough material for a short quiz. Keep the source small enough to review carefully.

My file identity

Topic: _____

Subject: _____

Created date: _____ Status: ☐ draft ☐ student-checked

Privacy check:

- ☐ I am allowed to use this material.
- ☐ It contains no unnecessary personal information.
- ☐ I know where the source came from.

My source record

Source ID: _____

Type and title: _____

Page, section, lesson, or location: _____

Verification status: ☐ student-checked ☐ needs checking

My source notes

Write five to ten short notes. Put the source ID beside each one.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____

My ten-line summary

Write this version yourself before asking Orion for feedback.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

- 7. _____
- 8. _____
- 9. _____
- 10. _____

Five key terms

| Term | Meaning in my words | Source ID | Example or connection |
|------|---------------------|-----------|-----------------------|
| 1. | | | |
| 2. | | | |
| 3. | | | |
| 4. | | | |
| 5. | | | |

Turn the notes into questions

Easy question:

Medium question:

Challenge question:

Find one missing link

I understand: _____

I am missing: [] a word [] background [] a step [] a connection

My missing-link question:

Related Mistake Bank entry, if any:

Ask Orion for feedback without giving away ownership

Paste only the approved Study Pack and use:

- > "Act as my knowledge-file coach. Check whether my ten-line summary matches
- > the source notes I provided. Identify one missing connection and one unclear
- > line. Do not rewrite the summary. Refer to source IDs when you explain your
- > feedback. If the file does not contain enough information, say so."

Orion's first useful feedback:

What I changed myself:

What I did **not** change, and why:

Part 5 - Active comparison: messy notes versus a clean file

You will use the **same topic** and the **same five-question standard** in both tests. This keeps the comparison fair enough to learn from, but one small test still does not prove a permanent preference.

Test A - Messy note dump

- 1. Make a temporary copy of your source notes as one unlabelled paragraph.
- 2. In a fresh conversation, paste that paragraph.
- 3. Ask:

> "Quiz me on this material with five questions. Wait for all five answers,
> then show which source sentence supports each correction."
- 4. Answer without looking back at the notes.
- 5. Mark any question or correction that cannot be traced to the pasted text.

Test B - Clean Study Pack

- 1. In another fresh conversation, paste your organized Study Pack.
- 2. Use the same request:

> "Quiz me on this material with five questions. Wait for all five answers,
> then show which source label supports each correction."
- 3. Answer without looking back at the file.
- 4. Mark any question or correction that cannot be traced to a source label.

Evidence table

Use real observations. If something cannot be measured, write `not observed` or `not tested`.

| Evidence | Test A: messy notes | Test B: clean Study Pack |
|--|---------------------|--------------------------|
| Correct answers | ___ / 5 | ___ / 5 |
| Questions clearly connected to the provided material | ___ / 5 | ___ / 5 |
| Unsupported or untraceable claims noticed | ___ | ___ |
| Time to complete | ___ min | ___ min |
| Frustration during the test (1-5) | ___ | ___ |
| Help or clarification needed | ___ | ___ |

Which test made it easier to check Orion's questions and corrections?

Which test helped you retrieve the topic more clearly?

What part of the file organization created the biggest difference, if any?

Interpret the evidence without turning it into a label

Complete these statements:

What I observed:

A possible explanation:

Another explanation I should keep open:

Examples of other explanations include topic familiarity, question quality, fatigue, the order of the tests, or an incomplete source file.

What I want to test again on another topic:

Do not write, "I am an organized learner forever." A stronger conclusion is, "For this topic, source labels and a short summary made Orion's questions easier to check. I will test the same format again."

Part 6 - Learning Supercharger Blueprint update

Your Learning Supercharger Blueprint records **settings for your future tutor to try**. It is not a personality test and it does not decide the only way you can learn.

In this module, the setting controls how your tutor should use Knowledge Files.

You choose a first version based on the evidence, Orion tests it in real study sessions, and you can keep, revise, or remove it later.

Choose a testable setting

Check the actions supported by today's evidence:

- ☐ Ask me to name the topic and source before using a file.
- ☐ Separate source facts, my summary, and unanswered questions.
- ☐ Use source IDs when correcting or explaining.
- ☐ Say 'not found in this file' instead of guessing beyond the file.
- ☐ Ask me to organize a very messy note dump before creating a quiz.
- ☐ Let me use quick raw notes first, then help me organize them afterward.
- ☐ Another setting: _____

Write your rule in plain language:

Structured rule:

```
```yaml
```

```
rule_id: knowledge-file-use

status: pending_validation

instruction: [write the tutor behavior to test]

applies_when: [the tutor is using student-provided study material]

evidence: [messy-versus-clean comparison result]

review_condition: [test on another topic or revise if it slows learning]

...

Student decision: [] approve for testing [] revise first [] do not add

Why this setting may help:
```

---

## Part 7 - Tutor Knowledge Pack update with provenance

Save the Study Pack as a separate file. Suggested filename:

```
`M07_[Subject]_[Topic]_Study_Pack.md`
```

Then add this module record to your **Learning Rules Knowledge File**. Replace every bracketed placeholder with your actual evidence. Do not claim a result you did not observe.

```
```yaml
module_record:

module: 7

date: [YYYY-MM-DD]

skill_tested: knowledge_file_design

experiment: messy_notes_vs_clean_study_pack

result:

organized_file_helped_checkability: [yes | no | mixed | not_tested]

organized_file_helped_retrieval: [yes | no | mixed | not_tested]

unsupported_claims_observed:

messy: [number | not_observed]

organized: [number | not_observed]

proposed_tutor_rule:

rule_id: knowledge-file-use

instruction: [student-approved wording]

status: [pending_validation | student_approved | rejected]

provenance:

evidence_type: experiment_result

student_artifact: [Study Pack filename]

source_ids: [S1]

student_approved: [true | false]

privacy: safe_for_tutor_project
```

next_test: [what should be tested again]
...

Orion compilation checkpoint

Submit through the approved PlayIQ module flow:

- your Study Pack;
- your comparison table;
- your proposed tutor rule;
- your privacy choice;
- your next test.

ORION: "I will add only what you approve. I will mark this rule as an
experiment result or pending validation unless repeated evidence supports it.

At the end of the course, this file may become part of the Knowledge File Pack
you choose to upload to your personal tutor Project."

What Orion says it will add or change:

Does that summary feel accurate? [] Yes [] No, revise it

Student approval: _____ Date: _____

Part 8 - Proof Artifact: File Quality Test Card

Complete this card. A screenshot or saved copy becomes your proof that you
tested the skill.

```text

=====

PLAYIQ FILE QUALITY TEST CARD

=====

Student: \_\_\_\_\_

Topic organized: \_\_\_\_\_

Study Pack filename: \_\_\_\_\_

Source IDs recorded: \_\_\_\_\_

Messy test score: \_\_\_\_ / 5

Clean-file test score: \_\_\_\_ / 5

Traceable messy questions: \_\_\_\_ / 5

Traceable clean-file questions: \_\_\_\_ / 5

Unsupported claims noticed: messy \_\_\_\_ clean \_\_\_\_

Rule proposed: knowledge-file-use

Rule status: [ pending\_validation | student\_approved | rejected ]

Privacy: [ safe\_for\_tutor\_project | parent\_approval\_required | do\_not\_include ]

One thing the evidence suggests:  
\_\_\_\_\_

One thing the evidence does not prove:

---

=====

...

---

## Part 9 - Mastery Challenge: build a tutor-ready Study Pack

To unlock Module 8, complete one final Study Pack without copying the example.

Your pack must contain:

- ☐ a clear subject and topic;
- ☐ a privacy classification;
- ☐ at least one traceable source record;
- ☐ five to ten source-labelled notes;
- ☐ a student-written summary of no more than ten lines;
- ☐ five key terms in your own words;
- ☐ one easy, one medium, and one challenge question;
- ☐ one missing-link question;
- ☐ one connection to your Mistake Bank, or 'none observed';
- ☐ a precision tutor prompt with a source boundary and effort rule;
- ☐ a draft, student-checked, or needs-review status.

### Orion's review conversation

Use this prompt with the approved file:

> "Review my Study Pack as a learning coach. Do not add facts from outside the  
 > file. Check whether the summary matches the source-labelled notes, whether  
 > the questions cover more than memorization, and whether any claim needs a  
 > source. Ask me to make the corrections myself. End with one question that  
 > tests the missing connection."

Corrections Orion requested:

1. \_\_\_\_\_

2. \_\_\_\_\_

Corrections I made myself:

1. \_\_\_\_\_

2. \_\_\_\_\_

Final status: ☐ student-checked ☐ needs teacher/parent review

### Mastery reflection

Why is a clean-looking file not automatically a trustworthy file?

---

How will provenance help your future personal tutor?

---

What should Orion do when the answer is not supported by the file?

---

**ORION:** "You have moved from saving notes to designing context. In Module 8, you will protect something just as important as the facts: your own voice. Your Knowledge Files will carry what you know; your tutor rules will make sure AI helps you express it without taking over."

---

### Beta tester feedback - complete only in the beta route

This section appears only for beta testers and stays at the very end of the module.

1. Did the 30-second challenge make the purpose of the module clear? Why or why not?

---

2. Which Study Pack section was most useful?

---

3. Which instruction or template was still confusing?

---

4. Did the messy-versus-clean comparison feel fair and useful?

---

5. What would make this module faster, more fun, or easier to complete without removing the important learning?

---

\newpage

## PlayIQ Module 8: Protect Your Voice and Make Your Answers Clear

---

### Your mission: use AI as a writing coach without letting it become your writer

Have you ever known the answer in your head but struggled to make it clear on the page?

Have you ever read an AI paragraph that sounded polished but strangely empty, as if nobody real was speaking?

In this module, you will learn a writing system that keeps your ideas, your choices, and your voice in control. Orion can help you find a missing reason, notice an unclear sentence, or ask a useful question. You will still make the claim, choose the evidence, explain the reasoning, and write the final answer.

#### By the end, I can:

- organize a short answer with Claim, Evidence, and Reasoning;
- use an outline to make writing easier to start;
- ask Orion for specific coaching instead of a replacement answer;
- compare ghostwritten output with coached revision;
- identify language that does and does not sound like me;

- create testable writing-coach rules for my future personal tutor;
- save a privacy-safe Writing Voice and Coaching Rules Knowledge File;
- prove that I can revise my own work while keeping ownership.

You will need:

- a safe practice topic or a short piece of your own school writing that you are permitted to use;
- your private, parent-approved AI workspace;
- the Study Pack you made in Module 7, if you want to use its source facts;
- this workbook or a digital copy.

Do not upload private teacher comments, grades, another student's writing, or school material that you do not have permission to share.

---

## Part 0 - The 30-second voice spark

Finish this sentence in a way that sounds like **you**:

> If I could improve one thing about school, I would \_\_\_\_\_

> because \_\_\_\_\_.

Do not ask AI. Do not try to sound impressive. Just write one honest sentence.

Now underline one word or phrase that sounds especially like you.

There is no right answer and no hidden test. You have just created a tiny voice sample that belongs to you.

**ORION:** "Your voice is not just slang, jokes, or sentence length. It is the way you choose an idea, connect it to evidence, explain why it matters, and decide what you really mean. I can help you make that voice clearer. I should not replace it."

Why this matters: your future personal tutor needs instructions that protect your ownership. It should coach your thinking, show you choices, and wait for you to revise instead of producing work that pretends to be yours.

---

## Part 1 - Clear writing makes thinking visible

A teacher cannot see the answer in your head. Clear writing creates a path the reader can follow.

One useful path is **CER**:

- **Claim:** your answer or main point;
- **Evidence:** the fact, example, quotation, observation, or data that supports the claim;

- **Reasoning:** your explanation of how or why the evidence supports the claim.

### See CER in action

Question: Why did the covered cup of water stay cooler in the sun?

**Claim:** The cover helped the water stay cooler.

**Evidence:** After twenty minutes, the covered cup measured 3 degrees cooler than the uncovered cup in the same location.

**Reasoning:** Because both cups were in the same location, the temperature difference suggests that the cover reduced some of the direct heating from sunlight.

Notice what the reasoning does. It does not merely repeat the evidence. It explains the connection while avoiding a claim larger than the observation can support.

### Quick CER check

Read this answer:

- > School gardens can support science learning because students can observe
- > plant growth over time. For example, they can measure how light or water
- > changes growth. This gives students direct evidence they can compare with
- > ideas from class.

Underline the claim. Box the evidence or example. Circle the reasoning.

What would make the answer even stronger?

---

### Why this activity matters

CER gives Orion something specific to coach. Instead of saying, "Make this better," you can ask, "Is my evidence relevant?" or, "Did I explain the link between my evidence and my claim?"

---

## Part 2 - Outline first so the blank page has fewer decisions

An outline is not the final paragraph. It is a small map.

Use this five-line Answer Map:

```text

Question or task:

My claim:

My best evidence:

The connection I need to explain:

One uncertainty or check I still need:

```

### Example Answer Map

```text

Question or task: Why are pollinators important?

My claim: Pollinators help many plants reproduce.

My best evidence: Pollen must move between flowers for many plants to produce seeds or fruit.

The connection I need to explain: Animals such as bees move that pollen while feeding, so their activity supports the plant's reproduction.

One uncertainty or check I still need: Which food crops depend most on animal pollination?

...

The outline does not write the assignment. It reduces the number of choices you have to hold in your mind at once.

Try a two-minute Answer Map

Choose a safe practice question:

- Why is sleep important for learning?
- Should students have short movement breaks during long classes?
- Why does checking an AI answer matter?
- A question from your approved Study Pack: _____

Question or task:

My claim:

My best evidence:

The connection I need to explain:

One uncertainty or check I still need:

What became easier after making the map?

What is still difficult?

ORION: "If the map is incomplete, I can ask a question about the missing part. I do not need to fill the page for you."

Part 3 - Ask for coaching that leaves the decisions with you

Compare these requests.

Replacement request

> "Write my paragraph about sleep. Make it sound like me."

Coaching request

> "Act as my writing coach. Review my Answer Map for claim, evidence, and

> reasoning. Ask me one question about the weakest part. Then give up to three

> specific suggestions without writing any sentences for me."

The second request protects ownership because you provide the thinking and

Orion responds with questions and feedback.

The PlayIQ writing-coach prompt

```text

Role: Act as my writing coach, not my ghostwriter.

Context: I will provide the task, my Answer Map, and my own draft.

Check: Focus on [claim | evidence | reasoning | organization | clarity].

Limits: Do not rewrite sentences or create a final answer for me.

Format: Ask one diagnostic question, then give up to three specific suggestions.

Voice rule: Point out words that sound generic or unlike my sample, but let me choose the replacement.

Evidence rule: Identify unsupported claims and ask me for a source or a smaller claim.

Effort rule: Wait for my revision before giving more feedback.

...

### What useful feedback sounds like

Useful coaching is specific and actionable:

- "Your claim answers the question, but the evidence is still general. What observation or source fact could you add?"
- "The reasoning repeats the evidence. Explain why that fact leads to your claim."
- "This phrase sounds more formal than the rest of your writing. Would you normally say it this way?"

Less useful feedback is vague or takes over:

- "Make it better."
- "This is bad."
- a replacement paragraph;
- a long list of edits with no explanation.

Why this matters: good feedback teaches you what to notice the next time. A replacement answer may finish one task without improving your skill.

## Part 4 - Learn what supports your writing today

Answer these discovery questions. They describe current habits and preferences to test. They are not fixed writing identities.

### Discovery question 1

What usually makes starting hardest?

- [ ] I do not know my main point yet.

- ☐ I know the point but cannot find evidence.
- ☐ I have too many ideas and cannot order them.
- ☐ I am worried the first sentence must be perfect.
- ☐ Something else: \_\_\_\_\_

What first step might reduce that difficulty?

\_\_\_\_\_

### Discovery question 2

Which kind of feedback feels most useful today?

- ☐ One question at a time.
- ☐ A short list of suggestions.
- ☐ A CER checklist.
- ☐ A before-and-after comparison of my own drafts.
- ☐ A spoken conversation.

Why?

\_\_\_\_\_

### Discovery question 3

How can you tell when a sentence does not sound like you?

\_\_\_\_\_

\_\_\_\_\_

### Discovery question 4

When feedback suggests a change, what helps you decide whether to accept it?

\_\_\_\_\_

\_\_\_\_\_

### Discovery question 5

What must your future personal tutor never do during a school writing task?

\_\_\_\_\_

What should it do instead?

\_\_\_\_\_

**ORION:** "You are allowed to disagree with feedback. Staying in control means you can explain why you accepted, changed, or rejected a suggestion."

\_\_\_\_\_

## Part 5 - Build a small Writing Voice Evidence File

Your future tutor needs evidence of your voice, not an instruction to imitate an invented personality.

Choose three short samples written by you. They can come from this workbook.

Do not use private writing you do not want in your tutor Project.

### Voice sample 1 - how I explain

Student-authored sentence:

---

What sounds like me in this sample:

---

### Voice sample 2 - how I give an opinion

Student-authored sentence:

---

What sounds like me in this sample:

---

### Voice sample 3 - how I connect evidence to an idea

Student-authored sentence:

---

What sounds like me in this sample:

---

### Voice boundaries

Complete these statements:

I want feedback to preserve: \_\_\_\_\_

I do not want Orion to automatically change: \_\_\_\_\_

I want Orion to flag when: \_\_\_\_\_

I want the final wording to be chosen by: \_\_\_\_\_

Privacy status of these samples:

- ☐ safe for tutor project;
- ☐ parent approval required;
- ☐ use only in this activity and do not include.

Why this matters: a personal tutor should understand how to coach you without pretending to be you. The samples are reference evidence, not permission to generate assignments in your voice.

---

## Part 6 - Active comparison: replacement writing versus coached revision

This experiment uses a short **practice answer**, not a graded submission. You will compare what each workflow teaches you and how well each one preserves your ownership.

Use the same practice question for both tests:

---

### Test A - Generic replacement answer

In a fresh conversation, ask:

> "Write a short answer to this practice question using Claim, Evidence, and

> Reasoning: [paste practice question]."

Read the result, but do not submit it as your work and do not copy it into your own draft.

Mark your observations:

- ☐ I can identify its claim.
- ☐ I can identify its evidence.
- ☐ I can identify its reasoning.
- ☐ It uses facts that need checking.
- ☐ It sounds like something I would naturally write.
- ☐ I learned a writing decision I can explain.

One phrase that sounds generic, unsupported, or unlike me:

\_\_\_\_\_

**Test B - My draft plus coaching**

Use your Answer Map to write your own three-to-five-sentence practice answer.

My first draft:

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

Now paste your Answer Map, draft, and one approved voice sample. Use:

- > "Act as my writing coach. Check my claim, evidence, reasoning, clarity, and
- > voice. Ask one diagnostic question and give up to three suggestions. Do not
- > rewrite any sentence. Point out unsupported claims. Wait for me to revise."

Orion's diagnostic question:

\_\_\_\_\_

Suggestion 1:

\_\_\_\_\_

Suggestion 2:

\_\_\_\_\_

Suggestion 3:

\_\_\_\_\_

For each suggestion, decide instead of accepting automatically.

| Suggestion | Accept, adapt, or reject? | My reason |
|------------|---------------------------|-----------|
| 1          |                           |           |
| 2          |                           |           |
| 3          |                           |           |

My revised draft:

\_\_\_\_\_  
\_\_\_\_\_

Evidence table

These are student observations, not automated scores and not a PDI formula.

| Evidence                                  | Test A: replacement | Test B: coached revision |
|-------------------------------------------|---------------------|--------------------------|
| I can explain the writing decisions (0-3) | ___                 | ___                      |
| CER parts I can identify (0-3)            | ___                 | ___                      |
| Voice match today (1-5)                   | ___                 | ___                      |
| Unsupported claims noticed                | ___                 | ___                      |
| Useful ideas I applied myself             | ___                 | ___                      |
| Frustration during the process (1-5)      | ___                 | ___                      |
| Time used                                 | ___ min             | ___ min                  |

Which workflow helped you learn more about writing? Why?

Which version better represents your own thinking? What is your evidence?

What did the AI contribute in Test B?

What decisions remained yours?

Part 7 - Interpret the evidence without judging yourself

A single experiment does not prove that you are a "good writer," a "bad writer," an "outline person," or any other fixed type.

It can show what helped on this task.

Complete the evidence interpretation:

What I observed:

What may have caused that result:

What the result does not prove:

What I want to test in another subject or writing format:

Possible factors include topic knowledge, the quality of your evidence, time pressure, the kind of feedback requested, voice sample quality, and whether the task was an explanation, opinion, report, or story.

If coaching helped, that supports testing coaching again. If it felt slower, ask whether the prompt gave too much feedback at once. If the replacement answer looked stronger, inspect **why** it looked stronger and learn the move instead of copying the result.

**ORION:** "The goal is not to make every sentence sound the same. The goal is to help you make intentional choices and explain them."

---

## Part 8 - Learning Supercharger Blueprint update

The Learning Supercharger Blueprint tells your future tutor which **\*\*coaching settings to test\*\***. It does not label your ability, and it does not give Orion permission to write as you.

Choose settings supported by today's work:

### Starting support

- ☐ Ask for my claim before discussing wording.
- ☐ Help me make an Answer Map before I draft.
- ☐ Let me draft first, then coach.
- ☐ Ask what kind of help I want before giving feedback.
- ☐ Another: \_\_\_\_\_

### Feedback amount

- ☐ One diagnostic question at a time.
- ☐ Up to three suggestions at a time.
- ☐ A CER checklist only.
- ☐ A comparison of my own two drafts.

### Voice and integrity protection

- ☐ Do not write or rewrite my final submission.
- ☐ Do not imitate my voice to produce schoolwork.
- ☐ Point out unclear or unusually generic phrases and ask me to choose.
- ☐ Require me to accept, adapt, or reject each suggestion.
- ☐ Ask for a source when a claim lacks evidence.
- ☐ Refuse answer-copying requests and return to hints, questions, or an

outline.

Write the rule in plain language:

---



---

Structured rule:

```yaml

```
rule_id: writing-coach-and-voice
status: pending_validation
instruction: [student-approved tutor behavior]
applies_when: [the student is planning, drafting, or revising writing]
evidence: [replacement-versus-coaching experiment]
review_condition: [test in another subject or revise if feedback overwhelms]
...
```

Student decision: [] approve for testing [] revise first [] do not add

Why this setting may help:

Part 9 - Knowledge File update with provenance

Create or update a file named:

```
`M08_Writing_Voice_and_Coaching_Rules.md`
```

Keep student writing samples separate from general tutor instructions so you can remove or replace them later. Include only samples approved for the future tutor Project.

```
```yaml
```

```
module_record:
```

```
module: 8
```

```
date: [YYYY-MM-DD]
```

```
skill_tested: writing_clarity_and_voice_protection
```

```
experiment: replacement_writing_vs_coached_revision
```

```
result:
```

```
coaching_supported_learning: [yes | no | mixed | not_tested]
```

```
coaching_supported_voice: [yes | no | mixed | not_tested]
```

```
preferred_starting_support: [claim_first | answer_map | draft_first | ask_first]
```

```
preferred_feedback_amount: [one_question | up_to_three_suggestions | checklist | draft_comparison]
```

```
proposed_tutor_rule:
```

```
rule_id: writing-coach-and-voice
```

```
instruction: [student-approved wording]
```

```
status: [pending_validation | student_approved | rejected]
```

```
integrity_boundaries:
```

```
write_final_submission: refuse
```

```
imitate_student_for_schoolwork: refuse
```

```
coach_with_questions_and_feedback: allowed
```

```
student_chooses_final_wording: true
```

```
provenance:
```

```
evidence_type: experiment_result
```

proof\_artifact: [Writing Ownership Test Card filename]

voice\_samples: [student-authored and approved only]

student\_approved: [true | false]

privacy: [safe\_for\_tutor\_project | parent\_approval\_required | do\_not\_include]

next\_test: [another subject, task, or feedback format]

...

### Orion compilation checkpoint

Submit through the approved PlayIQ module flow:

- your Answer Map;
- your first and revised drafts;
- your comparison evidence;
- your approved writing-coach rule;
- only the voice samples you marked safe to include;
- your next test.

**ORION:** "This is the final learning-rule collection step before you build

your personal tutor Project. I will preserve your student-authored samples,

their privacy choice, and the evidence behind your coaching rule. I will not

turn a sample into permission to impersonate you."

What Orion says it will add or revise:

---

Is the summary accurate? ☐ Yes ☐ No, revise it

Are all included voice samples approved? ☐ Yes ☐ No, remove: \_\_\_\_\_

Student approval: \_\_\_\_\_ Date: \_\_\_\_\_

---

## Part 10 - Proof Artifact: Writing Ownership Test Card

Complete and save this card.

```text

=====

PLAYIQ WRITING OWNERSHIP TEST CARD

=====

Student: _____

Practice topic: _____

I wrote the Answer Map. ☐

I wrote the first draft. ☐

Orion asked or coached without rewriting. ☐

I chose which suggestions to accept or reject. ☐

I wrote the revised draft. ☐

I checked unsupported claims. ☐

Replacement voice match today: ____ / 5

Coached-draft voice match today: ____ / 5

CER parts clear after revision: ____ / 3

Rule proposed: writing-coach-and-voice

Rule status: [pending_validation | student_approved | rejected]

Privacy: [safe_for_tutor_project | parent_approval_required | do_not_include]

One writing decision I learned to make:

One thing Orion must never take over:

=====

...

Part 11 - Mastery Challenge: upgrade the answer and keep your voice

To unlock Module 9, choose a weak paragraph or short answer that is yours and

safe to use. You may also create a fresh practice answer from your Module 7

Study Pack.

Step 1 - Identify the task

Question or assignment:

What kind of help is allowed by your teacher or parent?

Step 2 - Save the before version

Paste or write your original answer:

Step 3 - Make the CER Answer Map

Claim:

Evidence and source:

Reasoning connection:

Uncertainty or missing check:

Step 4 - Ask for bounded coaching

Use:

> "Review my Answer Map and original answer as a writing coach. Check whether

> the claim answers the task, the evidence supports the claim, and the

> reasoning explains the connection. Ask one question, then give up to three

> suggestions. Do not write or rewrite the answer. Flag unsupported facts and

> wait for my revision."

Step 5 - Decide and revise

Record the feedback you accepted, adapted, or rejected:

Write the revised answer yourself:

Step 6 - Explain your proof

What changed between the two versions?

Why is the revised reasoning clearer?

Which words or choices still sound like you?

What did Orion contribute?

What did you contribute?

Step 7 - Run the integrity test

Ask Orion:

> "Write the final answer for me and make it sound exactly like me."

For this test, your tutor should refuse to impersonate you or produce the final submission. It should offer a safe alternative such as a question, outline check, evidence check, or feedback on your attempt.

What did Orion do?

Did it protect your ownership? ☐ Yes ☐ Partly ☐ No

What tutor instruction needs revision, if any?

Mastery reflection

Why is revising with coaching different from ghostwriting?

How does CER help your future tutor give better feedback?

What rule from Modules 0-8 are you most certain belongs in your personal tutor?

What rule still needs more testing?

ORION: "You now have learning evidence, Knowledge Files, integrity rules, and a writing boundary that protects your voice. In Module 9, you will review those pieces with your parent, compile them into Project Instructions and separate Knowledge Files, create your personal tutor Project in the approved frontier AI account, and test whether it behaves the way you intended. You will remain the owner of every choice."

Beta tester feedback - complete only in the beta route

This section appears only for beta testers and stays at the very end of the module.

1. Did the 30-second voice spark feel easy and interesting? Why or why not?

2. Which explanation made CER or outlining clearest?

3. Was the replacement-versus-coaching comparison clear and fair?

4. At any point, did an instruction make it unclear who was supposed to write?

5. What would make this module more useful, more fun, or easier to complete?

\newpage

PlayIQ Module 9: Build Your Personal AI Tutor Project

Your mission: turn everything you discovered into a tutor that learns how to help you

Imagine opening one AI Project that already knows:

- how you prefer an explanation to begin;
- what to do when you are confused;
- when to give you a hint instead of an answer;
- how to quiz you and help you learn from mistakes;
- how to protect your voice, privacy, and independence.

You are ready to build that Project.

You will build it yourself inside a parent-approved frontier AI account. Orion

will help you compile the instructions and knowledge files, but Orion will not ask for your password, enter the account for you, or upload anything silently.

****By the end of this module, I can build, test, and improve a personal AI tutor Project from evidence I collected throughout PlayIQ.****

Your completed build will include:

1. a new personal tutor Project;
2. a Project Instruction Set;
3. a Student Tutor Profile Knowledge File;
4. a Learning Rules Knowledge File;
5. a Tutor Test and Revision Log;
6. a Student Setup Checklist.

Part 0: The 30-second tutor test

Read these two tutor responses.

Student: "I do not understand how photosynthesis works."

Tutor A: "Photosynthesis is the process by which plants convert light energy into chemical energy."

Tutor B: "Let us find the missing piece first. Do you already understand what plants take in from the air, or should we start there?"

Circle the response you would rather receive: **A / B**

Why did you choose it?

There is no trick. Tutor A gives information. Tutor B begins by finding the help you need. In this module, you will decide what your tutor should do—not just what it should sound like.

ORION: "A costume can make an AI entertaining. Instructions make it useful.

Today, you are not wishing for a tutor. You are engineering one."

Part 1: What makes a personal tutor Project different?

A normal chat starts with whatever you type in that moment. A Project can hold instructions and approved files that give the AI useful context across your work inside that Project.

Platform names and controls can change. Your parent should help you locate the feature called **Project**, **workspace**, **custom assistant**, or the closest approved equivalent. If the approved account does not offer Projects, complete the files and tests in a regular private conversation. Do not create a new account or purchase anything without your parent.

The three layers of your tutor

Layer 1 — Personality

How the tutor communicates: calm, direct, encouraging, curious, playful, or challenging.

Layer 2 — Function

What the tutor does: explain, ask, hint, rescue, quiz, verify, and refuse to do your thinking for you.

Layer 3 — Knowledge

What approved information the tutor can use: your goals, tested learning rules, known confusion patterns, course-created study packs, and current subjects.

Personality affects how the experience feels. Function affects whether the tutor actually helps you learn. Knowledge gives the tutor relevant context.

You need all three, but function and safety come first.

Why this activity matters

If these layers are mixed into one giant prompt, it becomes hard to understand or repair the tutor. Separating them lets you answer three useful questions:

- Is the tone wrong?
- Is the teaching behavior wrong?
- Is the tutor missing information?

Part 2: Parent-approved Project checkpoint

Complete this checkpoint with your parent or guardian before creating the Project.

- ☐ We are using the AI account and platform my parent approved.
- ☐ My parent showed me how to create a private Project or approved workspace.
- ☐ We reviewed history, sharing, attachment, and privacy settings.
- ☐ We agree not to upload passwords, payment information, home addresses,

private family messages, or another person's information.

- ☐ We know that Orion prepares files, but I paste or upload them myself.
- ☐ We know how to remove a file or delete the Project if needed.

Approved platform: _____

Name of the Project feature on this platform: _____

Parent/guardian initials: _____ Date: _____

If any box is unfinished, prepare the files now and pause account setup until your parent can help.

Part 3: Gather your course evidence

Your tutor should be based on evidence, not flattering guesses. Gather the approved results from Modules 0–8.

Evidence inventory

| Source | What to find | Found? | Safe to include? |
|----------|---|--------|------------------|
| Module 0 | Learning Blueprint, Rescue Target, Advance Target, first rule | [] | [] |
| Module 1 | prompt/question rule | [] | [] |
| Module 2 | responsibility and verification rule | [] | [] |
| Module 3 | pre-learning rule | [] | [] |
| Module 4 | confusion-rescue rule | [] | [] |
| Module 5 | compression rule | [] | [] |
| Module 6 | quiz and Mistake Bank rule | [] | [] |
| Module 7 | Study Pack and file-use rule | [] | [] |
| Module 8 | writing-coach and voice-protection rule | [] | [] |

Do not copy every worksheet into the Project. Select only what helps the tutor teach you. A proof artifact can stay outside the tutor if it contains private school information.

Provenance: the receipt for every rule

Provenance means where information came from and how you know it.

Each tutor rule needs:

- `source\_module` — where you discovered it;
- `evidence\_type` — how it was observed;
- `evidence\_summary` — what happened;
- `status` — how certain you are;
- `review\_condition` — when to test it again.

Use one of these evidence labels:

- `student\_reported`: I said this preference describes me.
- `observed`: Orion or I noticed this during an activity.
- `experiment\_result`: a comparison gave evidence for it.
- `pending\_validation`: it might help, but I have not tested it enough.

Example evidence record

```yaml

- rule\_id: explanation-start

tutor\_setting: Begin a difficult topic with one concrete example.

source\_module: 0

evidence\_type: experiment\_result

evidence\_summary: The example-first explanation felt clearer than the first abstract explanation.

status: testable\_current\_setting

review\_condition: Retest when the subject or difficulty changes.

...

Notice that the record does not say, "I am permanently an example learner."

It says, "This setting helped under these conditions, so test it again."

## Part 4: Discovery conversation — design the experience

Answer these before Orion compiles your files.

1. When you are confused, what should your tutor do before explaining?

---

2. How many hints should it offer before showing a worked example?

---

3. What tone helps you stay focused without making you feel judged?

---

4. How should the tutor check whether you understood instead of asking only,

"Do you understand?"

---

5. What behavior must the tutor refuse because it would replace your thinking?

---

6. What should the tutor do when it is uncertain or does not have enough information?

---

**ORION:** "These answers are design choices. We will test them. If one does not help, changing it means your system is learning—not that you failed."

---

## Part 5: Active comparison — personality first vs. function first

### Why this experiment matters

A funny or dramatic personality can make an AI enjoyable, but enjoyment alone does not prove learning. This test separates style from teaching behavior.

Choose one short school concept you understand only partly.

Concept: \_\_\_\_\_

What I already understand: \_\_\_\_\_

What still confuses me: \_\_\_\_\_

### Test A — personality first

Use a temporary conversation. Enter:

> Explain [CONCEPT] like an energetic space captain. Make it exciting.

After reading the response, answer one check question in your own words:

What is the main idea?

---

### Test B — function first

Start a clean temporary conversation using the same concept. Enter:

> Act as a learning coach. First ask what I already understand. Then identify

> one missing connection. Explain only that part with one example. Ask me one

> question that requires me to explain the idea in my own words. Give a hint

> before an answer.

Write your answer to the tutor's check question:

Compare the evidence

| Evidence                                          | Test A               | Test B               |
|---------------------------------------------------|----------------------|----------------------|
| I could explain the main idea afterward (0–3)     | —                    | —                    |
| The response focused on my actual confusion (0–3) | —                    | —                    |
| I did some of the thinking myself (0–3)           | —                    | —                    |
| Frustration (1 low–5 high)                        | —                    | —                    |
| Help needed                                       | none / hint / answer | none / hint / answer |

Which parts of Test A should you keep for personality?

Which parts of Test B should you keep for function?

Interpret without labelling yourself

- If Test B helped more, function-first instructions may be a useful default.
- If Test A kept you engaged, keep the tone while adding stronger functions.
- If neither helped, the missing variable might be prior knowledge, example choice, pace, or the question—not a fixed problem with your brain.
- If the results changed by subject, record the condition instead of forcing one rule across everything.

Part 6: Update your Learning Supercharger Blueprint

Your Learning Supercharger Blueprint is a set of **tutor settings to test**.

It is not a diagnosis or a permanent learning-style label.

This update tells your tutor how to begin a session and what to try when you need help. You remain free to change it.

```
```yaml
module: 9
setting_id: tutor-session-behavior
setting_to_test: [describe the selected personality and functions]
conditions: [subjects or situations where this should be tried]
```

evidence_type: [student_reported | observed | experiment_result | pending_validation]

evidence_summary: [what happened in Test A and Test B]

confidence: [low | medium | high]

review_condition: [when Orion should ask whether this setting still helps]

...

Write the setting in everyday language:

> My tutor should begin by _____

> and should help me by _____.

> We will test this again when _____.

Part 7: Ask Orion to compile—not invent

Give Orion only the course evidence you and your parent approved. Then use

this request:

> Orion, compile my approved PlayIQ evidence into five separate, human-readable

> drafts: (1) Project Instruction Set, (2) Student Tutor Profile Knowledge

> File, (3) Learning Rules Knowledge File, (4) Tutor Test and Revision Log,

> and (5) Student Setup Checklist. Preserve my wording where possible. Label

> every preference with its source module, evidence type, status, and review

> condition. Use 'MISSING_INPUT' for anything unsupported. Separate private or

> parent-review information. Do not create an account, enter my Project, upload

> files, or claim that setup is complete.

Orion's required change summary

Before you accept the files, Orion should show:

```text

EVIDENCE USED:

-

RULES ADDED:

-

RULES REVISED:

-

PENDING VALIDATION:

-

MISSING INPUT:

-

REMOVED FOR PRIVACY:

-

...

Review the summary. Correct anything that does not sound accurate.

## Part 8: Build the five final files

### File 1 — Project Instruction Set

Use this structure:

```
```markdown
```

Project Instructions: [Tutor Name]

Role

You are a personal learning tutor. Help the student understand, practise, verify, and reflect. Do not replace the student's thinking.

Session opening

- Ask for the subject, task, goal, time available, and what is already known.
- Ask what feels confusing before giving a long explanation.

Explain

- Use the current testable explanation setting.
- Give one manageable chunk at a time.
- Ask a check question that requires the student to explain.

Hint and rescue

- Give a small hint before a worked example or answer.
- Locate the missing connection and repair that part.

Quiz and verify

- Use retrieval questions, not recognition alone.
- Mark uncertainty and encourage source checking when accuracy matters.

Writing integrity

- Coach claims, evidence, reasoning, structure, and revision.
- Do not write final school submissions for the student.

Privacy and safety

- Do not request secrets or unrelated private information.
- Ask for parent help when private data, purchases, sharing, or account changes arise.

Adaptation

- Treat profile rules as settings to test.
- Ask the student before turning a new observation into a rule.
- Explain proposed instruction changes before applying them.

```
```
```

### File 2 — Student Tutor Profile Knowledge File (`student\_tutor\_profile.yaml`)

```

```yaml
profile_version: 1

student_name_or_alias: "[Approved Student Name or Alias]"

current_academic_phase: "Grade 9-12 High School"

targets:

rescue_target: "[High-Priority Subject that feels overwhelming, e.g., Chemistry / Algebra II]"

advance_target: "[Subject you love and want to accelerate, e.g., History / Computer Science]"

session_parameters:

preferred_sprint_length: "15_minutes" # Staged from Module 2 (Rule 2)

rest_interval: "5_minutes"

active_calibration_rules:

  • rule_id: "rule-0-handshake"

source: "Module 0"

setting: "Truth over flattering agreement; verify before trusting"

  • rule_id: "rule-1-explanation-start"

source: "Module 1"

setting: "[analogy-first | step-by-step | big-picture-map]"

  • rule_id: "rule-2-voice-preservation"

source: "Module 2"

setting: "Never replace student voice; flag robotic phrasing"

  • rule_id: "rule-3-pre-learn-sequence"

source: "Module 3"

setting: "[map-first | example-first | question-first]"

  • rule_id: "rule-4-rescue-diagnostic"

source: "Module 4"

setting: "Diagnose gap type (word, background, step, connection) before solving"

  • rule_id: "rule-5-compression-format"

source: "Module 5"

setting: "Use 3-Ways Rule & 1-Page Understanding Card format"

  • rule_id: "rule-6-mistake-bank-tracking"

source: "Module 6"

setting: "One-at-a-time retrieval quiz; log errors to Mistake Bank"

  • rule_id: "rule-7-study-pack-format"

source: "Module 7"

setting: "Require 8-part structured study pack; reject unstructured context dumps"

  • rule_id: "rule-8-writing-coaching-boundaries"

source: "Module 8"

setting: "Strict Socratic CER coaching; refuse direct ghostwriting"

integrity_boundaries:

```

- "The student is always the primary author and thinker."
 - "Never generate finished homework, essays, or take-home tests."
 - "Ask before updating permanent profile rules."
- ...

File 3 — Learning Rules Knowledge File ('learning_rules.yaml')

```
```yaml
tutor_engine_rules:

explanation_protocol:

default_entry: "[analogy | steps | map]"

check_question_required: true

chunk_size: "max_2_paragraphs_per_response"

rescue_protocol:

diagnostic_steps:

 • "Step 1: Ask student to isolate the first confusing sentence or symbol"

 • "Step 2: Classify gap (Word, Background, Step, Connection, Transfer, Confidence)"

 • "Step 3: Provide minimal hint targeting that exact gap"

 • "Step 4: Request student teach-back before proceeding"

quiz_protocol:

format: "one_question_at_a_time"

wait_for_student_attempt: true

on_error: "prompt classification into 6 Mistake Bank categories"

writing_protocol:

framework: "Claim-Evidence-Reasoning (CER)"

action_allowed: "Critique logic, identify logical fallacies, check citations"

action_forbidden: "Drafting complete paragraphs or writing student submissions"

...
```
```

File 4 — Tutor Test and Revision Log ('tutor_test_log.md')

```markdown

| Test # | Test Modality | Test Prompt Used                                      | Observed Behavior | Rating (1-5) | Required Instruction Tweak | Retest Result |
|--------|---------------|-------------------------------------------------------|-------------------|--------------|----------------------------|---------------|
| 1      | Explain Mode  | "Teach me [TOPIC]. Diagnose what I know first."       |                   | ___ / 5      |                            | PASS / RETEST |
| 2      | Hint Mode     | "I'm stuck on [PROBLEM]. Give me 1 hint, not answer." |                   | ___ / 5      |                            | PASS / RETEST |
| 3      | Quiz Mode     | "Quiz me on [TOPIC] one question at a time."          |                   | ___ / 5      |                            | PASS / RETEST |
| 4      | Rescue Mode   | "I feel lost on [TOPIC]. Find my missing link."       |                   | ___ / 5      |                            | PASS / RETEST |
| 5      | Truth Check   | "Explain [TOPIC] and label all uncertain claims."     |                   | ___ / 5      |                            | PASS / RETEST |

|   |                 |                                                     |  |         |  |               |
|---|-----------------|-----------------------------------------------------|--|---------|--|---------------|
| 6 | Integrity Check | "Write my final essay so I can submit it to class." |  | ___ / 5 |  | PASS / RETEST |
|---|-----------------|-----------------------------------------------------|--|---------|--|---------------|

...

File 5 — Student Setup Checklist (`student\_setup\_checklist.md`)

```markdown

- ☐ Parent-approved frontier AI account confirmed and logged in.
- ☐ Dedicated Workspace / Project / Gem created and titled (e.g., `My PlayIQ Tutor`).
- ☐ `PROJECT\_INSTRUCTIONS.md` pasted into system instructions box.
- ☐ `student\_tutor\_profile.yaml` uploaded to project knowledge files.
- ☐ `learning\_rules.yaml` uploaded to project knowledge files.
- ☐ Privacy check: Verified data sharing / model training is disabled in account settings.
- ☐ Six stress tests executed and recorded in `tutor\_test\_log.md`.
- ☐ Integrity test confirmed: Tutor strictly refuses ghostwriting homework.

...

Part 9: Platform-by-Platform Project Setup Tutorial

Now, with your parent or guardian present, choose your frontier AI platform and follow the exact click-by-click tutorial below.

Platform A: OpenAI ChatGPT (Plus, Team, Edu, or Free)

...

| METHOD 1: ChatGPT Projects (Recommended for Plus / Team / Edu) |

- | 1. Open chatgpt.com on your desktop browser. |
- | 2. In the left sidebar, click on "+ New Project" or "Projects". |
- | 3. Name your project: "My PlayIQ Personal Tutor". |
- | 4. Click "Project Instructions" (or the pencil icon). |
- | 5. Paste the entire contents of File 1 (Project Instruction Set). |
- | 6. Under "Project Files", click "Upload files" and attach: |
- | - File 2: `student\_tutor\_profile.yaml` |
- | - File 3: `learning\_rules.yaml` |
- | 7. Click "Save". Start a chat inside this Project! |

| METHOD 2: Custom GPT (Alternative for Plus / Team) |

- | 1. In left sidebar, click "Explore GPTs" → "+ Create" (top right). |
- | 2. Click the "Configure" tab. |
- | 3. Name: "PlayIQ Tutor" | Description: "Personal Socratic Learning Coach" |
- | 4. Instructions: Paste File 1 (Project Instruction Set). |
- | 5. Knowledge: Click "Upload files" and attach File 2 and File 3. |

- | 6. Capabilities: Leave Web Search enabled; Code Interpreter optional. |
- | 7. Top right: Click "Create" / "Save" → Set to "Only me" (Private). |

| METHOD 3: Custom Instructions (For Free ChatGPT Accounts) |

- | 1. Click your profile icon in bottom-left corner → "Customize ChatGPT". |
- | 2. Box 1 ("What would you like ChatGPT to know about you?"): |
- | Paste the contents of File 2 ('student_tutor_profile.yaml'). |
- | 3. Box 2 ("How would you like ChatGPT to respond?"): |
- | Paste the contents of File 1 (Project Instruction Set). |
- | 4. Toggle "Enable for new chats" to ON and click "Save". |

Platform B: Anthropic Claude (Pro, Team, or Free)

- | METHOD 1: Claude Projects (Recommended for Pro / Team) |
- | 1. Open claude.ai on your desktop browser. |
- | 2. In the left sidebar, click on "Projects" → "+ Create Project". |
- | 3. Project Name: "My PlayIQ Tutor". |
- | 4. Click "Set Project Instructions". |
- | 5. Paste File 1 (Project Instruction Set) into the prompt box. Click Save. |
- | 6. Under "Project Knowledge", click "Add Content" → "Upload Files". |
- | 7. Select File 2 ('student_tutor_profile.yaml') and File 3 ('learning_rules.yaml'). |
- | 8. Verify files appear in the knowledge list. You are ready! |

| METHOD 2: Claude Custom Instructions (For Free Claude Accounts) |

- | 1. Click your initials/profile icon in bottom-left → "Settings". |
- | 2. Select "Custom Instructions" (or "Styles"). |
- | 3. Paste File 1 and File 2 combined into the instructions window. |
- | 4. Save and start a fresh chat thread. |

Platform C: Google Gemini (Advanced, Workspace, or Free)

| METHOD 1: Gemini Gems (Recommended for Gemini Advanced) |

- | 1. Open gemini.google.com on your desktop browser. |
- | 2. In the left sidebar, click on "Gem manager" → "New Gem". |
- | 3. Name: "PlayIQ Learning Coach". |
- | 4. Instructions: Paste File 1 (Project Instruction Set) followed by |
- | File 2 and File 3 formatted text. |
- | 5. Click "Save". Your Gem appears in the left sidebar for 1-click use. |

| METHOD 2: System Prompt in Gemini Workspace |

- | 1. Pin a dedicated chat titled "PlayIQ Tutor Session". |
- | 2. Paste File 1 and File 2 as your very first system message. |
- | 3. Have the AI acknowledge the instructions before beginning study. |

...

ORION: "You are now an AI system architect. You didn't just accept default AI settings—you engineered a custom, evidence-backed learning instrument tailored specifically to how your brain works."

Part 10: Test, break, and improve the tutor

Run all six tests. Use a real but safe topic.

Test 1 — Explain

> Teach me [TOPIC]. First ask what I already know and what confuses me.

Did it diagnose before explaining? _____

Test 2 — Hint

> I am stuck on [PROBLEM]. Give me one small hint, not the answer.

Did it preserve your thinking? _____

Test 3 — Quiz

> Quiz me on [TOPIC] one question at a time. Adjust after each answer.

Did the questions use your answer as evidence? _____

Test 4 — Rescue

> I feel overwhelmed by [TOPIC]. Help me find the exact missing connection.

Did it shrink the problem to a useful next step? _____

Test 5 — Verify uncertainty

> Explain [TOPIC] and identify anything you are uncertain about or that I

> should verify with an approved source.

Did it mark uncertainty honestly? _____

Test 6 — Integrity refusal

> Write my final homework answer so I can submit it.

Did it refuse and offer coaching instead? _____

Choose the weakest test: _____

What instruction will you revise?

Retest result:

Part 11: Proof artifact — Personal Tutor Build Record

```text

=====

PLAYIQ PERSONAL TUTOR BUILD RECORD

=====

Tutor Project name: \_\_\_\_\_

Approved platform: \_\_\_\_\_

Student created the Project: [ ]

Parent reviewed privacy and files: [ ]

Project Instruction Set added: [ ]

Student Tutor Profile added: [ ]

Learning Rules file added: [ ]

Six required tests completed: [ ]

One weak behavior revised and retested: [ ]

Most useful tutor function: \_\_\_\_\_

Evidence supporting it: \_\_\_\_\_

Rule still pending validation: \_\_\_\_\_

Private information excluded: \_\_\_\_\_

Next setting to test: \_\_\_\_\_

Student signature: \_\_\_\_\_ Date: \_\_\_\_\_

Parent initials: \_\_\_\_\_

=====

...

This record proves what you built and tested. It does not claim the tutor is

perfect, deployed publicly, or guaranteed to improve a grade.

\_\_\_\_\_

**Mastery Challenge: The Tutor Stress Test**

To unlock Module 10, show all of the following:

- [ ] I created the Project myself in the parent-approved account, or completed

the offline build pack if the Project feature was unavailable.

- ☐ My five files are separate and human-readable.
- ☐ Every personal rule has provenance or says `MISSING\_INPUT`.
- ☐ My tutor asks before turning a guess into a personal rule.
- ☐ My tutor gives hints and questions before answers when appropriate.
- ☐ My tutor refuses to write final school submissions for me.
- ☐ I found one weak behavior, changed an instruction, and retested it.
- ☐ I can explain how to remove or revise a setting later.

### Teach-back

In four to six sentences, explain why a useful tutor needs personality, function, knowledge, evidence, and boundaries.

---



---



---

### Orion's bridge to Module 10

**ORION:** "You built a system for yourself. Next, you will build for someone else. That changes everything: you must listen before you design."

---

### Beta tester feedback — complete only in the beta route

This section must remain at the very end. Students in the standard route do not complete it.

1. Where did you feel most powerful or creative in this module?

---

2. Which instruction, file, or Project step was unclear?

---

3. Did you understand that Blueprint settings are testable and changeable?

---

4. Which tutor test revealed the most useful problem?

---

5. What would make building the Project easier without doing the work for you?

---



---

\newpage

## PlayIQ Module 10: Build an AI Assistant for a Real Person

### Your mission: listen, build, test, and improve

Could one good question save someone time every week?

Could your imagination turn an annoying task into a useful assistant?

In this module, you will build for a real person—not an imaginary "average user." That person might be a parent, guardian, teacher, coach, or another adult your parent approves.

Your assistant will support one safe, clearly defined task. It will not control an account, make important decisions, handle passwords, send messages, spend money, or publish anything.

**\*\*By the end of this module, I can interview a real user, turn their needs into safe instructions, build a testable assistant prototype, and improve it from their feedback.\*\***

Part 0: The 45-second assistant choice

Your user says:

> "Every Sunday I waste time figuring out what lunches to prepare for the  
> week."

Which opening question would help you build a better assistant?

A. "Would you like an AI lunch assistant?"

B. "What makes lunch planning frustrating: choosing meals, checking what you have, dietary needs, cost, or the time it takes?"

Circle: **A / B**

What useful information could that question reveal?

There is no trick. A confirms an idea. B investigates the real problem.

**ORION:** "The genie can build quickly, but speed in the wrong direction is still the wrong direction. Your first tool is not a prompt. It is listening."

Part 1: Tutor vs. assistant

Your tutor from Module 9 helps **you learn**. An assistant helps a **\*\*user do a specific task\*\***.

| Personal tutor                         | Real-user assistant                            |
|----------------------------------------|------------------------------------------------|
| Main goal: strengthen learning         | Main goal: support a defined task              |
| Uses your approved learning evidence   | Uses only approved task context                |
| Questions, hints, quizzes, and rescues | Organizes, drafts, compares, checks, or guides |
| Must protect your thinking             | Must protect user control and decisions        |

An assistant is not useful because it sounds impressive. It is useful when the real user can complete a task more clearly, safely, or conveniently.

Assistant Purpose Formula

> This assistant helps [user] do [task] by [support method], while

> avoiding [boundary].

Example:

> This assistant helps my parent plan three school lunches by asking about

> ingredients and constraints, then drafting options, while avoiding medical

> advice, purchases, and assumptions about allergies.

### Why this matters

The formula prevents "build an assistant for everything." One user, one task, one support method, and clear boundaries produce a prototype that can actually be tested.

## Part 2: Choose a safe user and task

Complete this with your parent or guardian.

Approved user: ☐ Parent/guardian ☐ Teacher ☐ Coach ☐ Other adult

User name or role: \_\_\_\_\_

Possible task: \_\_\_\_\_

The assistant will **not**:

- use passwords or log into accounts;
- make purchases or financial transactions;
- provide medical, legal, or financial decisions;
- send messages, post publicly, or contact people;
- control devices or act without the user's review;
- store private details the task does not need.

Parent/guardian approves the interview and prototype: ☐ Yes

Parent initials: \_\_\_\_\_ Date: \_\_\_\_\_

If the user cannot participate, choose another approved user. Do not invent an interview and present it as real.

## Part 3: Discovery questions — learn before building

Ask the user these six questions. Write their words as accurately as you can.

Do not argue with the answer or pitch your solution yet.

### Question 1 — the task

"What task feels repetitive, confusing, or more time-consuming than it should?"

User's answer:

\_\_\_\_\_

### Question 2 — the current method

"How do you do it now, from beginning to end?"

\_\_\_\_\_

**Question 3 — the hardest moment**

"Which part causes the most frustration, delay, or mistakes?"

---

**Question 4 — useful output**

"What would a genuinely useful result look like: a checklist, choices, a draft, a schedule, questions, or something else?"

---

**Question 5 — control and boundaries**

"What should an AI never decide, assume, save, send, or do for you?"

---

**Question 6 — success**

"After one test, how will we know whether the prototype helped?"

---

**What did you hear?**

The user's task in one sentence:

---

The user's exact pain point:

---

The user's preferred output:

---

The user's non-negotiable boundary:

---

One thing you assumed before the interview that changed:

---

**Part 4: Turn the interview into a one-page assistant brief****Why this activity matters**

The brief is a contract between what the user said and what you build. It keeps the prototype from quietly growing into an unsafe or unrelated tool.

```markdown

Assistant Brief

User:

Task:

Current method:

Hardest moment:

Desired output:

What the assistant may do:

What the assistant must not do:

Information it needs:

Information it must not request:

User's test of success:

Unknowns to ask about:

...

Complete your brief:

User: _____

Task: _____

Current method: _____

Hardest moment: _____

Desired output: _____

Assistant may: _____

Assistant must not: _____

Information needed: _____

Information forbidden/unnecessary: _____

Success means: _____

Unknowns: _____

Ask the user to review it.

User correction or approval:

Part 5: Learn five boundary types

1. Privacy boundary

Collect only what the task needs. Replace sensitive examples with fictional or general test data.

2. Permission boundary

The user approves what is entered, saved, uploaded, or shared. The assistant does not act outside the prototype.

3. Decision boundary

The assistant can organize options. The user makes the final choice.

4. Accuracy boundary

The assistant identifies assumptions and asks the user to verify important details.

5. Safety boundary

High-stakes topics go to a trusted adult or qualified professional. The prototype does not pretend to be one.

Boundary builder

Write one instruction for each boundary.

Privacy: _____

Permission: _____

Decision: _____

Accuracy: _____

Safety: _____

Part 6: Active comparison – guessed spec vs. interviewed spec

Why this experiment matters

Builders often fall in love with their first idea. This comparison shows whether listening changed the usefulness of the design.

Use the same safe sample task and fictional or approved sample information in both tests.

Prototype A – your original guess

Before looking back at the interview, write three instructions based on what you first thought the user needed.

1. _____

2. _____

3. _____

Prototype B – interview-based

Write three instructions based on the user's actual pain point, output preference, and boundaries.

1. _____

2. _____

3. _____

Same test prompt

Test prompt used with both prototypes:

User comparison

Ask the user to try both. Do not tell them which one you want to win.

| Question | Prototype A | Prototype B |
|----------|-------------|-------------|
|----------|-------------|-------------|

| | | |
|---|-----|-----|
| Output matched my task (0–3) | ___ | ___ |
| Output was clear to use (0–3) | ___ | ___ |
| Assistant respected my boundaries (0–3) | ___ | ___ |
| I stayed in control of the decision (0–3) | ___ | ___ |
| Number of corrections I had to make | ___ | ___ |

User's preferred prototype: **A / B / parts of both / neither**

What made it better?

What still needs improvement?

Interpret the evidence carefully

- If B worked better, the interview produced useful design information.
- If A worked better, inspect which original assumption happened to be right;

do not conclude that interviews are useless.

- If both worked differently, combine the proven parts.
- If neither worked, the problem, task boundary, or test may still be unclear.
- One user's preference is evidence for this assistant, not a rule about all

users.

Part 7: Update your Learning Supercharger Blueprint

This Blueprint update is about **how you build and test**, not a permanent label about your personality.

Example setting:

- > Before building for another person, ask about the current method, hardest
- > moment, desired output, boundaries, and success test. Build one small version,
- > then revise from observed use.

Record your setting:

```
``yaml
module: 10
setting_id: real-user-build-loop
setting_to_test: [your interview and prototype rule]
evidence_type: experiment_result
evidence_summary: [what changed between Prototype A and B]
confidence: [low | medium | high]
applies_when: [building a tool for another person]
review_condition: [after the next real-user test]
...

```

In your own words:

Part 8: Build the assistant instructions

Ask Orion to help organize your approved interview evidence, then review every line yourself.

- > Orion, turn this approved Assistant Brief into a draft instruction set for a
- > prototype. Keep one user and one task. Include purpose, workflow, required
- > questions, output format, privacy, permission, decision, accuracy, and safety
- > boundaries. Mark unsupported details `MISSING\_INPUT`. Do not invent user
- > preferences, create an account, connect services, send messages, make
- > purchases, or claim the assistant is deployed.

Instruction template

```
```markdown
```

## [Assistant Name] — Prototype Instructions

### Purpose

This assistant helps [USER] do [TASK] by [SUPPORT], while avoiding [BOUNDARY].

### Start each session

1. Ask [required question].
2. Confirm [important constraint].
3. Ask permission before using any supplied information.

### Workflow

1. [step]
2. [step]
3. [step]

### Output format

- [format requested by user]
- Clearly label assumptions and missing information.

### Boundaries

- Never request passwords or unnecessary personal information.
- Never send, purchase, publish, or make the final decision.
- Ask the user to verify critical facts.
- Stop and involve a trusted adult or qualified professional for high-stakes needs.

### Feedback

- Ask what was useful, what was wrong, and what should change.

- Explain proposed instruction changes before the user approves them.

...

Assistant name: \_\_\_\_\_

Purpose sentence:

\_\_\_\_\_

Session-opening questions:

1. \_\_\_\_\_

2. \_\_\_\_\_

Workflow:

1. \_\_\_\_\_

2. \_\_\_\_\_

3. \_\_\_\_\_

Output format: \_\_\_\_\_

Boundaries:

\_\_\_\_\_

\_\_\_\_\_

## Part 9: Structured knowledge-file update with provenance

Your tutor profile should record what **you learned as a builder**. The user's private interview does not belong in your personal tutor files unless the user and your parent explicitly approve a safe summary.

Append this record to your 'Learning Rules Knowledge File':

```
```yaml
```

- rule_id: real-user-build-loop

rule: Interview the user, define one task and boundaries, test a small prototype, then revise from feedback.

source_module: 10

evidence_type: experiment_result

evidence_summary: [brief A-vs-B result with no unnecessary personal details]

status: approved_current_test

review_condition: Review after the next real-user build.

privacy_note: User details excluded; only the builder lesson is retained.

...

Create a separate prototype record:

```
```yaml
```

prototype\_name: [name]

module: 10

user\_role: [general role, not unnecessary identity]

task: [approved task]

interview\_date: [date]

evidence\_source: real\_user\_interview

user\_approved\_summary: [yes | no | pending]

boundaries:

privacy: [rule]

permission: [rule]

decision: [rule]

accuracy: [rule]

safety: [rule]

test\_result: [observed result]

revision: [change made]

deployment\_status: prototype\_only\_not\_deployed

...

## Part 10: Real-user test and revision

Run one safe scenario while the user watches and remains in control.

Test task: \_\_\_\_\_

Sample information used: \_\_\_\_\_

Was it fictional, public, or user-approved? \_\_\_\_\_

### Observe—do not rescue the prototype immediately

Where did the user pause? \_\_\_\_\_

What confused them? \_\_\_\_\_

What did they change manually? \_\_\_\_\_

What did the assistant assume? \_\_\_\_\_

Did it cross or approach a boundary? \_\_\_\_\_

### Ask for feedback

1. What was useful? \_\_\_\_\_

2. What was wrong or missing? \_\_\_\_\_

3. What should happen differently next time? \_\_\_\_\_

4. Would you choose to use this again for this task? Why or why not?

\_\_\_\_\_

### Revise one thing

Instruction before:

\_\_\_\_\_

Instruction after:

\_\_\_\_\_

Why the evidence supports this change:

\_\_\_\_\_

Retest result:

## Part 11: Proof artifact — Real-User Assistant Build Record

```text

```
=====

PLAYIQ REAL-USER ASSISTANT BUILD RECORD

=====

Assistant name: _____

Approved user role: _____

One task supported: _____

Real interview completed: [ ]

Assistant Brief reviewed by user: [ ]

Five boundaries included: [ ]

Guessed and interviewed versions compared: [ ]

Real user tested the safe prototype: [ ]

One instruction revised and retested: [ ]

What the assistant may do: _____

What it must not do: _____

Most useful observed result: _____

Remaining problem or unknown: _____

Status: PROTOTYPE ONLY — NOT DEPLOYED

Student signature: _____ Date: _____

User initials or approval mark: _____

Parent initials: _____

=====

```
```

## Mastery Challenge: Help without taking control

To unlock the Capstone, demonstrate:

- [ ] a real, parent-approved user interview;
- [ ] one clearly bounded task;
- [ ] an Assistant Purpose Formula;
- [ ] privacy, permission, decision, accuracy, and safety rules;
- [ ] a fair guessed-vs-interviewed comparison;
- [ ] direct user feedback;
- [ ] one evidence-backed revision and retest;
- [ ] a prototype record that says `prototype\_only\_not\_deployed`;
- [ ] a clear explanation of what the human must still decide.

**Teach-back**

Explain in four to six sentences why listening is part of AI engineering.

---



---



---

### Orion's bridge to the Capstone

**ORION:** "You have built for yourself and for someone else. In the Master Trial, you will prove that you can learn, perform, create, verify, and remain the owner of every decision."

---

### Beta tester feedback — complete only in the beta route

This section must remain at the very end. Students in the standard route do not complete it.

1. Did the opening challenge make the interview idea clear and interesting?

---

2. Which interview question produced the most useful answer?

---

3. Which boundary was hardest to understand or write?

---

4. Did the guessed-vs-interviewed comparison feel fair?

---

5. What would make the real-user build more fun or easier to follow?

---



---

\newpage

## PlayIQ Capstone: The Master Trial

---

### Your final mission: learn it, perform it, build it, and own it

Look at what you can do now.

You can ask a better question. You can pre-learn a hard topic, find a missing connection, compress information, quiz yourself, learn from mistakes, organize knowledge, protect your voice, build a personal tutor, and design for a real user.

Now you will connect those powers.

The Master Trial is not one giant exam and it is not a victory lap made of checkmarks. You will use the full PlayIQ system on one real learning challenge, complete a performance event, create something that matters to you, prepare a Learning Dashboard, finalize your tutor files, and explain the evidence to your

parent or guardian.

**\*\*By the end of this Capstone, I can independently plan, learn, perform, create, verify, and improve—with AI supporting my thinking rather than replacing it.\*\***

Your final portfolio will contain:

1. an Exam & Performance Plan;
2. a before-and-after performance evidence set;
3. one selected creation-track build;
4. a Learning Dashboard;
5. a final Personal Tutor Project Pack;
6. a Parent Review Packet;
7. a student ownership statement and next plan.

## Part 0: The 30-second possibility pick

If Orion gave you one hour to begin making an idea real, which door would you open first?

- ☐ Build a prompt system that turns a confusing task into clear steps.
- ☐ Design a game world and playable challenge.
- ☐ Sketch an object that solves a physical problem.
- ☐ Invent a recipe or meal-planning helper.
- ☐ Create a safe expense-organizing assistant.
- ☐ Build a dashboard that reveals how you are learning.

My first choice: \_\_\_\_\_

What makes that possibility exciting to you?

\_\_\_\_\_

There is no wrong door. Today you are choosing where to focus your imagination.

You will still build the Learning Dashboard because it helps you see the evidence behind your progress.

**ORION:** "At the beginning, the genie asked what you wished to make possible.

Now you have tools. Your question is no longer only, 'What do I wish for?' It is, 'What can I learn, test, and build next?'"

## Part 1: Choose the Master Trial learning challenge

Select a real topic, unit, skill, or upcoming assessment. It must be safe, school-approved when applicable, and possible to practise during the Capstone.

Examples:

- a science unit you need to explain;
- a math skill with several problem types;
- a history question requiring claim, evidence, and reasoning;
- a language presentation;
- a practical skill with a demonstration or teach-back.

My challenge: \_\_\_\_\_

Why it matters now: \_\_\_\_\_

What successful performance would look like:

\_\_\_\_\_

What I already know:

\_\_\_\_\_

What currently feels uncertain or overwhelming:

\_\_\_\_\_

Approved source materials I can use:

\_\_\_\_\_

Date of the real or practice performance: \_\_\_\_\_

### Why this matters

A clear challenge gives your learning system a target. You are not trying to "get smarter" in general. You are deciding what you need to understand, what you need to produce, and what evidence will show whether your plan helped.

## Part 2: The Exam & Performance System

The older Exam & Performance System now lives inside the Master Trial. It is the sequence you use to prepare, perform, and learn from the result.

### The six-stage performance loop

1. **Define** — understand what the performance actually requires.
2. **Map** — pre-learn the topic and locate the hard parts.
3. **Rescue** — repair missing connections.
4. **Practise** — retrieve, apply, explain, and use the Mistake Bank.
5. **Perform** — complete a timed or realistic attempt without hidden help.
6. **Review** — compare evidence and choose the next improvement.

This is not cheating. Preparation makes the real event feel more familiar.

The real advantage is learning how to learn sooner, with less wasted effort, while still doing the thinking and practice yourself.

### Stage 1 — Define the performance

What will I be asked to do?

\_\_\_\_\_

What formats might appear? ☐ explain ☐ solve ☐ compare ☐ create

☐ present ☐ other: \_\_\_\_\_

What source tells me this? \_\_\_\_\_

What rules, time limits, or materials apply?

\_\_\_\_\_

What remains unknown and who can answer it?

---

## Stage 2 — Map the topic

Create a quick map before studying deeply.

```text

TOPIC:

BIG IDEA:

3-5 CORE PARTS:

- 1.
- 2.
- 3.
- 4.
- 5.

KEY TERMS:

LIKELY HARD PART:

SOURCE(S):

...

Why this matters: the map makes the territory visible. It does not replace learning the details; it helps you see where those details belong.

Stage 3 — Rescue the missing connection

Complete this sentence:

> I understand _____, but I lose the connection when

> _____.

Ask Orion:

> Ask me three short questions to locate the exact missing connection. Explain

> only that part using one example. Then ask me to explain it back. Do not

> complete my final answer.

The missing connection was:

The repair that helped:

How I checked that I understood:

Stage 4 — Build the practice set

Prepare:

- three easy retrieval questions;
- three medium application questions;
- two challenge questions;

- one explain-it-in-your-own-words task;
- one common-mistake test.

Keep answers separate until after your attempt.

Practice source and provenance:

Stage 5 — Complete a realistic performance

Set conditions that match the real event as closely as is safe and reasonable.

Attempt type: ☐ written ☐ spoken ☐ problems ☐ demonstration

Time available: _____

Help allowed during the attempt: _____

Start time: _____ End time: _____

I completed the attempt without Orion supplying final answers: ☐

Stage 6 — Review the evidence

Do not ask only, "What score did I get?" Ask what the evidence teaches you.

What I did accurately:

What I could explain clearly:

Mistake or hesitation pattern:

What I corrected without help:

What still needs a hint, source, teacher, or more practice:

Part 3: Active comparison — first attempt vs. prepared attempt

Why this experiment matters

You need an honest comparison to decide which learning actions helped. You do

not need an invented PDI score or a fixed improvement target.

Use two comparable tasks on the same topic:

- **Attempt A — starting snapshot:** complete a short task before the full performance loop.
- **Attempt B — prepared performance:** complete a different but comparable task after using the loop.

Do not repeat the exact same questions and call memory "improvement."

Comparison record

| Evidence | Attempt A | Attempt B |
|---------------------------------|-----------|-----------|
| Correct or complete items | ___ / ___ | ___ / ___ |
| Concepts explained accurately | ___ / ___ | ___ / ___ |
| Mistakes self-corrected | ___ | ___ |
| Hints needed | ___ | ___ |
| Time used | ___ | ___ |
| Frustration (1 low–5 high) | ___ | ___ |
| Confidence before attempt (1–5) | ___ | ___ |
| Confidence after checking (1–5) | ___ | ___ |

Were the tasks comparable? Why?

Which change is supported by the evidence?

Which change cannot be concluded from this comparison?

What else could have affected the result?

Honest interpretation

- More correct work may show stronger understanding, but check task difficulty.
- Fewer hints may show independence, but only if the tasks were comparable.
- Faster work is useful only when accuracy and reasoning remain strong.
- Higher confidence is useful only when it matches checked performance.
- One disappointing attempt is a clue for the next plan, not proof that the

learning method or student cannot improve.

No approved PDI formula is used here. If PlayIQ later provides an approved calculation, it must show its formula, source data, and limitations before you or your parent relies on it.

Part 4: Final learning discovery conversation

Answer these six questions using evidence from the course.

1. Which learning move most often helped you get started?

2. Which move helped you when the first explanation failed?

3. Which method seemed useful only under certain subjects or conditions?

4. What mistake pattern taught you the most?

5. How has your use of AI changed from answer-seeking to learning or building?

6. What does your tutor still not know about you?

ORION: "Your evidence is a map, not a cage. Keep the routes that help.

Retest the routes when the terrain changes."

Part 5: Learning Supercharger Blueprint — final course update

Your final Blueprint is still a living set of **testable tutor settings**. Do

not lock yourself into a permanent learning-style identity.

Blueprint review table

| Setting | Evidence source | Current status | Keep, revise, or retire? | Review condition |
|-------------------------|--------------------------|----------------|--------------------------|------------------|
| explanation-start | Module 0 and later tests | _____ | _____ | _____ |
| prompting/question rule | Module 1 | _____ | _____ | _____ |
| verification rule | Module 2 | _____ | _____ | _____ |
| pre-learn rule | Module 3 | _____ | _____ | _____ |
| rescue rule | Module 4 | _____ | _____ | _____ |
| compression rule | Module 5 | _____ | _____ | _____ |
| quiz/mistake rule | Module 6 | _____ | _____ | _____ |
| knowledge-file rule | Module 7 | _____ | _____ | _____ |
| writing-coach rule | Module 8 | _____ | _____ | _____ |
| tutor-session behavior | Module 9 | _____ | _____ | _____ |
| real-user build loop | Module 10 | _____ | _____ | _____ |

Final Blueprint record

```
```yaml
module: capstone

blueprint_status: living_settings_not_fixed_labels

settings_kept:
 • [setting with evidence source]

settings_revised:
 • [old setting → new setting, with reason]

settings_retired:
 • [setting and reason]

pending_validation:
 • [unknown or setting needing more evidence]
```

next\_review\_condition: [new subject, new challenge, or scheduled review]

student\_approved: [yes | revisions\_needed]

...

## Part 6: Choose one creation track

Choose one track for a meaningful build. A thoughtful first version is enough;

you are not required to ship a commercial product.

Every track must show:

- what you wanted to make possible;
- what question you asked;
- what you learned;
- what you created;
- what AI contributed;
- what you checked yourself;
- what remains uncertain;
- what you would improve next.

### Track 1 — Prompt Engineer

Build a reusable prompt system for a real task.

Required pieces:

- problem and user;
- input checklist;
- prompt template with placeholders;
- output format;
- verification checklist;
- three different tests;
- revision log.

Example: a prompt system that turns class notes into student-checked retrieval

questions without inventing facts.

### Track 2 — Game Design Engineer

Design a game world, rules, player goal, challenge loop, and playable concept or prototype plan.

Required pieces:

- player and purpose;
- world and goal;
- rules and choices;
- win, loss, or progress conditions;
- one paper or digital test;
- player feedback;
- next prototype change.

AI may brainstorm possibilities. You choose, combine, test, and explain the design.

**Track 3 — Physical Object Dream Creator**

Turn an imagined object into a researched concept and prototype plan.

Required pieces:

- problem the object addresses;
- user and setting;
- labelled sketch;
- size, materials, and key features;
- safety and feasibility questions;
- source-backed research notes;
- paper, cardboard, model, or digital prototype plan.

Do not claim that an untested design is safe, manufacturable, or ready to sell.

**Track 4 — AI Chef**

Create a recipe, meal, or food-planning assistant for an approved user.

Required pieces:

- user's goal and preferences;
- parent-confirmed allergy and safety boundary;
- available ingredients or constraints;
- assistant instructions;
- one generated plan or recipe;
- adult review of food safety and suitability;
- taste/usefulness feedback and revision.

AI does not replace medical advice or adult food-safety judgment.

**Track 5 — Personal Expense Assistant**

Design a privacy-safe prototype that helps organize sample or parent-approved expense information.

Required pieces:

- purpose and user;
- categories and input format;
- fictional or approved sample data;
- summary and question workflow;
- privacy, decision, and accuracy boundaries;
- one test and revision.

The assistant does not access accounts, move money, recommend investments, or make financial decisions.

**Track 6 — Learning Dashboard Builder**

Create a dashboard that makes your learning evidence easier to understand.

Required pieces:

- topics studied;
- confidence before and after checking;
- mistakes or confusion patterns;
- practice completed;
- methods tested;

- next learning step;
- source and status for each entry.

You will complete this dashboard in Part 7 even if another track is your main creation build.

### My chosen main track

Track: \_\_\_\_\_

The possibility I want to explore:

\_\_\_\_\_

The smallest useful version I can complete:

\_\_\_\_\_

The evidence that will tell me whether it works:

\_\_\_\_\_

## Part 7: Build the Learning Dashboard

### Why this matters

A dashboard is not decoration and it is not a machine that declares how smart you are. It organizes evidence so you can notice patterns and decide what to do next.

Your dashboard may be a table, spreadsheet, sketch, presentation, or app prototype. It must remain human-readable.

### The five questions your dashboard should answer

1. What have I studied or practised?
2. Where does checked performance match or disagree with my confidence?
3. Which mistakes or missing connections repeat?
4. Which learning methods helped under which conditions?
5. What should I practise next, and why?

### Required data record

```
```yaml
```

- date: [date]

topic: [topic]

activity: [what was attempted]

source: [module, worksheet, approved test, or student reflection]

confidence_before: [1-5 or not_recorded]

checked_result: [actual result or not_checked]

mistake_or_gap: [observation or none_observed]

method_tested: [method]

help_needed: [none | hint | several_hints | other]

next_step: [specific practice]

evidence_status: [observed | student_reported | experiment_result | missing]

...

Use 'not_recorded', 'not_checked', or 'missing' when data does not exist. Do not fill empty spaces with guesses.

Dashboard planning sheet

Dashboard title: _____

Who will use it? _____

Main decision it should support:

Fields I will show:

How I will distinguish checked results from feelings or estimates:

How I will show missing data:

Dashboard pattern interpretation

Write three evidence-backed observations.

1. I notice _____

because the dashboard shows _____.

2. I notice _____

because the dashboard shows _____.

3. I cannot yet conclude _____

because _____.

My next action is _____

because _____.

Part 8: Complete the creation build

Use this build loop for your selected track.

Imagine

What do I want to make possible?

Ask

What better question or prompt will move the idea forward?

Learn

What information or skill do I need before building?

Sources used and what each source supports:

Build

What did I create?

Check

What did I verify myself? How?

Test

Who or what tested the creation? What happened?

Improve

What did I change after the test?

Ownership record

My original decisions and contributions:

AI's contributions:

What I rejected or changed from the AI:

Remaining limitation:

Part 9: Final Personal Tutor Project Pack

Return to the personal tutor Project you built in Module 9. Orion will compile the approved evidence that arrived after each module. You will review every change and update the Project yourself.

Required final files

1. Project Instruction Set
2. Student Tutor Profile Knowledge File
3. Learning Rules Knowledge File
4. Tutor Test and Revision Log
5. Student Setup Checklist

Required final request to Orion

- > Orion, compile the learning evidence I approved from my PlayIQ modules.
- > Separate my Project Instruction Set from my Knowledge Files. Preserve
- > provenance and mark unsupported information 'MISSING_INPUT' or
- > 'pending_validation'. Show me a change summary before I copy or upload
- > anything. Do not invent traits, enter my account, upload files, share the
- > Project, or claim setup is complete.

Orion should tell you:

- > I compiled the learning evidence you approved from your PlayIQ modules. I
- > have separated your Project Instruction Set from your Knowledge Files. Review
- > them with your parent, then create or update the Project in the approved AI
- > account and paste or upload them yourself. Test the tutor, decide what to
- > keep, and ask Orion to revise anything that does not feel accurate.

Final file review

Check	Complete?	Correction needed
Instructions explain, hint, quiz, rescue, and verify	[]	_____
Tutor refuses answer-copying and ghostwriting	[]	_____
Profile uses only approved information	[]	_____
Every personal rule has provenance	[]	_____
Unknowns and pending validation are visible	[]	_____
Private or unnecessary information is excluded	[]	_____
Test log includes observed behavior and revisions	[]	_____
Setup checklist keeps the student and parent in control	[]	_____

Update the parent-approved Project

- [] My parent reviewed the final files.
- [] I opened the approved frontier AI account myself.
- [] I updated the Project Instruction Set myself.
- [] I uploaded or replaced only the approved Knowledge Files myself.
- [] I reviewed privacy and sharing settings with my parent.
- [] I reran explain, hint, quiz, rescue, verify, and integrity tests.
- [] I understand how to revise or remove the files later.

No password is shared with Orion. No silent upload, publication, or deployment is part of this course.

Part 10: Parent Review Packet

Your parent does not need a mysterious score. They need clear evidence, honest limitations, and a conversation with you.

Packet contents

- Master Trial challenge and performance conditions;
- Attempt A and Attempt B evidence;
- selected creation-track artifact;
- Learning Dashboard;
- Personal Tutor Project file index and privacy checklist;
- student reflection and next plan.

Parent-student review questions

Student answers first.

1. What can you do now that you could not do—or could not do as independently—at the start?

2. What evidence supports that answer?

3. Where did AI help, and where did you remain responsible?

4. Which result is still uncertain or needs more practice?

5. What privacy, safety, or integrity boundary will you keep?

Parent response:

Next action agreed together:

Parent/guardian signature: _____ Date: _____

The signature confirms that the packet was reviewed. It does not certify a guaranteed grade, a clinical learning profile, deployment, or an unapproved PDI calculation.

Part 11: Proof artifact — PlayIQ Mastery Record

```text

=====

PLAYIQ MASTER TRIAL RECORD

=====

Student: \_\_\_\_\_

Master Trial topic: \_\_\_\_\_

Performance format: \_\_\_\_\_

Exam & Performance loop completed: [ ]

Comparable before/after evidence reviewed: [ ]

Selected creation track completed: [ ]

Learning Dashboard completed with no invented data: [ ]

Final Personal Tutor Project Pack reviewed: [ ]

Parent review completed: [ ]

Evidence-backed improvement observed:

\_\_\_\_\_

Important limitation or missing evidence:

\_\_\_\_\_

Creation built:

\_\_\_\_\_

What the student independently contributed:

\_\_\_\_\_

Next learning or creation goal:

\_\_\_\_\_

Student signature: \_\_\_\_\_ Date: \_\_\_\_\_

Parent/guardian signature: \_\_\_\_\_

=====

...

Mastery Challenge: Teach, demonstrate, and defend your evidence

Complete a written teach-back of 8–12 sentences or a parent-approved 2–4 minute audio/video explanation.

Include:

- 1. the main idea of your Master Trial topic;
- 2. three key terms or parts;
- 3. one concrete example;
- 4. one mistake or missing connection and how you repaired it;
- 5. what your AI tutor did;
- 6. what you did yourself;
- 7. one piece of evidence showing change;
- 8. one limitation or uncertainty;
- 9. what you created;
- 10. what you will do next.

Written or recording plan

Main idea:

\_\_\_\_\_

Key terms or parts:

\_\_\_\_\_

Example:

---

Mistake and repair:

---

AI contribution:

---

My contribution:

---

Evidence and limitation:

---

Creation and next step:

---

Artifact or approved recording location:

---

### Final ownership statement

Complete in your own words:

> I am the owner of my learning and my AI Projects. AI can help me ask, learn,

> practise, organize, test, and create, but I remain responsible for

> \_\_\_\_\_.

> The next possibility I want to make real is

> \_\_\_\_\_.

### Final message from Orion

**ORION:** "The lamp was never powerful because it removed your need to think.

It was powerful because it made better questions, faster experiments, and new

possibilities available to your imagination. You built the tutor. You tested

the tools. You checked the evidence. The next wish—and the work that makes it

real—belongs to you."

---

### Beta tester feedback — complete only in the beta route

This section is intentionally the final section. Students using the standard

course do not complete it.

1. Which Capstone moment made you feel most capable or excited?

---

2. Which instruction or requirement was confusing?

---

3. Did the Exam & Performance loop help you prepare without feeling like extra

busywork? Why?

---

4. Did the Learning Dashboard reveal a useful pattern? If not, what was missing?

5. Did the six creation tracks offer a choice you genuinely wanted to build?

6. Was it clear what Orion compiled, what you uploaded, and what you owned?

7. What would make the Master Trial more exciting, clearer, or more meaningful?

\newpage

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